[abstract rewritten: citations moved to the introduction per PRD style (task 0.8); priority concessions retained in words. Change history in ACTION_PLAN.md and git.]

Gravitational Waves from Primordial Turbulence: What the Source Determines, and What It Does Not

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The gravitational-wave (GW) background sourced by turbulence in the early Universe is computed from three inputs: the spatial spectrum of the fluid, the unequal-time correlator (UETC) of the source, and the dissipation physics that terminates the cascade. These inputs are not on an equal footing, and the pipeline separates in its *asymptotes*: the strict $k \to 0$ infrared slope is fixed by the spatial model alone—a known causality theorem—and the ultraviolet by the temporal model alone, in a form whose aeroacoustic original predates the cosmological literature. What we add is the demarcation: every intermediate feature, the peak and the break alike, responds to both, and the observable band is intermediate. Two consequences follow. First, the causal infrared $\Omega_{\rm GW} \propto k^3$ is a property of the *quadratic* source and not of the fluid spectrum: it survives changing the infrared slope of the input, its bandwidth, the Mach number, the choice of stationary temporal model, and the removal of the temporal sector altogether. That the stress self-convolution floors at white noise is already established; we verify it through the GW kernel rather than through a spectral estimate, and with the temporal sector retained, by a band-split experiment in which the Kolmogorov range alone—an input with identically zero infrared power—still radiates a clean k^3. The k^3-versus-k^1 tension between analytic predictions and simulations therefore cannot be resolved on the spatial axis. It is not a disagreement about physics but about band: k^3 is the asymptotic law and k^1 the finite-duration regime above a break at the inverse source duration, a structure also obtained in the sound-shell and constant-source treatments. In observed frequency that break falls below LISA's energy-density window for an electroweak source over most of the parameter space, so that the causal asymptote usually lies below the instrument as well as below every simulation box; the exception is short-lived, small-scale sources, for which the break rises above the window and the causal side becomes the observable. Second, the displacement of the GW peak to the source scale requires only that the source correlator have a kink at zero time lag—a condition met by any source of finite lifetime and specific to no decorrelation model, since all cusped correlators collapse onto a single prediction. That the transform of a cusped correlator decays as ω^{-2} is a standard Abelian result, and its consequences for GW spectra, including for the peak, have been drawn before for coherent envelopes; what is new here is that the same argument applies at zero lag of a stationary, decorrelating UETC. This remains conditional on a measurement not yet made: the hydrodynamic UETC is Gaussian, and the magnetic anisotropic-stress UETC of a decaying, magnetically dominated cosmological source has not been measured. We close by arguing that the remaining freedom lies almost entirely in the third input: the dissipation microphysics of the early Universe is neither Navier–Stokes nor of unit magnetic Prandtl number, and no existing calculation or simulation of the GW signal uses the correct one. We identify that gap and set out how to close it, but do not close it here.

CONTENTS

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I. INTRODUCTION

A turbulent or magnetically turbulent early Universe radiates gravitational waves, and the shape of that background is one of the few observables that carries information about the fluid state of the plasma at the electroweak or QCD epoch [1–4]; more broadly, a stochastic background is one of the few probes of the first second that is not screened by the photon last-scattering surface [5–7]. The question has acquired an observational edge: pulsar-timing arrays now report evidence for a stochastic background in the nanohertz band [8], for which a cosmological turbulent source at the QCD epoch is among the candidate interpretations [9], while LISA is expected to probe the millihertz band where an electroweak-scale source would peak [10]. Independently, the case for a primordial magnetic field capable of driving such turbulence rests on blazar observations [11] and is bounded from above by the cosmic microwave background [12]; the generation and subsequent evolution of such fields is reviewed in Refs. [13–15], with the QCD crossover a candidate epoch for their production [16]. Turbulence is one of three sources a first-order transition is expected to produce, alongside bubble collisions and the longer-lived acoustic phase [17]; it is the least well characterised of the three, and the only one whose prediction depends on a closure assumption about the source statistics. Extracting that information requires a calculation with three separable inputs. One must specify the *spatial* spectrum of the source, the *temporal* correlation of the source with itself at unequal times, and the *dissipation* physics that sets where the cascade terminates. The literature has treated these three very unequally: the spatial spectrum is argued over in detail, the temporal correlator has been modelled by a coherent (separable) kernel [18, 19], by Kraichnan's random-sweeping approximation [2, 20–22], and most recently measured directly in simulations [23]; whereas dissipation is almost universally reduced to a single number ν entering through a cutoff wavenumber whose precise form is then argued to be unimportant [24], or dropped altogether [23].

This paper is an attempt to say precisely which features of the GW spectrum each input controls, and—more usefully—which features no choice of input can change. The organising result, developed in Sec. II D and verified model by model throughout, is that the pipeline factorises in its *asymptotes*: the strict $k \to 0$ infrared slope is determined by the spatial model alone, the far ultraviolet by the temporal model alone, and every intermediate feature—the peak and the break alike—responds to both. That statement is not a modelling convenience but a consequence of the source being quadratic in the fields, and it converts several open disagreements in the literature into questions with definite answers. The qualification matters and is not a hedge: a source of finite duration moves the infrared by two powers over a band [9, 25, 26], and every accessible band lies inside it, so the factorisation is a statement about limits rather than about what a detector sees. ~~and the peak is the only feature that responds to both~~ [the abstract and the conclusions were qualified to asymptotes in the first pass; the introduction still carried the unqualified form]

The most prominent of those disagreements is the infrared slope. Analytic treatments give a causal $\Omega_{\rm GW} \propto k^3$ below the peak, a slope which follows from the analyticity of the stress correlator at long wavelength and is in that sense model-independent [4, 18, 27]; the simulations of Ref. [28] reports a much flatter k^1. We show in Sec. V that both sides in fact use the same relation between the GW and the stress spectrum, so the disagreement cannot be about the gravitational-wave step at all; it is about the stress. That the stress cannot inherit a spectrum steeper than white noise, because it is a self-convolution of the fluid spectrum, is not new: it is the explanation Ref. [28] themselves give, it is derived for stress spectra in Ref. [29] and modelled through the stress evolution in Ref. [30], and it is already implicit in Refs. [18, 25]. What we add is a sharper test of it (Sec. V E) and the observation that it settles the stress without settling the gravitational waves: from the *same* white-noise stress the two branches of the finite-duration response give k^3 and k^1, so what is actually in dispute is which branch the observed band lies in. ~~and~~

~~it reduces to a single assumption, $\Omega_T \propto \Omega_M$, which is inconsistent with the stress being a self-convolution of the fluid spectrum.~~ [pre-empted (RoperPol:2019wvy, Brandenburg:2019uzj, Sharma:2022ysf) and retracted in Sec. V; the abstract and Conclusion 3 dropped this localisation in the first pass and the introduction did not] The convolution floors at white noise below the peak no matter how steep the fluid infrared is, and it does so even for a Gaussian field.

We then test that claim as directly as we can, by splitting the input spectrum into its causal k^4 band and its Kolmogorov $k^{-5/3}$ band and sourcing GWs from each alone (Sec. V E). The inertial band, whose input has no infrared power whatsoever, still radiates a clean k^3. Neither band's infrared slope, nor the bandwidth, nor the Mach number, nor the temporal model—nor deleting the temporal sector entirely—moves the exponent. The practical consequence is negative and worth stating plainly: no spatial modification of the source can bend the analytic k^3 towards the simulated k^1.

Sec. IV takes up the second disagreement, the location of the peak, where the sweeping prediction $\sim Mk_0$ and the simulated $\sim 2k_0$ differ by a factor $\sim 2/M$. We show that the displacement to the source scale is controlled by a single number—the slope of the source correlator at zero lag—and therefore that it requires only a kink there, a condition satisfied by any source with a finite lifetime. This considerably widens the circumstances under which the source-scale peak is expected, and sharpens the question that would overturn it.

Finally, Sec. VII argues that the residual freedom in the pipeline is concentrated in the input that has received the least attention. The early Universe is not a Navier–Stokes fluid at unit magnetic Prandtl number, which is what every GW calculation and every GW simulation assumes; it is a plasma whose momentum transport crosses between diffusive and free-streaming regimes, and whose magnetic Prandtl number exceeds unity by many orders of magnitude [31, 32]—a regime in which the magnetic field cascades far below the viscous scale of the velocity field [33]. We set out what that implies and what remains to be computed.

Throughout, Appendix B collects every derivation in this paper as a single uninterrupted chain of equations, in real space first and then in the spectral representation, so that any step can be checked without reading the surrounding prose. Appendix A is a table of symbols.

II. GRAVITATIONAL WAVES FROM A TURBULENT SOURCE

Before specializing to particular spectral and temporal models, we collect the standard analytic derivation of the gravitational-wave (GW) spectrum sourced by turbulence [1–4, 34]. This fixes the conventions used throughout and, in particular, shows why the remaining sections reduce the problem to computing the contracted source correlator $H_{ijij}(\mathbf{k},\omega)$ of Eq. (12) on the diagonal $\omega = k$.

A. Sourced wave equation

Write the tensor perturbations of a spatially flat FLRW background as

$$ds^2 = a^2(\tau)\left[-d\tau^2 + (\delta_{ij} + h_{ij})\, dx^i dx^j\right], \qquad \partial_i h_{ij} = h_{ii} = 0, \tag{1}$$

with h_{ij} transverse-traceless (TT). The linearized Einstein equations give

$$h''_{ij} + 2\frac{a'}{a}\, h'_{ij} - \nabla^2 h_{ij} = 16\pi G\, a^2\, \Pi^{\mathrm{TT}}_{ij}, \tag{2}$$

where $' = d/d\tau$ and Π^{TT}_{ij} is the TT part of the anisotropic stress. For an incompressible, non-relativistic turbulent fluid the source is the Reynolds stress,

$$\Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau) = \Lambda_{ij,lm}(\hat{k})\, T_{lm}(\mathbf{k},\tau), \qquad T_{lm} = w\, v_l v_m, \qquad \Lambda_{ij,lm} = P_{il}P_{jm} - \tfrac{1}{2}P_{ij}P_{lm}, \tag{3}$$

with $w = \bar{\rho} + \bar{p}$ the enthalpy density, $P_{ij} = \delta_{ij} - \hat{k}_i\hat{k}_j$, and Λ the TT projector. In the magnetohydrodynamic case the same expression holds with the Reynolds stress replaced by, or added to, the Maxwell stress B_iB_j; the conditions under which the primordial plasma admits an MHD description are set out in Ref. [35]. The double TT projection together with the Gaussian (Wick) reduction of the quartic velocity correlator, the standard quasi-normal closure of homogeneous isotropic turbulence [36–38], collapses the two-point function of (3) onto the contracted form $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ that appears in Eq. (12).

Turbulence acts on eddy-turnover times short compared with a Hubble time [1, 18], so over the source lifetime we drop the expansion ($a \approx 1$, friction negligible) and Eq. (2) becomes, in Fourier space, a forced oscillator,

$$h''_{ij}(\mathbf{k},\tau) + k^2 h_{ij}(\mathbf{k},\tau) = 16\pi G\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau). \tag{4}$$

B. Green-function solution and the GW spectrum

The retarded solution of Eq. (4) and its time derivative are

$$h_{ij}(\mathbf{k},\tau) = \frac{16\pi G}{k}\int_{\tau_i}^{\tau} d\tau'\, \sin[k(\tau-\tau')]\,\Pi_{ij}^{\rm TT}(\mathbf{k},\tau'), \qquad \dot h_{ij}(\mathbf{k},\tau) = 16\pi G\int_{\tau_i}^{\tau} d\tau'\, \cos[k(\tau-\tau')]\,\Pi_{ij}^{\rm TT}(\mathbf{k},\tau'). \tag{5}$$

With the Isaacson energy density $\rho_{\rm GW} = \langle \dot h_{ij}\dot h_{ij}\rangle/(32\pi G)$, valid for wavelengths short compared with the background curvature scale [34], and the unequal-time correlator (UETC) of the TT stress,

$$\langle \Pi_{ij}^{\rm TT}(\mathbf{k},\tau_1)\,\Pi_{ij}^{\rm TT*}(\mathbf{k}',\tau_2)\rangle = (2\pi)^3\,\delta^{(3)}(\mathbf{k}-\mathbf{k}')\,\Pi(k;\tau_1,\tau_2), \tag{6}$$

inserting (5) and averaging the product $\cos[k(\tau-\tau_1)]\cos[k(\tau-\tau_2)] \to \frac{1}{2}\cos[k(\tau_1-\tau_2)]$ over the fast GW oscillations gives the master formula [1, 2, 18]

$$\frac{d\rho_{\rm GW}}{d\ln k} \simeq \frac{2G\,k^3}{\pi}\int d\tau_1 \int d\tau_2\, \cos[k(\tau_1-\tau_2)]\,\Pi(k;\tau_1,\tau_2). \tag{7}$$

The GW spectrum is thus the temporal Fourier transform of the source UETC, evaluated at the on-shell frequency $\omega = k$. For a statistically stationary (or quasi-stationary) source, $\Pi(k;\tau_1,\tau_2) = \Pi(k;\tau_1-\tau_2)$ and the double time integral factorizes into the source duration $T_{\rm src}$ times the temporal transform $\widetilde{\Pi}(k,\omega) = \int d(\Delta\tau)\, e^{i\omega\Delta\tau}\Pi(k;\Delta\tau)$, which is precisely the spatio-temporal correlator $H_{ijij}(\mathbf{k},\omega)$ of Eq. (12). Hence

$$\frac{d\rho_{\rm GW}}{d\ln k} \;\propto\; k^3\, H_{ijij}(\mathbf{k},\omega)\big|_{\omega=k} \quad\Longrightarrow\quad \Omega_{\rm GW}(p) \;\propto\; p^3\, H(p,p), \qquad p = \frac{k}{k_0},\quad q = \frac{\omega}{k_0}, \tag{8}$$

the GW dispersion $\omega = k$ becoming the diagonal $q = p$ used in Sec. H (with characteristic strain $h_c^2(p) = p\,H(p,p)$). Everything downstream is therefore an evaluation of H_{ijij}, whose spatial part is a convolution of the velocity power spectrum (quadratic source, Eqs. (13)–(14)) and whose temporal part is the decorrelation model carried by $\Pi(k;\tau_1,\tau_2)$.

C. Spectral shape from the literature

For Kolmogorov inertial-range input the analytic results have three robust pieces:

- *Infrared ($k \ll k_0$):* causality forces the stress correlator to be analytic (white) as $k \to 0$, so $H \to$ const and $\Omega_{\rm GW} \propto k^3$ [3, 4].
- *Peak ($k \sim k_0$):* set by the largest eddies, with parametric amplitude $(d\Omega_{\rm GW}/d\ln k)_{\rm peak} \sim (H_*/k_0)\,\Omega_K^2$, i.e. the squared kinetic-energy fraction suppressed by the source-duration factor $H_*/k_0 \sim H_*\tau_{\rm eddy}$ [1, 3].
- *Ultraviolet ($k \gg k_0$): model-dependent*, fixed by the temporal decorrelation $\Pi(k;\tau_1,\tau_2)$. Kraichnan sweeping gives the Gogoberidze tail [2], while a δ-correlated or a decaying-turbulence kernel give different slopes.

The UV slope is precisely where the four UETC models of Appendices C and D (δ-spectrum vs. white-noise, Kraichnan vs. decay) differ. This asymmetry between the three regimes—universal infrared, source-set peak, model-dependent ultraviolet—is the organising fact of the paper, and Sec. II D makes it precise.

D. The model space, and a uniform notation for the kernels

Every kernel derived in this paper is a choice of two independent ingredients, and it is worth making the factorisation explicit at the outset because most of the disagreements in the literature—and every discrepancy discussed below—trace to one axis or the other rather than to the derivation itself.

a. Axis 1: the spatial source model S. What the fluid two-point function is taken to be. We use four:

- $S_{\rm K41}$: Kolmogorov inertial range, $E(k) \propto k^{-5/3}$ on $k_0 < k < k_d$, with $R = (k_d/k_0)^{4/3}$ the Reynolds number (Sec. III);
- $S_{\delta k}$: monochromatic, $E(k) = E_0\delta(k-k_0)$ (Sec. C 1);
- $S_{\delta r}$: real-space white noise, $R_{ij} \propto \delta^{(3)}(\mathbf{r})$, equivalently $E(k) \propto k^2$ (Sec. C 2);
- $S_{\rm full}$: a full input spectrum—a Batchelor or Saffman infrared band joined to the Kolmogorov inertial range (Sec. III D).

b. Axis 2: the temporal model T. What the unequal-time correlator (UETC) of the source is taken to be. We use four:

- $T_{\rm sw}$: Kraichnan sweeping, a *Gaussian* in the time difference, $f(\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$ with $\eta_k \propto \varepsilon^{1/3}k^{2/3}$;
- $T_{\rm dec}$: decaying turbulence following Brandenburg & Kahniashvili (BK2016) [39], a *power law* $f(\tau) = (1 + |\tau|/\tau_k)^{-2/3}$;
- $T_{\rm imp}$: impulsive, δ-correlated in time (Sec. D);
- $T_{\rm burst}$: a finite memory time Δt, independent of k —equivalently a source held coherent over a window of that length—with the triangular lag correlator $f(\tau) = (1 - |\tau|/\Delta t)\Theta(\Delta t - |\tau|)$ (Sec. E 2).

c. What the two axes do not *contain: the emission window.* Both axes describe the source *field*—S its equal-time spatial correlator, T its dependence on the time *lag*—and neither says how long the source acts. The standard formulation keeps the two apart: the unequal-time correlator is an envelope in the mean time multiplying a stationary correlator in the lag, and the duration enters only as the limits of the double time integral [9, 25, 26, 40]. Duration is therefore a third specification, not a third value of T. We nevertheless keep $T_{\rm burst}$ on the temporal axis, because in the one case where duration acts alone—a fixed spatial pattern switched on for a time Δt—the two descriptions coincide identically:

$$2\int_0^\infty d\tau\,\cos(\omega\tau)\Big(1 - \frac{|\tau|}{\Delta t}\Big)\Theta(\Delta t - |\tau|) = \frac{4\sin^2(\omega\Delta t/2)}{\Delta t\,\omega^2} = \frac{1}{\Delta t}\Big|\int_0^{\Delta t} d\eta\, e^{i\omega\eta}\Big|^2. \tag{9}$$

The window factor Eq. (47) *is* the temporal factor of an admissible stationary correlator of memory Δt—admissible because the triangle is the autocorrelation of the switched-on box—up to the normalisation Δt, which belongs to the amplitude. Two consequences follow. First, the two powers a finite window removes are a *band* effect, not a change of the infrared asymptote: the left-hand side tends to the constant Δt as $\omega \to 0$ and reaches $2/(\Delta t\,\omega^2)$ only for $\omega\Delta t \gg 1$. Second, the window is not a new mechanism: by Eq. (37) it is the universal $-2f'(0^+)/\omega^2$ tail with $f'(0^+) = -1/\Delta t$, and all that distinguishes $T_{\rm burst}$ from $T_{\rm dec}$ is that its coherence time is independent of k (Table VII). [Referee C: duration filed as a UETC made the grid look self-contradictory]

d. Notation. We write the resulting dimensionless kernel as

$$H[S,T](p,q;M,R), \qquad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \tag{10}$$

suppressing M or R where the kernel does not depend on them, and the GW spectrum on the sound-cone diagonal is always $\Omega_{\rm GW}(p) \propto p^3 H[S,T](p,p)$. Table I lists every combination derived here, its defining equation, and its analytic structure; Sec. G then compares them systematically. Two entries of the grid deserve a warning before any spectrum is read off them: $S_{\delta k}$ has zero spectral width and $S_{\delta r}$ has no infrared power at all, so both violate the hypotheses of the universal infrared theorem (Sec. V C) and neither should be expected to show k^3.

Kernel	Defining equation	Analytic structure	Sec.
$H[S_{\rm K41}, T_{\rm sw}]$	Eq. (23)	2-D wavevector integral, Gaussian temporal factor	III
$H[S_{\rm K41}, T_{\rm dec}]$	Eq. (35)	2-D wavevector × 1-D Feynman parameter	IV B
$H[S_{\delta k}, T_{\rm sw}]$	Eq. (C5)	closed form; M in the frequency scale only	C 1 a
$H[S_{\delta k}, T_{\rm dec}]$	Eq. (C6)	closed form, Eq. (C8)	C 1 b
$H[S_{\delta r}, T_{\rm sw}]$	Eq. (C13)	2-D wavevector integral; M-cutoff in the bounds	C 2
$H[S_{\delta r}, T_{\rm dec}]$	Eq. (C14)	2-D wavevector × Eq. (C16)	C 2
$H[S_{\delta k}, T_{\rm imp}]$	Eq. (D13)	closed form; no M or q dependence	D
$H[S_{\delta k}, T_{\rm burst}]$	Eq. (E9)	closed form; window factor $\sin^2$	E 2
$H[S_{\rm full}, T_{\rm sw}]$	Eq. (25)	as $S_{\rm K41}$ with a shape factor $S(k)$	III D

TABLE I. The model grid. Each kernel is a (spatial source, temporal UETC) pair in the notation of Eq. (10). The analytic structure column is what actually has to be evaluated: note that the two decaying entries are closed-form or single-integral rather than the frequency convolutions they are usually written as, by Eqs. (C8) and (C16).

III. KRAICHNAN–K41 MODEL (GOGOBERIDZE 2007)

We use the spatial/temporal Fourier pair $R_{ij}(\mathbf{r},\tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k},\tau)$ and $f(\eta_k,\tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k,\omega)$, and adopt the isotropic incompressible model of [2] (Eqs. 23–26),

$$F_{ij}(\mathbf{k},\tau) = A(k)\, P_{ij}(\hat{k})\, f(\eta_k,\tau), \qquad P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i\hat{k}_j, \tag{11}$$

with $A(k) = C_k\epsilon^{2/3}k^{-11/3}/4\pi$, $f(\eta_k,\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}}\epsilon^{1/3}k^{2/3}$.

The fourth-rank Fourier-space correlator $H_{ijkl}(\mathbf{k},\omega) = (2\pi)^{-4}\int d\tau\, d^3\boldsymbol{\xi}\, e^{i\omega\tau - i\mathbf{k}\cdot\boldsymbol{\xi}} R_{ijkl}(\boldsymbol{\xi},\tau)$ with the contraction $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ (Gogoberidze Eq. 28) reads, in real space,

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^4}\int d\tau\, d^3\mathbf{r}\, e^{i\omega\tau - i\mathbf{k}\cdot\mathbf{r}}\Big[R_{ii}(\mathbf{r},\tau)R_{jj}(\mathbf{r},\tau) + \tfrac{1}{3}R_{ij}(\mathbf{r},\tau)R_{ij}(\mathbf{r},\tau)\Big]. \tag{12}$$

Inserting the spatial Fourier representation of R_{ij}, the $\mathbf{r}$ integral produces a momentum-conserving $\delta^{(3)}(\mathbf{k}-\mathbf{q}-\mathbf{p})$. Renaming $\mathbf{q}\to\mathbf{k}_1$, $\mathbf{u} = \mathbf{k}-\mathbf{k}_1$,

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^7}\int d\tau\, e^{i\omega\tau}\, d^3k_1\big[F_{ii}(\mathbf{k}_1,\tau)F_{jj}(\mathbf{u},\tau) + \tfrac{1}{3}F_{ij}(\mathbf{k}_1,\tau)F_{ij}(\mathbf{u},\tau)\big]\,. \tag{13}$$

The temporal Fourier transform then converts the τ integral into a frequency δ-function, leaving the frequency convolution

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1\left[\tilde{F}_{ii}(\mathbf{k}_1,\omega_1)\tilde{F}_{jj}(\mathbf{u},\omega-\omega_1) + \tfrac{1}{3}\tilde{F}_{ij}(\mathbf{k}_1,\omega_1)\tilde{F}_{ij}(\mathbf{u},\omega-\omega_1)\right]. \tag{14}$$

Substituting Eq. (11) and using $P_{ii}(\hat{k}) = 2$ together with $P_{ij}(\hat{k}_1)P_{ij}(\hat{u}) = 1 + (\hat{k}_1\cdot\hat{u})^2$, the tensor contractions reduce Eq. (14) to

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1\, A(k_1)A(u)\,\tilde{f}(\eta_{k_1},\omega_1)\tilde{f}(\eta_u,\omega-\omega_1)\left[4 + \tfrac{1}{3}\big(1 + (\hat{k}_1\cdot\hat{u})^2\big)\right]. \tag{15}$$

This is the Fourier representation valid for all k; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \le \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

The law of cosines $\hat{k}_1\cdot\hat{u} = (k^2 - k_1^2 - u^2)/(2k_1u)$ turns the bracket in Eq. (15) into a homogeneous polynomial in (k,k_1,u); expanding and grouping gives the geometric kernel

$$\mathcal{K}(k,k_1,u) \equiv \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \qquad \text{(cf. Eq. 29 of [2]).} \tag{16}$$

The temporal transform $\tilde{f}(\eta_k,\omega) = (2/\eta_k)\exp(-\omega^2/\pi\eta_k^2)$ combines with $A(k)$ into the single spectral factor

$$g(k,\omega) \equiv A(k)\,\tilde{f}(\eta_k,\omega) = \frac{2E_k}{4\pi k^2\eta_k}\exp\!\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \qquad E_k = C_k\epsilon^{2/3}k^{-5/3}. \tag{17}$$

Going to spherical coordinates for $\mathbf{k}_1$ ($d^3k_1 = k_1^2\,dk_1\,d\phi\,d\mu$ with $\mu = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}}_1$), the azimuthal integral contributes 2π. Trading μ for $u = \sqrt{k^2 + k_1^2 - 2kk_1\mu}$ via $d\mu = u\,du/(kk_1)$, with $u \in [|k-k_1|, k+k_1]$,

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^7}\int_0^\infty \frac{k_1\,dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\,du\int_{-\infty}^{\infty} d\omega_1\, g(k_1,\omega_1)\, g(u,\omega-\omega_1)\,\mathcal{K}(k,k_1,u). \tag{18}$$

Switching to the dimensionless variables $q = k_1/k$, $s = u/k$, $\Omega = \omega/(\epsilon^{1/3}k^{2/3})$, $\Omega_1 = \omega_1/(\epsilon^{1/3}k^{2/3})$, g becomes $g(k,\omega) = (1/\sqrt{2\pi})\,C_k\epsilon^{1/3}k^{-13/3}\exp(-2\Omega^2)$. The product $g(k_1,\omega_1)g(u,\omega-\omega_1)$ thus separates into a dimensional prefactor $\propto k^{-26/3}$ times a manifestly dimensionless Gaussian envelope $\Phi(q,\Omega_1;s,\Omega-\Omega_1) = q^{-13/3}s^{-13/3}\exp[-2\Omega_1^2q^{-4/3} - 2(\Omega-\Omega_1)^2s^{-4/3}]$. Collecting all k-dependent factors gives the scaling form

$$H_{ijij}(\mathbf{k},\omega) = \frac{C_k^2\,\epsilon}{256\,\pi^8}\,k^{-5}\,\mathfrak{H}(\Omega), \qquad \mathfrak{H}(\Omega) = \int_0^\infty dq\,q\int_{-\infty}^{\infty} d\Omega_1\int_{|1-q|}^{1+q} ds\,s\;\Phi\,\widetilde{\mathcal{K}}(q,s), \tag{19}$$

with $\widetilde{\mathcal{K}}(q,s) = \mathcal{K}(k,k_1,u)|_{k_1=qk,u=sk}$ (homogeneous of degree zero). All physical scales are absorbed in the prefactor $C_k^2\epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Using the Gaussian-erfc identity (Eq. A.1 of [2], $\int_0^\infty dy\, e^{-Ay^2-B(y-x)^2} = \sqrt{\pi/[4(A+B)]}\, e^{-ABx^2/(A+B)}\, \mathrm{erfc}[-Bx/\sqrt{A+B}]$), the ω_1 integration in Eq. (18) can be done in closed form, yielding the doubly-bounded form

$$H_{ijij}(k,\omega) = \frac{\varepsilon}{24\pi(2\pi)^{3/2}k} \int_{k_0}^{k_d} dk_1\, k_1^{-10/3} \int_{\max(|k_1-k|,k_0)}^{\min(k_1+k,k_d)} du\, u^{-10/3}\,(k_1^{-4/3}+u^{-4/3})^{-1/2}$$
$$\times \left[27 - \tfrac{k^2}{k_1^2} - \tfrac{k^2}{u^2} + \tfrac{k^4}{2k_1^2u^2} + \tfrac{k_1^2}{2u^2} + \tfrac{u^2}{2k_1^2}\right] \exp\!\Big(-\tfrac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3}+u^{4/3}}\Big)\, \mathrm{erfc}\!\Big(-\tfrac{\sqrt{2}\,\varepsilon^{-1/3}\omega\, u^{-4/3}}{(k_1^{-4/3}+u^{-4/3})^{1/2}}\Big). \tag{20}$$

k_0 is the outer scale and k_d the dissipation scale; the integration window collapses near the boundaries of the inertial range. This integral is finite as $k \to 0$.

Define the dimensionless parameters

$$M = \frac{1}{c}\Big(\frac{\varepsilon}{k_0}\Big)^{1/3}, \quad \frac{k_d}{k_0} = R^{3/4}, \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{c\,k_0}, \qquad x = \big(\tfrac{k_1}{k_0}\big)^{-4/3}, \quad y = \big(\tfrac{u}{k_0}\big)^{-4/3}, \tag{21}$$

where $M = u_0/c = (\varepsilon/k_0)^{1/3}/c$ is the outer-scale eddy velocity in units of the speed of light, following Gogoberidze et al. [2] (this is normalised to c, not to the sound speed; see Sec. III E). We keep c explicit throughout and set $c = 1$ only when evaluating numerically. The Mach number enters the GW kernel through the k-dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{Mc}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, which couples frequency to wavenumber in the Gaussian factor of Eq. (20). As a result M sets the *effective* wavenumber support of the integral: the exponential $\exp[-2q^2/(M^2(z^{4/3}+y^{4/3}))]$ (with $z = k_1/k_0$) cuts off contributions with $z^{4/3}+y^{4/3} \lesssim 2q^2/M^2$, so the peak of the GW spectrum shifts with M. Note that the kernel depends on frequency and Mach number only through the ratio $q/M = \omega/(k_0u_0)$, in which c cancels: c appears in the *definitions* of q and M but not in the dimensionless kernel, so its numerical value is immaterial and setting $c = 1$ is exact rather than an approximation. The substitution $k_1 = k_0x^{-3/4}$, $u = k_0y^{-3/4}$ gives $dk_1\, k_1^{-10/3} = -(3/4)k_0^{-7/3}x^{3/4}dx$ and $(k_1^{-4/3}+u^{-4/3})^{-1/2} = k_0^{2/3}(x+y)^{-1/2}$; the integration limits become $x \in [R^{-1}, 1]$ and $y \in [y_{\min}(x), y_{\max}(x)]$ with

$$y_{\max}(x) = \min(|x^{-3/4}-p|^{-4/3},\, 1), \qquad y_{\min}(x) = \max((x^{-3/4}+p)^{-4/3},\, R^{-1}). \tag{22}$$

Plugging into Eq. (20) yields

$$H_{ijij}(p,q) = \frac{3M^3k_0^{-4}}{256(2\pi)^{3/2}\,p} \int_{R^{-1}}^{1} dx\, x^{3/4} \int_{y_{\min}}^{y_{\max}} dy\, y^{3/4}\,(x+y)^{-1/2}$$
$$\times \left[27 - p^2(x^{3/2}+y^{3/2}) + \tfrac{p^4}{2}x^{3/2}y^{3/2} + \tfrac{1}{2}(x^{-3/2}y^{3/2} + y^{-3/2}x^{3/2})\right]$$
$$\times \exp\!\big(-\tfrac{2xy}{x+y}\tfrac{q^2}{M^2}\big)\, \mathrm{erfc}\!\big(-\tfrac{\sqrt{2}\,q}{M\sqrt{x+y}}\big). \tag{23}$$

C. Mach-dependent peak of the stationary spectrum

Evaluating the kernel (23) on the GW dispersion $\omega = k$ (the diagonal $q = p$ of Eq. (8)) gives the dimensionless energy-density spectrum $\Omega_{\mathrm{GW}}(p) \propto p^3\, H_{ijij}(p,p;M,R)$ of stationary Kraichnan–K41 turbulence. The Mach number enters *only* through the sweeping rate $\eta_k \propto M\,k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y}\,q^2/M^2]$ and the accompanying erfc in Eq. (23). This factor is a soft cutoff that suppresses the integrand once $p \gtrsim M$, so the spectral peak tracks the Mach number, $p_{\mathrm{peak}} \sim M$, until it saturates near the outer scale as $M \to 1$ (the luminal ceiling $u_0 = c$; see Sec. III E). In the convention of Eq. (21) $M = u_0/c$, so curves with $M > 1$ correspond to $u_0 > c$ and are formal extrapolations beyond the luminal limit, unphysical for a relativistic plasma but retained throughout to expose the M-scaling and its aeroacoustic (U^8) connection. Figure 1 shows this directly: the peak slides from $p_{\mathrm{peak}} \simeq 0.05$ at $M = 0.03$ to $p_{\mathrm{peak}} \simeq 2.6$ at the formal $M = 3$, while the amplitude climbs steeply (the M^3 prefactor of Eq. (23) times the widening M-support of the integral). The infrared rise is the universal causal slope $\Omega_{\mathrm{GW}} \propto p^3$ (Sec. II C), common to all M; only the location and height of the peak respond to M.

The M-dependence of the IR amplitude is fixed entirely by the prefactor: for $p \ll M$ the Gaussian and erfc factors of Eq. (23) reduce to unity, so $H_{ijij}(p \to 0)$ is set by the M^3 prefactor alone and $\Omega_{\mathrm{GW}} \to C\,M^3p^3$ with C a universal (M-independent) constant. Equivalently the characteristic strain scales as $h_c = \sqrt{p\,H} \sim M^{3/2}\sqrt{p}$ in the IR. Compensating the spectrum by M^{-3} (the strain by $M^{-3/2}$) therefore collapses every curve onto the single universal p^3 line in the IR, isolating the genuinely M-dependent peak and UV; Fig. 1(b) confirms this collapse numerically (residual spread $< 1.5\%$ at $p \simeq 3 \times 10^{-4}$, and a measured IR exponent $d\ln H/d\ln M \to 3.00$ as $p \to 0$).

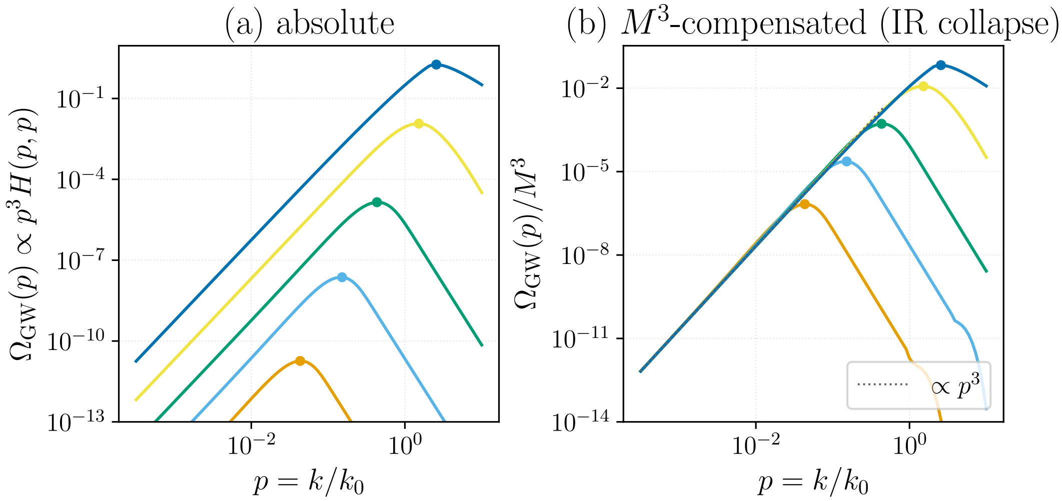


FIG. 1. Stationary Kraichnan–K41 (Gogoberidze 2007) GW spectrum $\Omega_{\rm GW}(p) \propto p^3 H_{ijij}(p, p; M, R)$, computed from Eq. (23) on the on-shell diagonal of Eq. (8), at $R = 10^4$. The five curves are $M = 0.03, 0.1, 0.3, 1, 3$, ordered by increasing amplitude and peak frequency (lowest, leftmost curve is $M = 0.03$; highest, rightmost is the formal $M = 3$); dots mark each peak. In the convention $M = u_0/c$ the curves with $M > 1$ lie beyond the luminal limit and are formal extrapolations (Sec. III E). (a) Absolute: shared IR slope p^3, amplitude growing $\sim M^3$, peak moving right with M ($p_{\rm peak} \sim M$ in the inertial range, bending over near the outer scale). (b) M^3-compensated, $\Omega_{\rm GW}/M^3$: since the IR amplitude scales as M^3 (strain as $M^{3/2}$), all curves collapse onto the universal p^3 line (dotted) in the IR and peel off only at their M-dependent peak. Generated by `Notebooks/stationary_mach_peak.py`.

D. Spectra in the $\Omega(k)/k$ convention: comparison with simulations

To compare the analytic kernel with the direct numerical simulations of Roper Pol et al. [28] we recast the spectra in their convention. There $\Omega(k)$ denotes the energy *per logarithmic wavenumber*, $\Omega(k) = d\rho/d\ln k$, so that $\Omega(k)/k = E(k)$ is the ordinary per-dk spectrum. Our master result $\Omega_{\rm GW}(p) \propto p^3 H_{ijij}(p, p)$ of Eq. (8) is precisely $d\rho_{\rm GW}/d\ln k$, hence

$$\frac{\Omega_{\rm GW}(k)}{k} \;\propto\; \frac{\Omega_{\rm GW}(p)}{p} \;=\; p^2\, H_{ijij}(p, p; M, R), \qquad p = \frac{k}{k_0}, \tag{24}$$

with H_{ijij} the stationary kernel of Eq. (23). To compare on the simulations' own footing we feed a *full* fluid input spectrum into this kernel, with both an infrared band and the Kolmogorov inertial range,

$$E(k) \;\propto\; \begin{cases} k^4, & k < k_0, \\ k^{-5/3}, & k \geq k_0, \end{cases} \tag{25}$$

joined continuously at k_0. The spectrum enters the Appendix-A integral Eq. (20) only through the amplitude $A(k) = E(k)/4\pi k^2$; writing $A(k) = S(k)\, A_{\rm Kol}(k)$ with the Kolmogorov law $A_{\rm Kol} \propto k^{-11/3}$ extrapolated to all k, the shape factor is $S(k) = 1$ for $k \geq k_0$ and $S(k) = (k/k_0)^{s+5/3}$ for $k < k_0$ (energy slope $E \propto k^s$, here $s = 4$). Every kinematic and temporal factor of the kernel is left untouched: we extend the k_1, u integrations down to an infrared cutoff $k_{\rm IR} = k_0/R_{\rm IR}$ and multiply the integrand by $S(k_1)\, S(u)$. Setting $R_{\rm IR} = 1$ recovers Eq. (23) identically.

Figure 2 shows the resulting GW spectrum at $M = 1$ with $R_{\rm IR} = 100$ (two infrared decades), together with the full fluid input spectrum and the two slopes the literature reports for the GW inertial range, drawn as reference guides: $k^{-11/3}$ from the simulation of [28] (run ini2) and $k^{-9/2}$ from [2]. We plot no simulation data points and defer a quantitative data-level comparison to future work. Both curves are *dimensionless*: the GW curve is the bare kernel with the source stress normalised to unity, so the GW–fluid vertical offset carries no physical meaning (the relic $\Omega_{\rm GW}$ acquires the additional suppression $\sim \Omega_{\rm M}^2\, (\mathcal{H}_*/k_*)$ and gravitational couplings that the kernel omits, which is why the simulated $\Omega_{\rm GW}/k \sim 10^{-13}$ lies far below the fluid spectrum); only slopes and peak locations are comparable, and the fluid curve is rescaled to share the GW peak.

Three features survive this honest treatment. *(i) The infrared stays causal.* The GW spectrum rises with the causal slope $\Omega_{\rm GW} \propto k^3$ ($\Omega_{\rm GW}/k \propto k^2$, fitted $k^{+2.0}$); it does *not* inherit the steeper fluid k^4 band, because the GW infrared is fixed by the analyticity of the stress rather than by the input spectrum. *(ii) The infrared band lifts the amplitude, not* *the inertial range.* Feeding the band raises the GW amplitude near and below the peak by a factor $\simeq 2.0$ and shifts the peak from $k \simeq 1.18\,k_0$ to $k \simeq 1.01\,k_0$ (both in this figure's $\Omega_{\rm GW}/k = p^2 H$ convention; the two ends had been quoted in different conventions, 1.55 being the $p^3 H$ value), a change that depends only weakly on the infrared slope (a further $\simeq 1.49$ between Batchelor k^4 and Saffman k^2) and leaves the inertial range untouched. *(iii) The ultraviolet tension is robust.* Above the peak the spectrum steepens *continuously*, and at $M = 1$ far more steeply than either reference: the measured post-peak slope is $\simeq k^{-5.0}$—steeper than both Gogoberidze's $k^{-9/2}$ and the simulation's $k^{-11/3}$—before the k-dependent sweeping decorrelation bends it below any single power law. The analytic stationary kernel therefore does *not* reproduce the extended $k^{-11/3}$ range seen in the simulation; at the simulations' Mach number it is markedly steeper, crossing that slope rather than following it, and this conclusion is unchanged by the infrared treatment. This high-k disagreement is precisely the simulation–analytic tension that motivates this paper—and that the decaying kernel of Sec. IV substantially relieves.

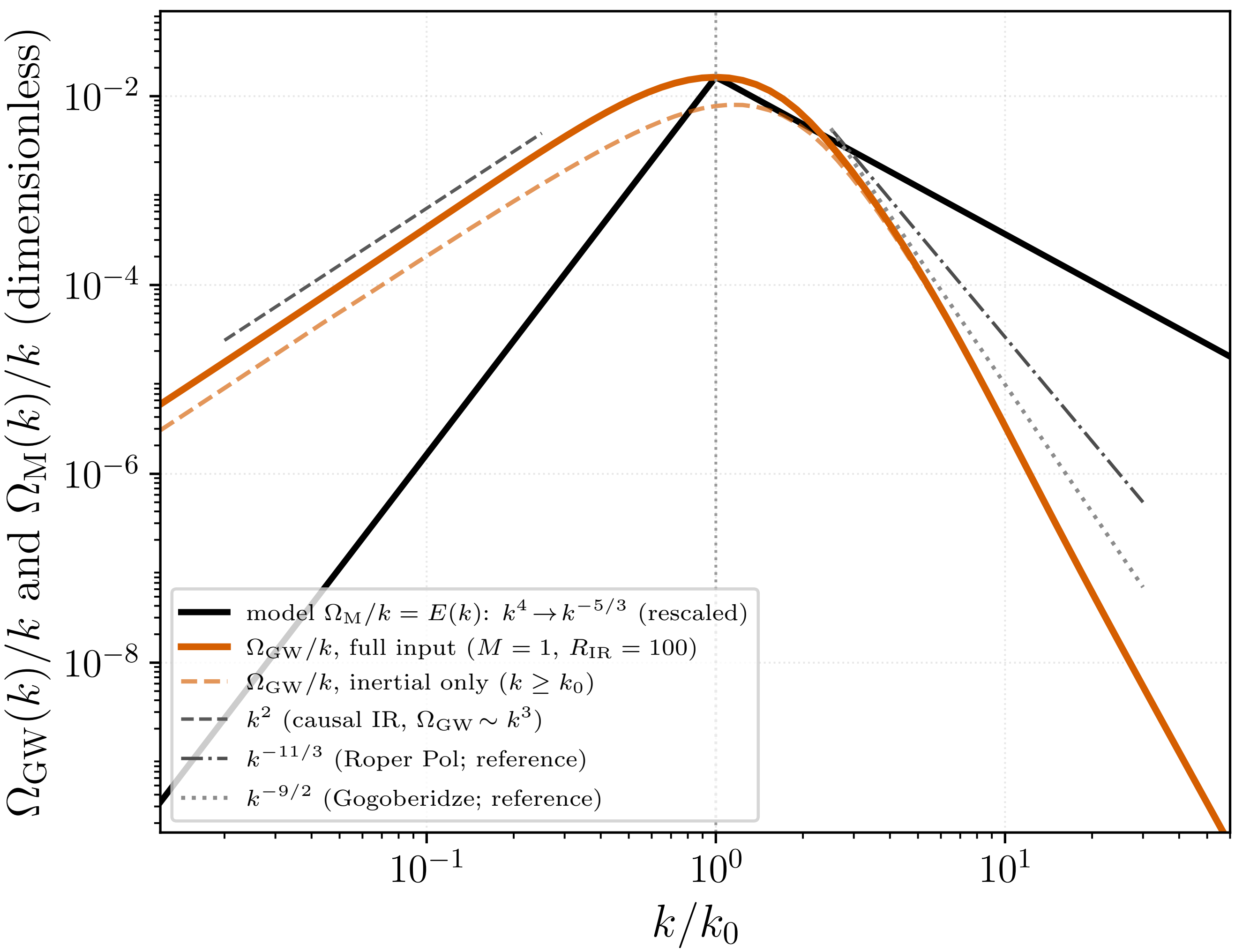


FIG. 2. Analytic spectra in the Roper Pol et al. [28] convention $\Omega(k)/k = E(k)$. Orange (solid): GW spectrum $\Omega_{\rm GW}(k)/k = p^2 H_{ijij}(p, p; M, R)$ [Eq. (24)] computed from a *full* fluid input spectrum—an infrared band $E \propto k^4$ ($k < k_0$) joined to the Kolmogorov range $E \propto k^{-5/3}$ ($k \geq k_0$), Eq. (25)—fed into the Appendix-A kernel Eq. (20) through the shape factor $S(k) = A/A_{\rm Kol}$ ($M = 1$, $R = 10^4$, infrared range $R_{\rm IR} = k_0/k_{\rm IR} = 100$). Orange (dashed): the same kernel restricted to the inertial range only ($R_{\rm IR} = 1$, i.e. core Eq. (23))—the infrared band raises the amplitude near the peak but leaves the ultraviolet unchanged. Black: the full fluid input spectrum, rescaled to share the GW peak (the GW–fluid vertical offset is unphysical; only shapes compare). Grey dashed: the causal infrared guide k^2 ($\Omega_{\rm GW} \sim k^3$). Dash-dot/dotted: the $k^{-11/3}$ (Roper Pol, run ini2) and $k^{-9/2}$ (Gogoberidze) slopes *reported* in the literature, as references only—no simulation data points are plotted. At $M = 1$ the GW spectrum steepens past the peak to $\simeq k^{-5.0}$, steeper than both references, independently of the infrared treatment. Generated by `Notebooks/roperpol_convention_spectra.py`.

E. Discussion: the Mach number, equivalent definitions, and the spectral shape

The preceding two subsections make the Mach number the single control parameter of the stationary spectrum, so it is worth stating precisely what M is, why the conflicting definitions in the literature are the *same* quantity, and what about the spectrum does and does not respond to it.

a. Explicit formulation. Throughout we use, following Gogoberidze et al. [2],

$$M = \frac{u_0}{c} = \frac{1}{c}\Big(\frac{\varepsilon}{k_0}\Big)^{1/3}, \qquad q = \frac{\omega}{c\,k_0}, \tag{26}$$

the outer-scale eddy velocity in units of the speed of light. We retain c in every expression and set $c = 1$ only for numerical evaluation; because the kernel depends on frequency and Mach number solely through $q/M = \omega/(k_0 u_0)$, in which c cancels (Eq. (21)), this choice is exact and not an approximation. The grouping is fixed by dimensions: with $[\varepsilon] = L^2T^{-3}$ and $[k_0] = L^{-1}$ one has $\varepsilon/k_0 = (L/T)^3$, so $(\varepsilon/k_0)^{1/3}$ is a velocity and the inverse grouping $(k_0/\varepsilon)^{1/3}$ — which appears occasionally in the literature — is an *inverse* velocity and cannot be a Mach number. M does not enter Eq. (20) as a free amplitude; it enters *only* through the k-dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{Mc}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, hence only through the Gaussian $\exp[-\frac{2xy}{x+y}q^2/M^2]$ and the accompanying erfc of Eq. (23).

b. Normalisation to c, not to the sound speed. Two points of convention need care, because both have caused numerical disagreements between papers for the same physical flow. First, our M is a velocity ratio to the *speed of light*, not to the sound speed. A relativistic plasma has $c_s = c/\sqrt{3}$, so the true acoustic Mach number is

$$M_s = \frac{u_0}{c_s} = \sqrt{3}\,\frac{u_0}{c} = \sqrt{3}\,M, \tag{27}$$

larger than M by $\sqrt{3}$. The transonic point $u_0 = c_s$ therefore sits at $M = 1/\sqrt{3} \simeq 0.58$ in our normalisation, and the luminal ceiling $u_0 = c$ at $M = 1$; a purely vortical velocity field can in principle approach the latter, whereas $M_s \sim 1$ (genuine transonic turbulence) corresponds to $M \simeq 0.58$. This is the convention of Gogoberidze et al. [2], who define $M = (\varepsilon/k_0)^{1/3}$ in units $c = 1$ and note separately that the plasma sound speed is $c/\sqrt{3}$; it is *not* the same as the acoustic M_s used elsewhere, and the two differ by the fixed factor $\sqrt{3}$.

Second, and independently of the first, $u_0 = (\varepsilon/k_0)^{1/3}$ is the single-eddy outer velocity, which differs from the integrated rms by an $\mathcal{O}(1)$ cascade constant: for a Kolmogorov spectrum $\int_{k_0}^{\infty} E\,dk = \frac{3}{2}C_K(\varepsilon/k_0)^{2/3}$, so $u_{\rm rms} = \sqrt{\frac{3}{2}C_K}\,(\varepsilon/k_0)^{1/3} \approx (1.5\text{–}2)\,(\varepsilon/k_0)^{1/3}$, depending on C_K and on whether one normalises $\int E\,dk$ to $\langle u^2\rangle$ or $\frac{1}{2}\langle u^2\rangle$. This $\mathcal{O}(1)$ factor and the $\sqrt{3}$ above are distinct and multiply: a paper quoting $M_s = u_{\rm rms}/c_s$ and one quoting our $M = (\varepsilon/k_0)^{1/3}/c$ describe the same flow with Mach numbers differing by $\sqrt{3}\times(1.5\text{–}2)$. Both identifications further *break down* when the cascade picture fails: for spectra whose energy integral is ultraviolet-dominated (the white-noise / Batchelor model of Sec. C 2, $E \propto k^2$), where $(\varepsilon/k_0)^{1/3}$ no longer measures $u_{\rm rms}$; for decaying turbulence, where $\varepsilon \to \varepsilon(t)$ and $M \to M(t)$ (Sec. F); and in MHD, where the relevant ratio is often the Alfvénic $M_A = u_{\rm rms}/v_A$ rather than the acoustic M [3, 41].

c. Shape versus peak versus amplitude. It is essential to separate three properties of the spectrum, only one of which is M-independent. *(i) Shape.* The dimensionless spectral form — the causal infrared slope $\Omega_{\rm GW} \propto p^3$ (Sec. II C) and the inertial power law — is universal: it is set by causality and by the source power law, not by M. This is the property that is legitimately called "M-independent," and it is what the M^{-3}-compensated collapse of Fig. 1(b) isolates. *(ii) Peak location.* Here the literature genuinely splits. In spatial-scale–anchored treatments the peak is pinned to the source scale k_0 (set by the largest eddy / bubble size), so $p_{\rm peak} = \mathcal{O}(1)$ and M controls only the amplitude [4]; in sweeping-anchored treatments the peak is tied to the eddy-turnover frequency $u_0k_0 \propto Mk_0$, so it shifts with M [2, 28]. Our kernel realises the second picture through $\eta_k \propto Mk^{2/3}$, giving $p_{\rm peak} \sim M$ (Sec. III C). The two are not in contradiction: a peak "at the stirring scale" is M-independent only because it is measured in units of k_0 and the source is assumed quasi-static over an eddy. *(iii) Total energy / amplitude.* This is never M-independent. In the stationary kernel the infrared amplitude scales as the M^3 prefactor of Eq. (23) ($\Omega_{\rm GW} \to CM^3p^3$, Sec. III C); more generally the quadrupole argument gives $\Omega_{\rm GW} \propto \Omega_K^2 \propto M^4$, steepening toward $M^{5\text{–}6}$ once the finite source duration is included [4], the extra power being absent from a single-time stationary kernel. We derive this progression explicitly for the decaying source in Sec. IV B: the full-spatial kernel gives an infrared amplitude $\propto M^4$ and a source-pinned peak amplitude $\propto M^{5.3}$ [Eq. (38), Fig. 8]. Consequently a stated "M-independence" of the GW signal can only refer to the shape (i); any claim that the peak position and the total energy are simultaneously M-independent is either a normalised-units artifact (dividing by Ω_K^2 and measuring p in units of k_0) or a conflation of the universal shape with the dimensional peak and amplitude.

IV. LOCATION OF THE SPECTRAL PEAK: SWEEPING VERSUS SOURCE SCALE

A striking feature of the stationary spectrum of Fig. 1 is that, for subsonic turbulence, the GW energy density peaks at wavenumbers *far below* the outer scale, $p_{\rm peak} = k_{\rm peak}/k_0 \ll 1$ (e.g. $p_{\rm peak} \simeq 0.04$ at $M = 0.03$). This is counterintuitive: the GW source is the quadratic stress $T_{ij} \sim u_i u_j$, and for a velocity field concentrated at the stirring scale k_0 one expects the stress—and hence the radiation—to peak near $k \sim k_0$, with spatial support extending to at most $k = 2k_0$ by the triangle inequality. We show here that the analytic peak is fixed instead by the *temporal* (sweeping) decorrelation, giving $p_{\rm peak} \simeq 1.49\,M$, and that the "$k \simeq 2k_0$" expectation is approached only as $M \to 1$ (the luminal limit) and is never reached for physical subsonic or transonic turbulence.

a. Self-similar Mach scaling. In the dimensionless kernel of Eq. (23) the Mach number enters *only* through the sweeping rate $\eta_k \propto M k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ (and its accompanying erfc) evaluated on the GW cone $q = p$ of Eq. (8); the geometric kernel and the triangle-inequality bounds carry no M. Together with the explicit M^3 prefactor of Eq. (23), the on-cone kernel must therefore collapse to a single function of the sweeping variable $\xi \equiv p/M = k/(Mk_0)$,

$$H_{ijij}(p,p;M) \;=\; M^3\,\widetilde{H}_{ijij}(\xi), \qquad \xi \equiv \frac{p}{M}, \tag{28}$$

where $\widetilde{H}_{ijij}$ is the *compensated kernel*: an infrared plateau $\widetilde{H}_{ijij}(0) \equiv A \simeq 2.1 \times 10^{-2}$ (the universal causal rise, Sec. II C) that rolls off once $\xi \gtrsim 1$, where the sweeping cutoff switches on. The energy density and characteristic strain follow as

$$\Omega_{\rm GW} = p^3 H_{ijij} = M^3 p^3\,\widetilde{H}_{ijij}(\xi), \qquad h_c^2 = p\,H_{ijij} = M^3 p\,\widetilde{H}_{ijij}(\xi). \tag{29}$$

b. Numerical proof of the collapse. Equation (28) is verified directly in Fig. 3(b): the compensated kernel $H_{ijij}(p,p;M)/M^3$ plotted against $\xi = p/M$ for $M = 0.03$–3 falls on the single curve $\widetilde{H}_{ijij}(\xi)$, with infrared plateau $A \simeq 2.1 \times 10^{-2}$ and $< 13\%$ residual spread dominated by the formal $M = 3$ case (superluminal, shown to expose the trend). The cutoff thus depends on p and M *only* through $\xi = p/M$—the unambiguous signature of the sweeping rate—and switches on at $\xi \sim 1$, i.e. $p \sim M$. (How this single scaling is best displayed—compensating one axis or both—is taken up in Sec. IV A.)

c. Peak condition and raw peak location. Differentiating $\Omega_{\rm GW} = M^3 p^3 \widetilde{H}_{ijij}(\xi)$ [Eq. (29)] on the cone,

$$\frac{d\ln\Omega_{\rm GW}}{d\ln p} \;=\; 3 + \frac{d\ln\widetilde{H}_{ijij}}{d\ln\xi} \;=\; 0 \quad\Longrightarrow\quad \left.\frac{d\ln\widetilde{H}_{ijij}}{d\ln\xi}\right|_{\xi_*} = -3, \qquad p_{\rm peak} = \xi_*\,M, \tag{30}$$

so the peak sits where the compensated kernel has logarithmic slope -3, at a fixed ξ_* independent of M. Reading the peak directly from each computed spectrum (Fig. 3c; sub-grid argmax, no model fit) gives a location linear in M across the entire physical range,

$$p_{\rm peak} \;=\; 1.515\,M^{1.007} \qquad (M \lesssim 1), \tag{31}$$

with $\xi_* = 1.488$ read from the collapsed shape, consistent with Eq. (30). [the erfc correction to the kernel moves this constant and its whole family: $\xi_* = 1.47 \to 1.488$, the fit $1.48M^{1.00} \to 1.515M^{1.007}$, and the two end-points below; every occurrence in the text, the figures and the generating scripts has been updated together, and the value is quoted as $1.49\,M$ wherever the law rather than the fit is meant] Only as $M \to 1$ (the luminal ceiling) does the peak begin to run into the outer scale and the linear law saturate: $p_{\rm peak}/M$ falls from 1.53 at $M = 1$ to 0.744 at the formal $M = 3.4$. Reaching the "source-scale" value $p \simeq 2$ would require $M \simeq 1.4$, i.e. $u_0 \simeq 1.4c$; the stationary sweeping peak therefore *never* reaches the source scale for any physical (subluminal) turbulence, and is far below it throughout the subsonic range where cosmological flows lie.

d. Resolution. The naive "$k \simeq 2k_0$" is a purely *spatial* statement: it is the largest wavenumber at which the quadratic stress has support, the triangle edge $\Theta(2-p)$ that appears explicitly in the monochromatic kernel [Eq. (C5)]. It is an ultraviolet *edge*, not a peak. The GW peak, by contrast, is fixed in *frequency*: gravitational waves are sourced on the cone $\omega = k$, while a turbulent eddy of wavenumber k decorrelates at the sweeping frequency $\eta_k \sim M k_0^{1/3} k^{2/3}$, which for subsonic flow is $\ll k$ for every $k \gtrsim k_0$. The source therefore has negligible temporal power at the frequency $\omega = k$ needed to radiate at wavenumber k, except at the lowest wavenumbers $k = \omega \lesssim M k_0$, and the spectrum is throttled down to $p_{\rm peak} \sim M$. This is the aeroacoustic (Lighthill) suppression of slow sources [42, 43]: subsonic turbulence radiates at its eddy-turnover frequency $\sim M k_0$, not at its spatial scale k_0. Whether the true peak follows this sweeping scale or remains pinned near k_0 (Sec. III E) is precisely the simulation–analytic tension this paper addresses; the stationary kernel commits unambiguously to the sweeping scale.

e. The factor two is kinematic, and the edge is also a maximum. The preceding paragraph is right that $2k_0$ is a support edge, but it understates the case, and the correction matters for how the two pictures should be told apart. Because every stress in this problem is *quadratic* in the fields—u_iu_j, B_iB_j alike—the stress in Fourier space is a self-convolution, $T(\mathbf{k}) = \int d^3q\, F(\mathbf{q})F(\mathbf{k}-\mathbf{q})$. With the field concentrated near $|\mathbf{q}| = k_0$ the reachable sums $\mathbf{q}_1+\mathbf{q}_2$ fill a ball of radius $2k_0$, so no stress mode can exceed twice the field wavenumber; but the spectral measure $E_\mathrm{T} = 4\pi k^2 P_\mathrm{T}$ rises as k^2, so the stress spectrum piles up *against* that upper edge and peaks there. The edge is therefore also the maximum. This statement involves a single instant—no emission time, no propagator—so it is duration-independent and holds for any quadratic source, and it is the ultraviolet end of the very same convolution whose infrared end supplies the white-noise floor $P_\mathrm{T}(k \to 0) = \int d^3q\, P(q)^2$ of Secs. C 2 and V: one integral, read at its two edges.

Two qualifications follow, both testable. The clean factor of two requires the field to be concentrated in a narrow band; for a broad inertial range the edge smears, which is why the inertial-range columns of Table II show no doubling at all. And the gravitational-wave spectrum is the stress divided by k^2, which pulls its maximum *below* the stress maximum—by an amount set by the source's spectral width, so that a GW peak somewhat under $2k_0$ is the expected outcome for a sharply peaked but not monochromatic source. Our digitization of the simulated spectrum gives $1.84\,k_0$ (Sec. IV C), consistent with that; the shell-resolved time series scatters about a slightly higher $2.4\,k_0$, the difference being within the wavenumber resolution at which k_0 itself can be located. Emission duration enters this picture only indirectly, through $k_0(t)$ drifting downwards under the helical inverse transfer, with $k_\mathrm{GW}(t) \simeq 2k_0(t)$ holding instantaneously throughout—which is exactly the behaviour of Fig. 10(a).

The reading of the factor two as the kinematic edge of the self-convolution, and the consequences drawn from it in this paragraph, are our interpretation; [28] report the $2k_0$ peak but do not diagnose the mismatch with the analytic $\sim Mk_0$ in these terms. Stated this way, though, the two predictions differ by a clean factor $\simeq 2/M$, and—since the two sensitivities are to *different* variables, the simulated peak to the decay history and the analytic peak to the Mach number—a simulation series that varies M at fixed initial spectrum separates them by about an order of magnitude in peak frequency. That is a far cleaner discriminant than comparing amplitudes, which is entangled with an acknowledged normalisation correction on the simulation side (Sec. IV C).

f. Caveat: stationarity versus the simulated peak. The scaling $p_\mathrm{peak} \simeq 1.49\,M$ derived above is specific to *stationary*, continuously stirred turbulence, in which the source at scale k carries temporal power only up to its eddy frequency η_k. Cosmological turbulence is not of this kind: it is sourced *impulsively*—e.g. at a phase transition—and then decays. A sudden onset of duration Δt injects broadband temporal power up to $\omega \sim 1/\Delta t \sim k_0$, which the single-time stationary kernel does not contain. Because the spatial stress is far more powerful at the outer scale k_0 than at Mk_0, this broadband component pins the GW peak near the *spatial* stress scale, $p \sim 1$–2, almost independently of M. This is what the direct simulations of [28] display, and it is the spatial-scale picture of Sec. III E. In the convention $M = u_0/c$ the sweeping peak $1.49M$ approaches the spatial band $p \sim 1$–2 only as $u_0 \to c$, and would coincide with $p \simeq 2$ only at the superluminal $M \simeq 1.4$; the two predictions therefore *diverge across the entire physical range* and converge only formally beyond the luminal limit (Fig. 5). Current simulations, far from sitting in a band where the pictures agree, are *deeply subsonic* ($M \simeq \sqrt{\Omega_\mathrm{M}} \sim 0.05$, Sec. V), which is exactly the regime of maximal divergence—and there the data follow the source-scale (impulsive/decaying) peak, not the sweeping one. This peak displacement is the spatial counterpart of the amplitude deficit noted in Sec. III E—the stationary kernel gives $\Omega_\mathrm{GW} \propto M^4$ rather than the $M^{5\text{–}6}$ recovered once the finite source duration is restored—both stemming from the broadband temporal power that an impulsive, finite-duration source has and the stationary kernel omits. We compute the decaying case directly in Sec. IV B below; it confirms a source-scale peak $p \simeq 2.4$, nearly independent of M, but traces it to the *shape* of the decaying temporal correlation rather than to the impulsive onset emphasised here.

A. Physical units and what the collapse hides

a. Dimensional dictionary. The kernel works in the dimensionless pair $\{p = k/k_0,\ M\}$; the observable spectrum is the characteristic strain $h_c(f)$ at the redshifted frequency f today. The two are connected by the standard cosmological map [2, 3]. A comoving mode is observed at

$$f \;=\; \frac{\hat\omega}{2\pi}\, f_H, \qquad f_H \simeq 1.6\times 10^{-5}\,\mathrm{Hz}\left(\frac{g_*}{100}\right)^{1/6}\frac{T_*}{100\,\mathrm{GeV}}, \tag{32}$$

with $\hat\omega = \omega/H_*$ and $f_H = H_*(a_*/a_0)$ the Hubble frequency today. The stirring scale in Hubble units is $\hat k_0 = 2\pi/\gamma$, where $\gamma \equiv l_0 H_*$ is the largest-eddy size in Hubble radii, so on the GW cone $\hat\omega = \hat k = \hat k_0\, p$ and

$$f \;=\; f_0\, p, \qquad f_0 \equiv \frac{f_H}{\gamma}, \tag{33}$$

i.e. $p = k/k_0$ is simply the frequency measured in units of the stirring-scale frequency f_0. Inverting, the abscissa of the standard strain figures, $(f/\mathrm{Hz})(g_*/100)^{-1/6}(\gamma/0.01)(T_*/100\,\mathrm{GeV})^{-1}$, equals p up to the fixed constant 1.6×10^{-3} and *carries no* M. For the amplitude, $\Omega_{\rm GW} = p^2 h_c^2$ with $h_c^2 = p\,H_{ijij}(p,p)$, while the physical strain (Kahniashvili Eq. 51) is $h_c = 2\times10^{-14}(100\,\mathrm{GeV}/T_*)(100/g_*)^{1/3}(\hat\omega\,\hat\tau_T\hat H_{ijij})^{1/2}$. Multiplying h_c by $(g_*/100)^{1/3}(T_*/100\,\mathrm{GeV})$ cancels the cosmological prefactor, and the further factors $(\gamma/0.01)^{-3/2}(\zeta/0.01)^{-1/2}$ remove the stirring-scale and source-energy (ζ) normalisations, leaving a quantity that depends only on M and p. Both axes of the published figure are therefore our dimensionless (p, h_c) with every *scenario* parameter $(g_*, T_*, \gamma, \zeta)$ scaled out; the single remaining knob is M.

b. Normalising the Mach dependence: one axis or two. With the scenario parameters removed, the residual Mach dependence of Eq. (29) can be scaled out in two inequivalent ways, which answer different questions.

(i) Amplitude only (y-axis). Divide the strain by $M^{3/2}$ (equivalently $\Omega_{\rm GW}$ by M^3) but keep the physical frequency $f \propto p$ on the abscissa. From Eq. (29), $h_c\,M^{-3/2} = \sqrt{p\,\widetilde H_{ijij}(p/M)}$, which is M-independent *only* in the infrared, where $\widetilde H_{ijij}\to A$: the causal rises stack, but the peaks stay at their true frequencies $f_{\rm peak} = f_0\,(1.49\,M)$ and *slide* with M. This is the convention of the published strain plots (the $h_c\,M^{-3/2}$ ordinate of [2]); it preserves the observable—where in the band each model peaks—needed to confront a detector.

(ii) Both axes. Additionally measure the frequency in sweeping units, $x \to \xi = p/M = f/(Mf_0)$. Substituting $p = M\xi$ in Eq. (29),

$$\frac{\Omega_{\rm GW}}{M^6} = \xi^3\,\widetilde H_{ijij}(\xi), \qquad \frac{h_c}{M^2} = \sqrt{\xi\,\widetilde H_{ijij}(\xi)}, \tag{34}$$

both functions of ξ *alone*: the infrared rise *and* the peak collapse onto one curve (Fig. 4). The compensation rises from M^3 to M^6 ($M^{3/2}\to M^2$ in strain) because the p^3 rise contributes an extra M^3 once the abscissa is measured in units of Mf_0 rather than f_0. This is the right diagnostic for the *theory*—it exhibits the self-similarity and proves the cutoff is a function of p/M only—but it necessarily *discards* the physical peak frequency that view (i) keeps. The two are complementary: y-only for observation, both-axis for the scaling argument.

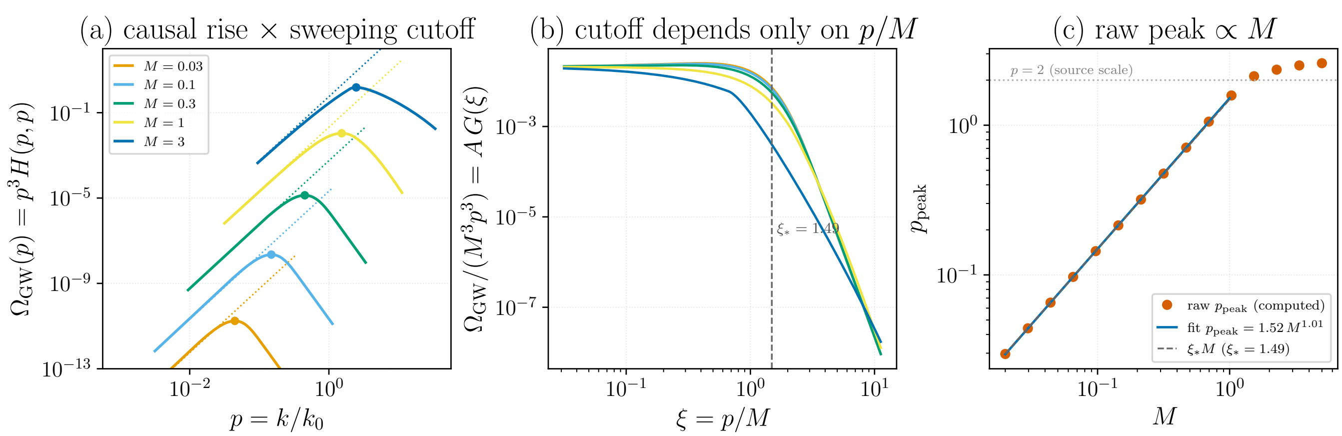


FIG. 3. Why the stationary GW spectrum peaks at $p \simeq 1.49\,M$ rather than at the source scale $p\sim2$, computed from $\Omega_{\rm GW}(p) = p^3 H_{ijij}(p,p;M,R)$ [Eq. (23), $R = 10^4$]. (a) Each spectrum (solid) follows its causal infrared asymptote $A\,M^3p^3$ (dotted) and rolls over at the peak (dot) where the sweeping cutoff switches on. (b) Collapse: $\Omega_{\rm GW}/(M^3p^3) = H_{ijij}(p,p;M)/M^3$ versus $\xi = p/M$ falls on the single compensated kernel $\widetilde H_{ijij}(\xi)$ for every M, proving the cutoff depends only on p/M [Eq. (28)]; the universal peak position is $\xi_* = 1.488$. (c) Peak location read directly from each computed spectrum (points) versus M: a power law $p_{\rm peak} = 1.515\,M^{1.007}$ (line) [Eq. (31)], coincident with ξ_*M (dashed), that bends below it toward the $p = 2$ source scale (dotted) only as $M\to1$ (the luminal limit; $M>1$ is superluminal). Generated by `Notebooks/stationary_peak_analysis.py`.

B. The decaying source pins the peak at the source scale

The caveat above is borne out quantitatively by repeating the peak analysis with the *decaying* temporal model in place of the stationary one. We keep the same full-spatial kernel—identical geometric factor and triangle-inequality integration bounds—but replace the Kraichnan Gaussian decorrelation on each stress leg by the BK2016 power law $(1+|\tau|/\tau_k)^{-2/3}$ of Sec. C 1 b, with the sweeping eddy time $\tau_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$.

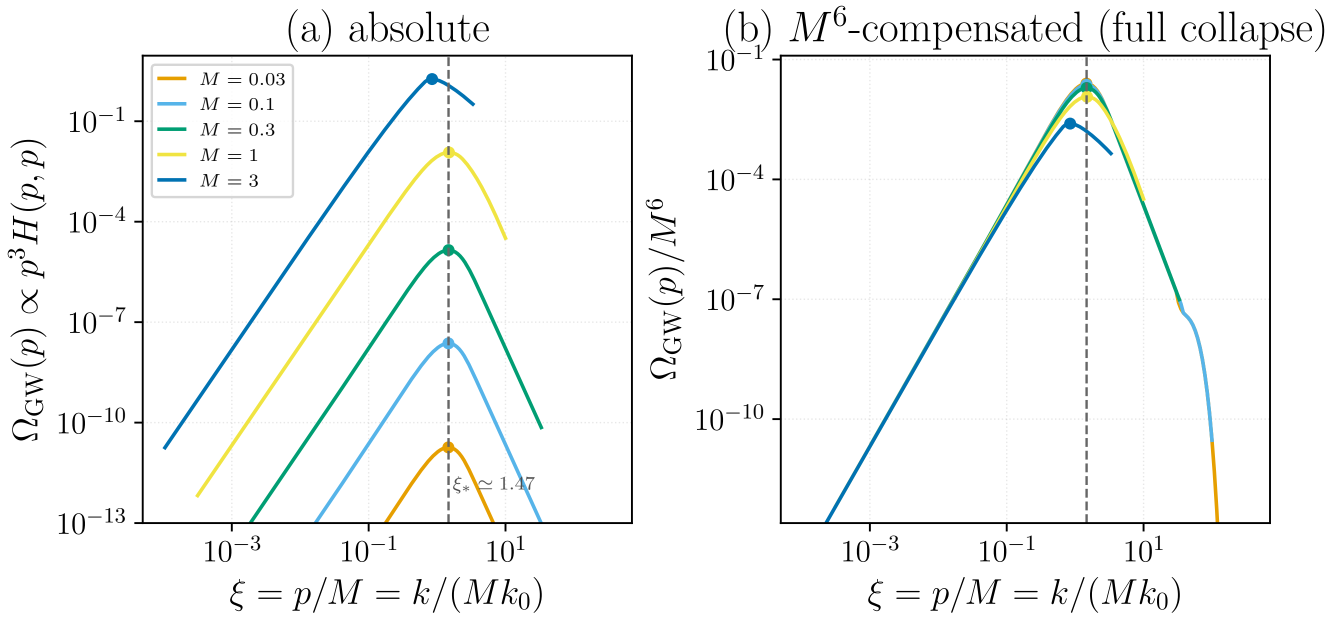


FIG. 4. Single-variable collapse of the stationary spectrum on the sweeping-scaled wavenumber $\xi = p/M = k/(Mk_0)$, computed from $\Omega_{\rm GW}(p) = p^3 H_{ijij}(p,p;M,R)$ $(R = 10^4)$. (a) Absolute spectra: the peaks of all Mach numbers stack on the vertical line $\xi_* \simeq 1.49$, only their amplitudes differ ($\propto M^3$). (b) M^6-compensated spectra $\Omega_{\rm GW}/M^6$ versus ξ collapse onto the single universal curve $\xi^3 \widetilde{H}_{ijij}(\xi)$ of Eq. (34)—infrared rise and peak together—because the p^3 rise contributes an extra M^3 on the ξ axis. The subluminal curves ($M = 0.03$–1) coincide; only the formal $M = 3$ curve (peak $\xi \simeq 0.85$; superluminal, $u_0 = 3c$) departs, where the sweeping cutoff yields to the inertial-range/triangle edge. Generated by `Notebooks/stationary_mach_peak_xM.py`.

a. Method: Fourier transform of the time-domain product. The temporal factor of the decaying kernel is the frequency convolution $\int d\omega_1\, \hat{g}(\omega_1 \tau_{k_1})\, \hat{g}((\omega - \omega_1)\tau_u)$ of the two legs. By the convolution theorem this equals the Fourier transform of the time-domain *product* of the two decorrelations,

$$\int d\omega_1\, \hat{g}(\omega_1\tau_1)\hat{g}((\omega-\omega_1)\tau_2) \;=\; \frac{2\pi}{\tau_1\tau_2}\int_0^\infty d\tau\, \cos(\omega\tau)\left(1+\tfrac{\tau}{\tau_1}\right)^{-2/3}\left(1+\tfrac{\tau}{\tau_2}\right)^{-2/3}, \tag{35}$$

with $\tau_1 = \sqrt{x}/M$, $\tau_2 = \sqrt{y}/M$. The right-hand side, unlike the convolution—whose integrand is integrably singular at both endpoints—is smooth and rapidly convergent; evaluated this way the kernel is some two orders of magnitude cheaper to compute and well converged.

b. Result: an M-independent, source-scale peak. On the GW cone the decaying spectrum $\Omega_{\rm GW}(p) = p^3 H^{\rm decay}_{ijij}(p,p;M,R)$ peaks at

$$p^{\rm decay}_{\rm peak} \;\simeq\; 2.4, \tag{36}$$

essentially independent of both the Mach and the Reynolds number: the peak moves only from 2.29 to 2.56 as M runs over 0.3–2, and from 2.40 to 2.60 as R runs over 10^3–10^5 (a peak set by the dissipation scale would scale as $R^{3/4}$). This is in sharp contrast to the stationary law $p_{\rm peak} = 1.49\, M$ of Eq. (31), which tracks the sweeping scale. The two are compared in Fig. 7.

c. Mechanism: the high-frequency tail. The peak is fixed by the high-frequency tail of the temporal correlation. The stationary Gaussian sweeping factor $\exp[-\frac{2xy}{x+y} q^2/M^2]$ cuts the source off sharply once $q \gtrsim M$, throttling the spectrum down to $p \sim M$. The power-law decorrelation of decaying turbulence instead has a *heavy* frequency tail, $\hat{g}(q) \sim q^{-1/3}$, so the source retains power at the frequency $\omega = k$ required to radiate at every wavenumber k; the spectrum then extends past the sweeping cutoff and its peak is fixed by the spatial structure of the stress near the source scale $p \simeq 2.4$, independently of M. This realises the spatial-scale picture of Sec. III E and matches the source-scale peak of the decaying simulations [28]; the apparent contradiction with the low Gogoberidze peak dissolves once one recognises that the two correspond to opposite limits of the temporal correlation—Gaussian (rapidly decorrelating, stationary) versus power law (slowly decorrelating, decaying).

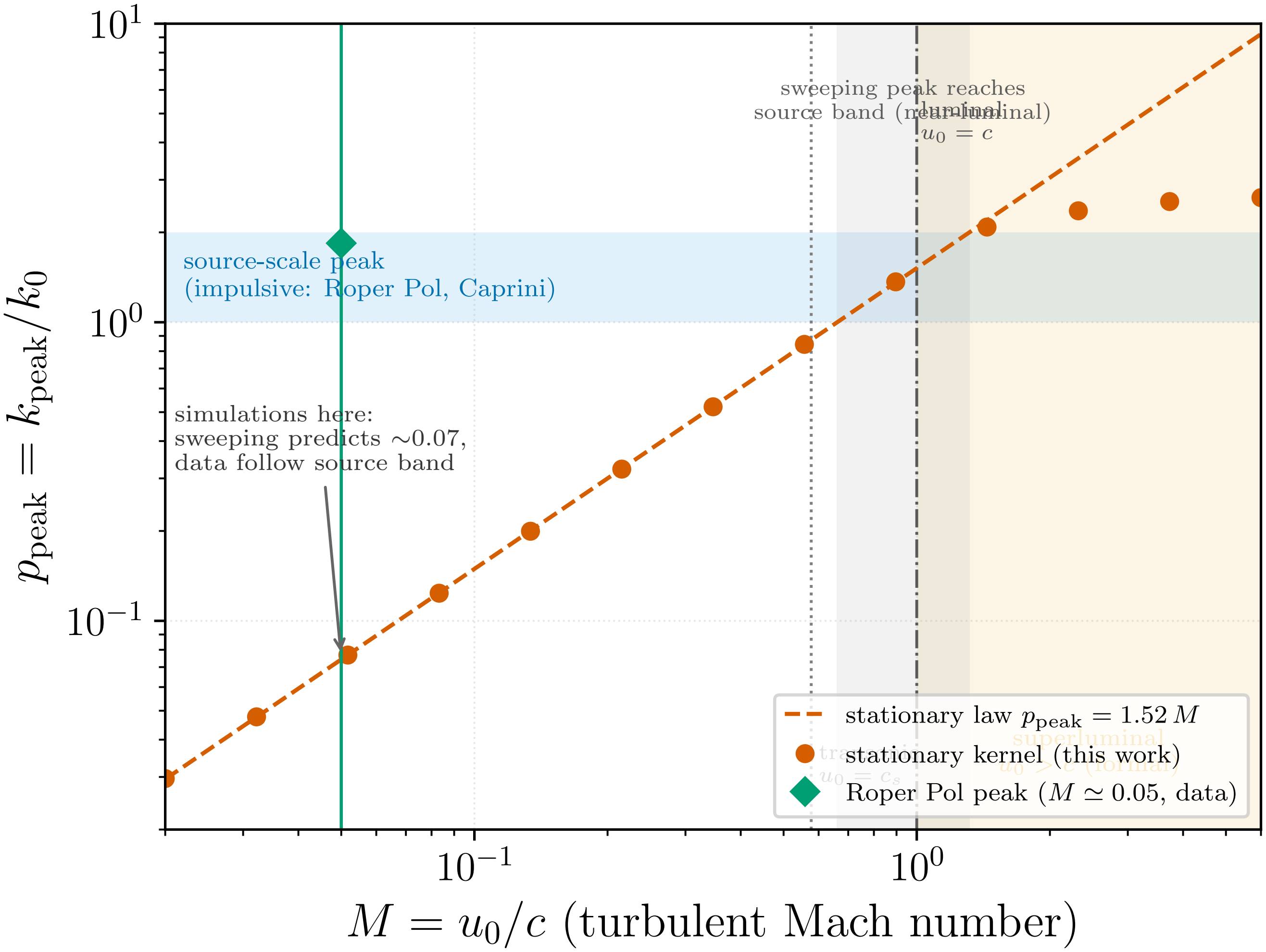


FIG. 5. Reconciliation of the stationary and impulsive pictures of the GW peak. Points and dashed line: the peak $p_{\rm peak} = k_{\rm peak}/k_0$ *computed* from the stationary kernel $\Omega_{\rm GW} = p^3 H_{ijij}(p, p; M, R)$ ($R = 10^4$), following $p_{\rm peak} \simeq 1.5\,M$. Shaded horizontal band ($p \sim 1$–2): the competing spatial-scale picture, in which an impulsively sourced, decaying flow radiates at the stress scale independently of M (Roper Pol et al. [28]; Caprini et al. [4])—a schematic, *not* computed here and not data. In the convention $M = u_0/c$ the sweeping law enters the source band only at $M \simeq 0.66$–1.32 (grey strip), i.e. near or beyond the luminal ceiling $M = 1$ (dash-dotted; $M > 1$ is superluminal, shaded); the transonic point $u_0 = c_s$ is at $M = 1/\sqrt{3} \simeq 0.58$ (dotted). The two pictures therefore diverge across the whole physical range. Current simulations sit at $M \simeq 0.05$ (green line/diamond, Roper Pol peak 1.84), deeply subsonic, where the sweeping law predicts $p \simeq 0.07$ but the data follow the source band—the regime of maximal divergence, not of coincidence. Generated by `Notebooks/peak_reconciliation.py`.

d. The decorrelation shape, not the onset. This pinning is already present in the *quasi-stationary* decaying kernel used here, in which the two-time correlation is taken at a single emission epoch and there is no impulsive switch-on. It is therefore the *shape* of the temporal decorrelation—its heavy frequency tail—rather than the broadband power of a sudden onset invoked in the caveat of Sec. IV, that is responsible for displacing the peak to the source scale. Restoring the finite source duration (the self-similar slow-time integration) modifies the amplitude and overall normalisation but leaves this peak displacement intact.

e. Caveat: the peak displacement rests on a cusp that simulations dispute. This mechanism is more fragile than it looks, and the fragility is sharply localised. The heavy tail is controlled by a *single* number—the one-sided slope of the squared correlator at zero lag. For any f with $f(0)$ finite the standard Fourier result is

$$T(\omega) = 2\int_0^\infty d\tau\, \cos(\omega\tau) f(\tau) \;\longrightarrow\; -\frac{2f'(0^+)}{\omega^2}, \tag{37}$$

so the ω^{-2} tail exists if and only if the correlator has a *cusp* at $\tau = 0$. The BK2016 power law does: $f'(0^+) = -4/3$ for the squared correlation, which reproduces the coefficient $8/3$ of Eq. (C9) exactly. A Gaussian does not: $f'(0^+) = 0$, and its tail then falls faster than any power.

This matters because Auclair et al. [23] find from direct hydrodynamic simulations that the unequal-time correlator of the velocity field is *Gaussian in the time difference*, $P_v \simeq P_v \exp[-\frac{1}{2}k^2 v_{\rm sw}^2(\tau-\zeta)^2]$ with $v_{\rm sw}^2 \simeq \bar{v}^2/3$, as predicted by Kraichnan's random-sweeping model, and explicitly reject power-law decorrelation built on scale-dependent eddy-turnover times. Under that UETC the tail is super-exponential and the peak-pinning argument of this section does not go through; the peak would revert to the sweeping scale of Sec. IV.

It should be stressed that combining the two does *not* resolve the question. One might expect a Gaussian sweeping factor to regulate the power law's cusp, but it cannot: a Gaussian has zero slope at the origin, so multiplying by it leaves $f'(0^+)$—and hence the ω^{-2} tail—unchanged (we verify slopes -1.97 to -2.13 for sweeping widths spanning a decade). The question is therefore irreducibly the one Eq. (37) poses: does the true unequal-time correlator of a freely decaying, genuinely non-stationary flow carry a $|\tau|$ kink at zero lag? The sweeping literature says no for the short-lag Eulerian correlation; whether the secular decay of the amplitude contributes one is, to our knowledge, unsettled, and the source-scale peak of Sec. IV B should be read as conditional on it. Two qualifications cut the other way, and both favour the cusp. Ref. [23] reports that its simulated spectra are reproduced by a source held constant in time and switched off abruptly after a few turnover times—a hard finite lifetime, which is precisely the mechanism of the next paragraph; and Ref. [9] obtains the source-scale peak from exactly that constant-in-time assumption. The measurement that would settle it is of the *magnetic* UETC, which has not been made: Ref. [23] measures the hydrodynamic one.

f. A hard finite lifetime supplies the same cusp—and the same peak. The condition Eq. (37) imposes is weaker than the BK2016 power law, and it is worth showing that something far cruder satisfies it. Take a source that is simply coherent for a time τ_c and gone afterwards. The naïve encoding, a top-hat $f(\tau) = \Theta(\tau_c - |\tau|)$, is not admissible: being idempotent it is its own square, and its transform $T(\omega) = 2\sin(\omega\tau_c)/\omega$ changes sign, so it violates Bochner's theorem [37]—a correlation function must have a non-negative transform; positive-definiteness of the UETC as a modelling constraint, and the Gibbs kernel as an admissible choice, are due to Ref. [23]—and delivers a *negative* gravitational-wave energy density over part of the spectrum, together with exact nulls at $\omega\tau_c = n\pi$. Measured on $\omega\tau_c \in (0, 40]$ it is negative on 47% of the axis, and the sourced spectrum inherits this, going negative over 20.5% and 25.0% of the plotted range at $\hat{\tau} = 10$ and 100 [Fig. 6(b)]. This is the same pathology on which Caprini et al. discard a candidate unequal-time correlator, and it is a useful reminder that "switch the correlation off after τ_c" is not a free modelling choice.

The admissible version is the triangle $f(\tau) = (1 - |\tau|/\tau_c)\Theta(\tau_c - |\tau|)$, which is the top-hat convolved with itself and therefore the genuine autocorrelation of a source switched on for a window τ_c; its transform $\tau_c\,\mathrm{sinc}^2(\omega\tau_c/2)$ is non-negative by construction, and we find it positive at every p and every $\hat{\tau}$ tested. What matters here is that it carries a kink, $f'(0^+) = -1/\tau_c$, so Eq. (37) applies and its transform falls as ω^{-2}: we measure an ultraviolet envelope slope -1.95 against the top-hat's -0.98, and $T\omega^2 \to 4.05$ against the $4/(\tau_c\omega^2)$ that the cusp of the *squared* tent predicts. The consequence is Fig. 6(d): the peak of the triangle-sourced spectrum stays between $p = 1.3$ and 2.1 as $\hat{\tau}$ runs over four decades, close to the $p \simeq 2.4$ of the decaying kernel, while the sweeping law $1.49M$ falls through four decades over the same range and predicts $p_{\rm peak} = 0.037$ at $\hat{\tau} = 100$, a factor 45 below what is measured.

The source-scale peak is therefore considerably more robust than Sec. IV B claims. It does not require the BK2016 power law, nor self-similar decay, nor any particular decorrelation shape: it requires only that the correlator have a kink at zero lag, which any source with a hard finite lifetime possesses. The Auclair et al. objection removes one realisation of the cusp, not the mechanism. What would remove the mechanism is a correlator that is smooth at the origin—a Gaussian is the obvious example—and the question posed after Eq. (37) is correspondingly sharpened: not whether decaying turbulence follows BK2016, but whether its correlator is differentiable at zero lag.

g. Amplitude: an extra power of M. The same kernel fixes the amplitude as cleanly as the peak. The Mach number enters the full-spatial decaying kernel in only two places: the prefactor $\propto M^3$ of Eq. (23) and the temporal factor Eq. (35). Rescaling $\tau \to \tau/M$ in the latter shows it equals $M\,F(\omega/M)$ with F independent of M (the geometric weight, kernel bracket and triangle bounds carry no M), so on the GW cone $\omega = k$ the spectrum factorises exactly as

$$\Omega_{\rm GW}(p; M) \;=\; M^4\, p^3\, \Psi(p,\, p/M)\,, \tag{38}$$

with Ψ the M-independent rescaled spatial integral. Two limits follow. In the infrared $p/M \to 0$ freezes $\Psi \to \Psi(p, 0)$, and the amplitude scales as the bare prefactor, $\Omega_{\rm GW} \propto M^4$ (measured: $M^{3.98}$)—one power steeper than the stationary M^3 of Eq. (23), the extra power being the single M from the temporal convolution. At the source-pinned peak $p \simeq 2.4$, by contrast, the argument p/M slides down the universal Ψ as M grows, and the scaling steepens to $\Omega_{\rm GW}^{\rm peak} \propto M^{5.3}$, with the band-integrated energy following the same $M^{5.3}$ (Fig. 8). This *derives*, rather than postulates, the $M^4 \to M^{5\text{–}6}$ progression quoted in Sec. III E: the M^4 is the genuine infrared prefactor, while the $M^{5\text{–}6}$ peak and energy steepening follows from the source-scale pinning acting on the p/M-dependent temporal weight—a mechanism already present

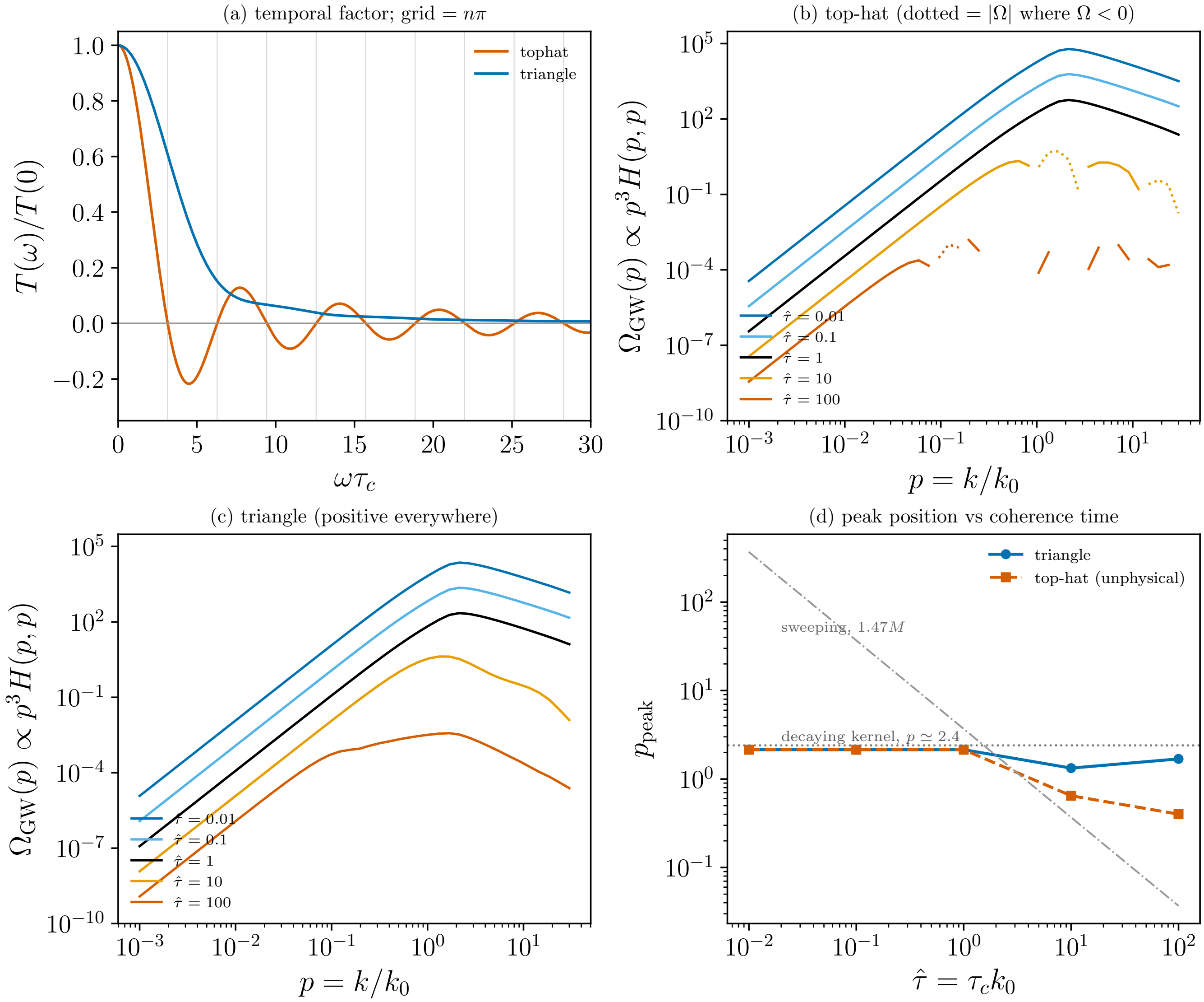


FIG. 6. A source with a hard finite coherence time τ_c, at $R = 10^4$, $R_{\rm IR} = 10^2$, full input spectrum, with $\hat\tau = \tau_c k_0$ as in Eq. (H2). *(a)* The two-leg temporal factors at a single scale, normalised to $T(0)$; vertical grid at $\omega\tau_c = n\pi$. The top-hat crosses zero at every $n\pi$ and is negative between the odd and even multiples; the triangle only touches zero and stays non-negative. *(b)* Top-hat-sourced spectra. Solid where $\Omega_{\rm GW} > 0$, dotted showing $|\Omega_{\rm GW}|$ where it is negative: the curve breaks into disconnected arcs separated by the nulls. This is unphysical output from an inadmissible correlator, shown as a worked example. *(c)* Triangle-sourced spectra, positive at every p and $\hat\tau$. *(d)* Peak position against $\hat\tau$, with the decaying kernel's $p \simeq 2.4$ (dotted) and the stationary sweeping law $1.49M = 1.49\sqrt{2\pi}/\hat\tau$ (dot-dashed) for comparison. The triangle pins the peak at the source scale across four decades of coherence time; the sweeping law does not describe it. Produced by `Notebooks/finite_coherence_gw.py`.

in the quasi-stationary kernel, distinct from and prior to the finite-duration correction. (For reference the stationary peak amplitude scales similarly, $\sim M^{5.4}$, because its peak rides outward to $p = 1.49M$; the amplitude discriminant between the two pictures is therefore the *infrared* slope, M^4 versus M^3, not the peak height.)

C. Confrontation with the Roper Pol simulation

We now test the two pictures against the relativistic MHD turbulence simulation of Roper Pol et al. [28] (run ini2), whose fluid and GW spectra (their Fig. 1) we digitize directly from the published figure by pixel extraction—not by eyeballed point-reading. The fluid spectrum $\Omega_{\rm M}/k$ shows the expected Batchelor infrared k^4 joined to a Kolmogorov $k^{-5/3}$ inertial range, peaking at $k_0 \simeq 673$ in the simulation's units; integrating it gives a turbulent energy fraction

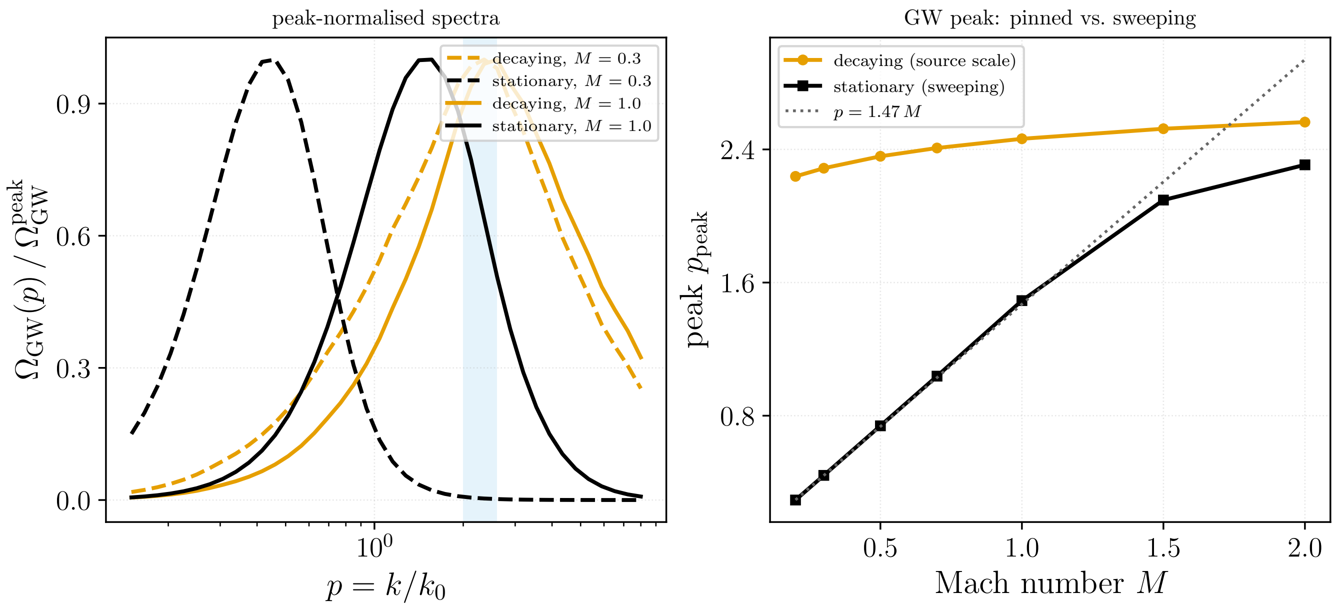


FIG. 7. The decaying source pins the GW peak at the source scale, in contrast to the stationary sweeping peak. (a) Peak-normalised spectra $\Omega_{\rm GW}(p) = p^3 H_{ijij}(p, p; M, R)$ ($R = 10^4$) for the decaying (power-law, orange) and stationary (Gaussian, black) kernels at $M = 0.3$ (dashed) and $M = 1$ (solid): the decaying peaks fall in the source-scale band $p \simeq 2$–2.6 (shaded) for both M, whereas the stationary peaks shift with M. (b) Peak location versus M: the decaying peak is flat at $p_{\rm peak} \simeq 2.4$ [Eq. (36)], while the stationary peak follows $p_{\rm peak} = 1.49\,M$ (dotted) until it saturates only as $M \to 1$ (the luminal limit). Computed with the full-spatial kernel via the FT-of-product Eq. (35); generated by `Notebooks/fullspatial_decay.py`.

$\Omega_{\rm M} = \int (\Omega_{\rm M}/k)\, dk \simeq 1.8 \times 10^{-3}$, i.e. a deeply subsonic effective Mach number $M \simeq \sqrt{\Omega_{\rm M}} \sim 0.05$. Two features of the simulated GW spectrum discriminate between the sweeping and source-scale pictures.

a. Spectral shape. Figure 9 overlays our analytic $\Omega_{\rm GW}/k = p^2 H_{ijij}(p, p)$ for both temporal models on the digitized data. Each prediction's amplitude is fixed not by a free fit but by the measured energy fraction—scaled so that its total GW energy equals the simulated $\Omega_{\rm GW} = (\mathcal{H}_*/k_0)\,\Omega_{\rm M}^2$. Normalised this way the three curves coincide near the peak, and the simulated ultraviolet slope ($\simeq k^{-11/3}$, the value reported in [28]) lies *between* the two kernels: the stationary Gaussian-sweeping kernel is too steep ($\simeq k^{-5.5}$, essentially M-independent across the subsonic range) while the decaying power-law kernel is too shallow ($\simeq k^{-2.6}$). The physical decorrelation is intermediate between the rapidly-decorrelating (stationary) and slowly-decorrelating (decaying) limits; neither idealization alone reproduces the inertial-range slope.

b. Peak location. The sharper discriminant is the peak. In the $\Omega_{\rm GW} = d\rho_{\rm GW}/d\ln k$ convention the simulated GW peak of Ref. [28] sits at $k_{\rm peak} \simeq 1.84\,k_0$—*above* the source scale, near $2k_0$, the "twice the source scale" value expected when the quadratic stress autocorrelates a source peaked at k_0. Figure 10 plots the GW peak against M for both kernels together with the time-resolved simulation track: because the run is *decaying*, its Mach number falls with time, and the measured peak stays pinned at the source scale ($\simeq 2.4\,k_0$) as $M(t)$ sweeps down rather than sliding along the sweeping law—a direct, time-domain discriminant between the two pictures. The decaying kernel pins the peak at $p_{\rm peak} \simeq 2.4$ for every M and passes through the simulation point; the stationary sweeping kernel instead gives $p_{\rm peak} \simeq 1.49\,M$, which at the simulation's Mach lies at $\simeq 0.07\,k_0$, more than an order of magnitude below both the source scale and the data. This conclusion is robust to the uncertain effective Mach: across the *entire* subsonic range $M < 1$ the sweeping peak never reaches k_0, whereas the simulated peak—like the decaying kernel—lies above it. The simulated GW spectrum therefore peaks where the decaying, source-scale picture predicts, not where the stationary sweeping picture does.

c. Amplitude and efficiency. The overall amplitude is a genuine efficiency, not a free normalisation: the saturated GW energy scales quadratically with the source energy, $\Omega_{\rm GW} \propto \Omega_{\rm M}^2$, times the square of the dominant length scale, so that on dimensional grounds $\Omega_{\rm GW} \sim \Omega_{\rm M}^2 (\mathcal{H}_*/k_0)^2$ up to an $\mathcal{O}(1)$–$\mathcal{O}(10^2)$ prefactor that depends on the driving. From the run `ini2` spectra we measure a dimensionless efficiency $\Omega_{\rm GW}/\Omega_{\rm M}^2 \simeq (2$–$6) \times 10^{-5}$ (Fig. 11b), consistent with the digitized late-time value 1.7×10^{-5} used to normalise Fig. 9. Roper Pol et al. find the efficiency itself depends strongly on the driving—forced acoustic and forced magnetic turbulence radiate $\sim 200\times$ and $\sim 10\times$ more than an initially imposed, freely decaying field of the same energy—and that LISA detectability of electroweak-scale

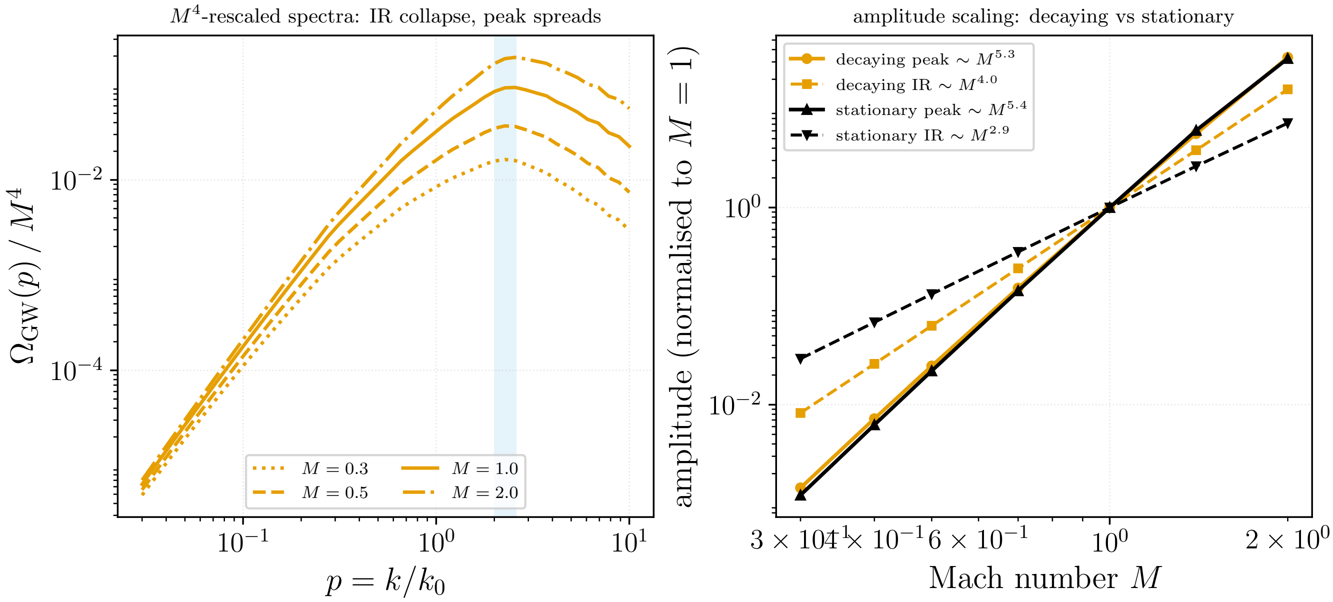


FIG. 8. Mach scaling of the full-spatial decaying GW amplitude, from the exact factorisation $\Omega_{\rm GW} = M^4 p^3 \Psi(p, p/M)$ [Eq. (38)]. (a) M^4-rescaled spectra $\Omega_{\rm GW}(p)/M^4$ for $M = 0.3$–2 ($R = 10^4$): the curves collapse in the infrared—confirming the M^4 prefactor—and spread upward through the source-scale band (shaded), the peak amplitude growing faster than M^4. (b) Amplitude versus M, normalised at $M = 1$, with fitted power laws: the decaying infrared amplitude scales as $M^{4.0}$ (one power above the stationary $M^{2.9}$), while the source-pinned peak amplitude steepens to $M^{5.3}$. Generated by `Notebooks/decaying_amplitude.py`.

turbulence needs ~ 0.1–10% of the plasma energy in motions or magnetic fields [28]. This is why the amplitude is set by an energy fraction and a length scale, and why a decaying source (our and their `ini2` case) is the least efficient of the drivings. One caution about using amplitudes to adjudicate between the analytic and simulated pictures: absolute normalisations on the simulation side were revised between preprint versions (the arXiv record for [28] notes a corrected normalisation error), and the analytic amplitude carries its own $\mathcal{O}(1)$–$\mathcal{O}(10^2)$ driving-dependent prefactor. The peak *frequency* carries neither ambiguity, which is why we treat it, and not the amplitude, as the discriminant.

d. Steady oscillatory state. A caveat on what "the" spectrum means here. After the magnetic source decays, GW *production* stops but the modes do not: each GW mode free-rings at its frequency, so the spectrum shown (theirs [28] and ours) is a *late-time time-average* over at least one oscillation period, once the energy fluctuates about a steady mean rather than growing. Our reanalysis makes this concrete (Fig. 11b): $\Omega_{\rm GW,tot}$ becomes steady to $\pm 4\%$ for $t > 1.15$ while $\Omega_{\rm M,tot}$ is still falling. The stationary and decaying kernels of this paper both target this time-averaged steady state, not an instantaneous snapshot.

e. Viscosity. The simulations run at magnetic Prandtl number $\nu/\eta = 1$ and at the smallest viscosity the resolution allows—far above the physical value—so their Reynolds number is modest. In our kernels this enters only as the inertial-range width $R = (k_d/k_0)^{4/3}$, which sets the dissipation cutoff k_d and hence the upper end of the $k^{-11/3}$ tail; it does *not* set the tail's slope, which is the Kolmogorov inertial-range value. Consistently, the decaying peak we compute drifts only from 2.40 to 2.60 as R runs over 10^3–10^5 (a dissipation-set peak would scale as $R^{3/4}$), confirming that the peak and the infrared are viscosity-insensitive and that viscosity is a purely ultraviolet, dissipation-range effect on the GW spectrum.

V. THE PRINCIPAL SIMULATION–ANALYTIC DISCREPANCY

Setting the two analytic treatments in the literature—the stationary Kraichnan–K41 model of Gogoberidze et al. [2] and the helical inverse-cascade model of Kahniashvili et al. [3]—beside the direct simulations of Roper Pol et al. [28] isolates one discrepancy that is both physically significant and shared by *every* analytic estimate, the kernels of this paper included. It concerns not the location of the peak (Sec. IV) but the slope of the spectrum *below* the peak.

a. Three temporal pictures, three peaks. The analytic works differ in how they model the source's unequal-time correlation, and that choice fixes their spectra. Both [2] and the direct-cascade stage of [3] adopt a Gaussian

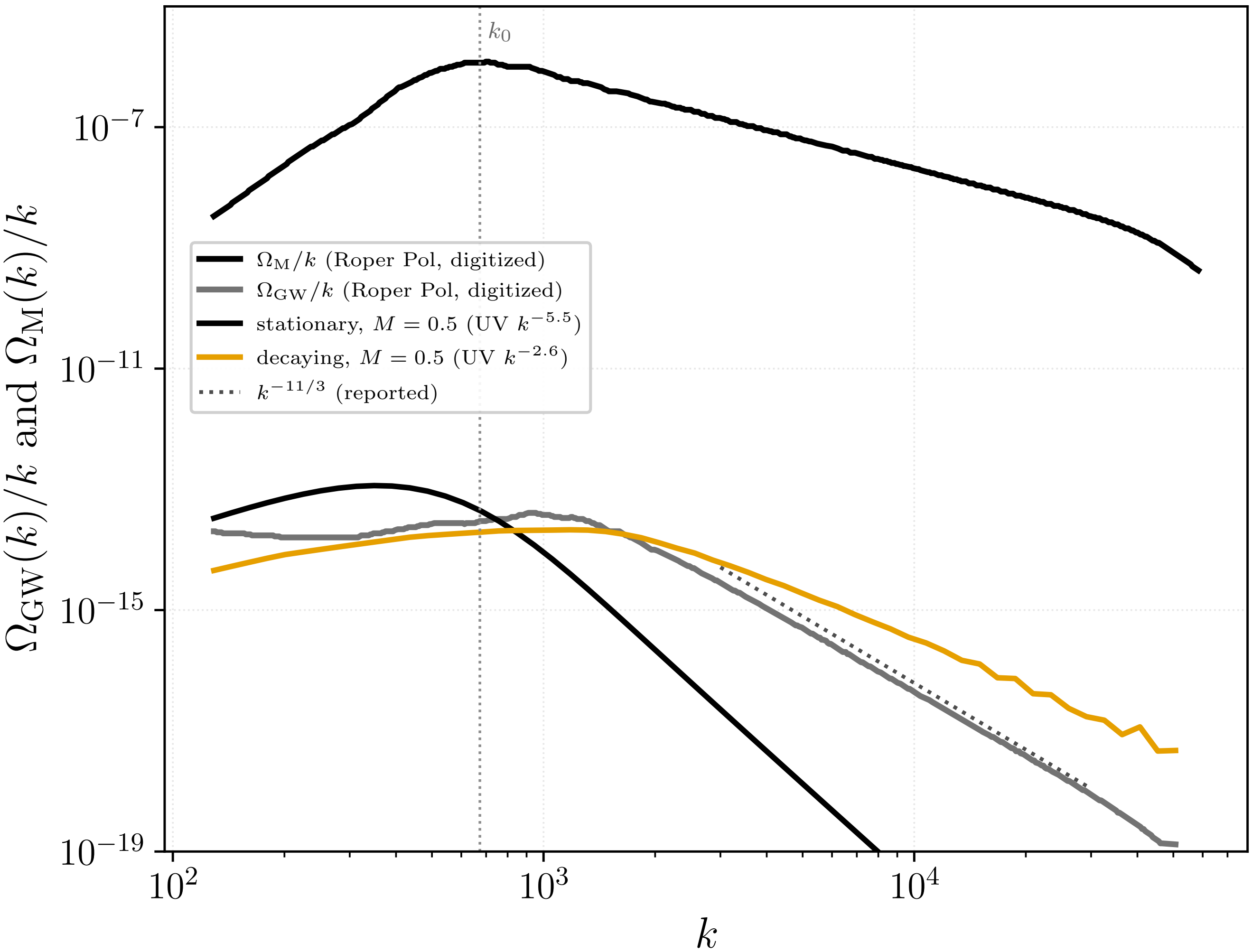


FIG. 9. Analytic GW spectra confronted with the digitized Roper Pol et al. [28] simulation (their Fig. 1, run ini2; pixel-digitized, no fabricated points). Black/grey: digitized fluid $\Omega_{\rm M}/k$ and GW $\Omega_{\rm GW}/k$. Orange/thin black: our decaying and stationary $\Omega_{\rm GW}/k = p^2 H_{ijij}(p,p)$ at $M = 0.5$ (the ultraviolet slope is M-insensitive in the subsonic range), each scaled so its total GW energy matches the simulated $\Omega_{\rm GW} = (\mathcal{H}_*/k_0)\Omega_{\rm M}^2$ (energy-fraction normalisation, no free fit). The simulated UV slope ($\sim k^{-11/3}$, dotted) falls between the too-steep stationary and too-shallow decaying kernels; the dotted vertical marks the source scale k_0. Generated by `Notebooks/roperpol_comparison.py`.

decorrelation $f(\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$ with the eddy-turnover rate $\eta_k \propto \varepsilon^{1/3}k^{2/3}$, which radiates predominantly at the turnover frequency and places the GW peak near $\omega \sim Mk_0$, below the source scale for subsonic turbulence. Kahniashvili et al. add a second, dominant contribution from the helical *inverse* cascade: because magnetic helicity is conserved the energy-carrying eddy grows, and requiring its cascade time to reach the Hubble time drives the dominant peak down to the Hubble frequency $f_H \simeq 1.6\times10^{-5}$ Hz, essentially independent of the cascade model. The simulations [28] make no closure assumption—the source is freely decaying MHD turbulence—and find the GW peak at $k_{\rm GW} \simeq 2k_0$, twice the field scale, the elementary consequence of the source being the *quadratic* stress $T_{ij} \sim B_iB_j$ whose spectrum peaks at twice the field peak. These three pictures place the GW peak at Mk_0, at f_H, and at $2k_0$ respectively—spanning orders of magnitude (Table III); the $2k_0$ result is the one borne out in Sec. IV C.

b. The discrepancy: k^3 versus k. Beneath the peak the disagreement is sharper and universal. Every *quasi-stationary* analytic treatment—the aeroacoustic Gaussian-closure framework of [2, 3] and both the stationary and decaying single-emission-epoch kernels of this paper—gives the *causal* infrared slope

$$\Omega_{\rm GW}(k) \propto k^3 \qquad (k < k_{\rm peak}), \tag{39}$$

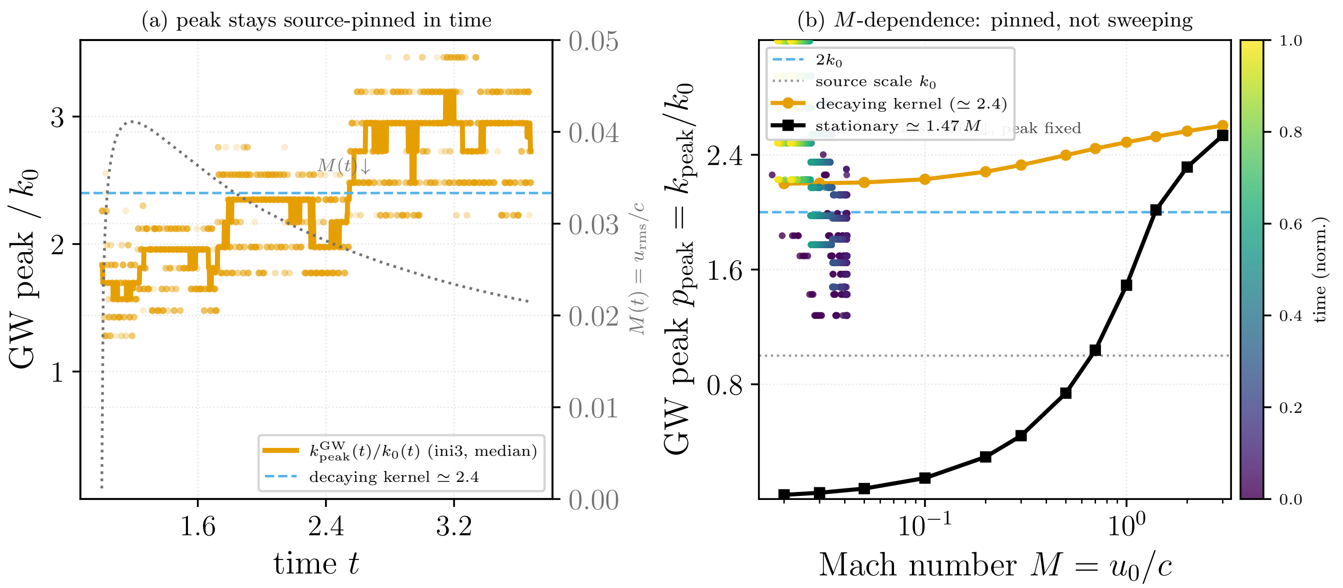


FIG. 10. The GW spectral peak $k_{\rm peak}/k_0$ (in the $\Omega_{\rm GW} = d\rho/d\ln k$ convention) *through time* and versus Mach number, from the public time-resolved data of the decaying runs `ini2`/`ini3` [28] (Zenodo 10.5281/zenodo.3692072). *(a)* The peak-to-source ratio $k^{\rm GW}_{\rm peak}(t)/k_0(t)$ (points, with a running median) holds at the source-scale value $\simeq 2.4$ (dashed) throughout the run, while the instantaneous Mach number $M(t) = u_{\rm rms}(t)/c$ (dotted, right axis) rises after switch-on and then *decays*; the absolute peak $k_{\rm peak}$ and k_0 both drift down together (helical inverse transfer), but their ratio is fixed. The scatter is the discrete-shell wavenumber resolution (k_0 at shells ~ 4–7). *(b)* The same peak versus M: analytic decaying (source-scale, $\simeq 2.4$) and stationary (sweeping, $\simeq 1.49\,M$) kernels, with the simulation *time-track* (points coloured by time) superposed. As the source decays the Mach number sweeps down through $M(t) \simeq 0.04 \to 0.02$, yet the peak stays on the decaying branch at $\simeq 2.4\,k_0$—roughly two orders of magnitude above the sweeping prediction ($1.49M \sim 0.03$) at the same M. The time evolution therefore discriminates the two pictures directly: a sweeping peak would slide *down* the black curve as $M(t)$ falls, whereas the simulated peak does not move. Generated by `Notebooks/gw_peak_vs_mach.py`.

which follows from the analyticity of the equal-time stress correlator at $k \to 0$. The simulations instead find the far shallower

$$\Omega_{\rm GW}(k) \;\propto\; k^1 \qquad (k < k_{\rm peak}), \tag{40}$$

i.e. a *flat* $\Omega_{\rm GW}(k)/k$. Roper Pol et al. single this out as their central new result—"the subinertial power law $\Omega_{\rm GW} \sim k$ is a novel result from our simulations that was not obtained in previous analytical estimates"—their Table II contrasting the analytic infrared slope 3 with the simulated slope 1 for the *same* magnetic spectrum $\Omega_{\rm M} \propto k^5$. Our own digitization of their Fig. 1 (Sec. IV C) confirms it: the measured GW infrared slope is $\simeq k^{+1.3}$, not k^3 [Fig. 9].

c. Localising the disagreement: it is not about the gravitational-wave step. That table carries more information than the single number usually quoted from it. Writing $n[X]$ for the logarithmic slope $d\ln X/d\ln k$, *every* entry of it—subinertial and inertial-range columns alike—is reproduced by two bookkeeping rules. The wave equation divides the source by k^2,

$$n[\Omega_{\rm GW}] \;=\; n[\Omega_{\rm T}] - 2, \tag{41}$$

and the strain follows from $\Omega_{\rm GW} \propto f^2 h_c^2$,

$$n[h_c] \;=\; \tfrac{1}{2}\big(n[\Omega_{\rm GW}] - 2\big). \tag{42}$$

Equation (42) is nothing but this paper's own $\Omega_{\rm GW} = p^2 h_c^2$ [Eq. (29)] read as a slope, so the kernels here sit on exactly the same bookkeeping; the conversion between the two conventions is $\Omega_{\rm GW} = (2\pi^2/3H_0^2) f^2 h_c^2$, and a quoted slope is meaningless without saying which of the two it refers to. Applied to Table II: $(3-2)/2 = 1/2$, $(1-2)/2 = -1/2$, $(-8/3-2)/2 = -7/3$, $(-14/3-2)/2 = -10/3$; and the inertial columns follow from the magnetic slope alone, Kolmogorov $E_{\rm M} \propto k^{-5/3} \Rightarrow n[\Omega_{\rm M}] = -2/3 \Rightarrow n[\Omega_{\rm GW}] = -8/3$, Golitsyn $E_{\rm M} \propto k^{-11/3} \Rightarrow -8/3 \Rightarrow -14/3$. In particular $\Omega_{\rm GW} \propto f^{-8/3}$ in the inertial range is *equivalent* to $h_c \propto f^{-7/3}$, not to any steeper strain law.

Two features of the table then settle where the disagreement can live. First, $n[\Omega_{\rm M}] = 5$ in *both* columns: both sides describe the same causal (Batchelor) magnetic field, so the disagreement is not about the turbulence. Second, the two inertial-range columns carry a single value each, with no analytic/simulation split—above the peak the two approaches agree. The disagreement is confined to one cell-pair, the subinertial $\Omega_{\rm GW}$, and since both sides use Eq. (41) they cannot be disagreeing about the gravitational-wave step at all. What they disagree about is the *stress* spectrum: the analytic side takes $\Omega_{\rm T} \propto \Omega_{\rm M} \propto k^5$, the simulation finds $\Omega_{\rm T} \propto k^3$, white noise.

d. Why the stress cannot be the whole story. ~~The single assumption at fault. That one substitution is the whole discrepancy, and it~~ [demoted: this substitution is pre-empted as an explanation (Refs. [28–30]) and, as the subsection on the branch diagram then shows, identifying it does not by itself resolve the gravitational-wave slope — so it is no longer presented as "the whole discrepancy"] That substitution cannot hold for a quadratic source. Because $T_{ij} \sim B_i B_j$ is quadratic in the field, its spectrum is the magnetic spectrum *convolved with itself*, $P_{\rm T}(\mathbf{k}) = \int d^3q\, P_{\rm M}(\mathbf{q}) P_{\rm M}(\mathbf{k}-\mathbf{q})$. For k below the field peak that integral is dominated by pairs $\mathbf{q}_1 \simeq -\mathbf{q}_2$ sitting *on* the peak, whose contribution does not vanish as $k \to 0$: $P_{\rm T}(0) = \int d^3q\, P_{\rm M}(q)^2$ is a finite constant, so $E_{\rm T} \propto k^2$ and $\Omega_{\rm T} \propto k^3$ *however steep* $\Omega_{\rm M}$ is at small k. A k^5 field cannot make a k^5 stress; the quadratic operation floors the slope at white noise, and the floor binds only below the peak—which is exactly why the inertial-range columns need no analytic/simulation split. This is the same convolution that manufactures large-scale correlation out of an uncorrelated field in Sec. C 2, read one power of k shallower.

The point worth stressing is that this is purely kinematic and survives every closure assumption in the analytic chain. It holds in particular for a perfectly Gaussian field, since Wick's theorem gives $\langle B^2 B^2 \rangle = 2\langle BB \rangle^2$, which *is* that self-convolution. Gaussianity, the sweeping ansatz, and stationarity are therefore all innocent of the infrared discrepancy on the stress. Our own kernels do not take the substitution—they carry the double convolution explicitly [Eqs. (13)–(14)] and so reproduce the white-noise stress floor—which is why the residual gap must be, and below is shown to be, a statement about the *temporal* response rather than about the stress. None of this is new, and it should not be read as a localisation of the discrepancy. The white-noise floor is the explanation Ref. [28] give themselves, it is derived for stress spectra in Ref. [29] and carried through the stress evolution in Ref. [30]; we restate it because the bookkeeping of Table II shows exactly where it enters, and because it is what makes the *next* step necessary. Once the stress is white noise on both sides, the same stress feeds both branches of Eq. (47), and the surviving question is entirely about the source's temporal history—which is what Sec. V A then computes. [Referee B and the prior-art sweep: presented as ours, this is claim 3 of the report, published and subsequently retracted in this very section]

slope of	analytic	simulation	Kolmogorov	Golitsyn
$\Omega_{\rm M}$	5	5	$-2/3$	$-8/3$
$\Omega_{\rm T}$	5	3	$-2/3$	$-8/3$
$\Omega_{\rm GW}$	3	1	$-8/3$	$-14/3$
h_c	1/2	$-1/2$	$-7/3$	$-10/3$

TABLE II. Logarithmic slopes $n[X] = d\ln X/d\ln k$ compiled by [28]: the two left columns are the subinertial range as estimated analytically and as measured in the simulation, the two right columns the inertial range for a Kolmogorov and a Golitsyn magnetic spectrum. The $\Omega_{\rm M}$, $\Omega_{\rm GW}$ and h_c rows are theirs; the $\Omega_{\rm T}$ row is not tabulated by them and is reconstructed here by inverting Eq. (41). Every entry follows from the $\Omega_{\rm M}$ row through Eqs. (41)–(42), so the disagreement reduces to the single reconstructed cell-pair $n[\Omega_{\rm T}] = 5$ versus 3.

e. Why it is significant. The discrepancy is two infrared powers of k, and it flips the sign of the strain slope: $h_c(f) \propto f^{+1/2}$ in the analytic estimates versus $f^{-1/2}$ in the simulations. Because the peak sits near the high-frequency end of the accessible band, this shallower-than-causal slope governs the *entire* sub-peak range—most of the observational band for an electroweak-scale source—so it changes the predicted low-frequency power, and hence the detectability, in the opposite direction to what the causal estimates assume. The stakes are sharpened by the fact that the two pictures also place the peak differently (Sec. IV): for an electroweak source with $k_0 \sim 10^2\,\mathcal{H}_*$ there is a band lying *above* the analytic peak $\sim Mk_0$ and *below* the simulated peak $\simeq 2k_0$—roughly 0.15–3 mHz—in which the analytic spectrum is already falling while the simulated one is still rising as f^1. The two predictions there have opposite *sign* of slope, not merely different normalisations. That decade overlaps LISA's region of best *energy-density* sensitivity: with the standard instrument noise the minimum of $\Omega_{\rm sens}(f)$ lies near 2.5 mHz and stays within a factor of two of that minimum over 1.0–4.6 mHz. (The often-quoted 3–10 mHz is the minimum of the *strain* sensitivity; the f^3 weighting moves the Ω minimum down.) The slope difference therefore bears on detectability and on reconstruction alike. The stake is higher still for pulsar timing, whose band lies below the peak—by up to about two decades, though for the largest processed eddies it sits immediately below it [9]—and which therefore measures little but this tail. [Referee A said 3–10 mHz; that is the strain minimum. Recomputed: Ω min = 2.52 mHz.] ~~That decade sits just below LISA's sensitivity minimum, which lies at 3–10 mHz, so the slope difference bears more on parameter reconstruction than on raw detectability.~~ (The rate at which the analytic prediction falls above its peak is model-dependent and is quoted inconsistently in the literature; only the sign of the slope is needed for the point.)

f. What finite duration can and cannot do. The causal k^3 of Eq. (39) is the strict $k \to 0$ asymptote and is unavoidable: the equal-time stress of a spatially compact source is analytic (white) at $k \to 0$, and our quasi-stationary kernels reproduce it (measured infrared slope 2.93 over $p = 10^{-3}$–10^{-2}). Could the *finite duration* of the source nonetheless open a shallower band above this asymptote? Carrying the unequal-time integration over a finite emission window $[0, T_{\rm em}]$ replaces the stationary frequency factor by the finite-duration kernel

$$\int_0^{T_{\rm em}}\!\!\int_0^{T_{\rm em}} dt_1\, dt_2\, e^{i\omega(t_1-t_2)} \;\longrightarrow\; \frac{4\sin^2(\omega T_{\rm em}/2)}{\omega^2} \;\xrightarrow[\text{ens. avg}]{\omega T_{\rm em}\gg 1}\; \frac{2}{\omega^2}, \tag{43}$$

This object appears three times in this paper and it is worth saying once that it is the same object each time: as Eq. (9) it is the temporal factor of the triangular correlator $T_{\rm burst}$, here it is the finite-window kernel, and as Eq. (47) it is condition (II) of the universal infrared theorem. One formula, read in three contexts. It tends to $T_{\rm em}^2$ (a constant, giving k^3) for $\omega T_{\rm em} \ll 1$ but to $2/\omega^2$ (giving k^1) for $\omega T_{\rm em} \gg 1$—*provided the source stays coherent across the window.* A source that is both coherent and sustained over $[0, T_{\rm em}]$ would thus carry a k^1 sub-peak band; this is the regime in which a source treated as statistically stationary over a finite duration reproduces the simulated slope.

g. Freely decaying turbulence does not realise it. Self-similar decaying turbulence —whose scaling laws in the early-Universe context were established in Refs. [32, 44]—fails the coherent-*and*-sustained condition. Its correlation time is the eddy-turnover (sweeping) time $\tau_c(k) = 1/(\varepsilon^{1/3}k^{2/3}) \propto 1/(Mk^{2/3})$, fixed by the velocity and *independent* of the decay time; Kolmogorov consistency ($\varepsilon \sim u^3/L \sim u^2/\tau_{\rm st}$) ties the decay time to the outer-scale eddy turnover, $\tau_{\rm st} \sim L/u$, so the dimensionless decay rate $\varepsilon_0 = \tau_{\rm eddy}/\tau_{\rm st}$ is of order unity, not a free parameter. As the flow decays the eddy time grows, but the source amplitude falls faster ($\Phi_{\rm eq} \propto (1+t/\tau_{\rm st})^{-34/21}$ per leg), so the eddies turn coherent only *after* they have decayed away: the growing coherence is cancelled by the decaying amplitude, and the coherent, sustained window required for k^1 is never reached for sub-peak modes. We verify this with a first-principles full-spatial self-similar kernel—the geometry of Sec. IV B with the temporal factor replaced by the genuine slow-time double-time integral over $[0, T_{\rm em}]$, which reduces to the quasi-stationary kernel as $T_{\rm em} \to \infty$ (cross-check ratio $\to 1$). Its infrared slope is causal ($\simeq 2.5$, steepening to 3 as $p \to 0$) and essentially *independent of the decay rate*—2.53, 2.57, 2.51 for $\varepsilon_0 = 0$, 1, 4 [Fig. 12]. The finite duration of a decaying source does not flatten the infrared.

It is worth reporting the branch parameters of that figure explicitly, because a null result of this kind is only meaningful if the calculation was in a position to see the effect it fails to find. The kernel runs at $T_{\rm em} = 40$ and $M = 1$ with $\tau_c(x) = \sqrt{x}/M$, so $T_{\rm em}/\tau_c$ runs from 40 at the outer scale to 4000 at the dissipation scale, and across the fitted band $p = 0.06$–0.45 the dimensionless frequency is $\omega T_{\rm em} = 2.4$–18. The band therefore lies *above* the window break $p = \pi/T_{\rm em} = 0.079$: a source held coherent over this window *would* have shown the linear branch here, and the measurement is that the decaying source does not. The test is a real one.

What it does not test is equally worth stating. The band lies above the window break but *below* the decorrelation break $p = \pi/\tau_c = 3.1$ at the outer scale, and it is the shorter of the two memories that controls: the source forgets itself long before the window closes, so the effective coherence is τ_c and not $T_{\rm em}$, and the band sits on the causal side of the break that matters. That is the mechanism behind the null result rather than a qualification of it—it is the same statement as "the growing coherence is cancelled by the decaying amplitude" above, read as a break location—but it does mean the figure bounds the *decorrelation* time and not the duration. [Referee C, Task 2.4: the figure might not have tested its own caption; the branch parameters are now reported so the reader can check that it does]

h. Roper Pol's own explanation, and a time-domain reconciliation. The white-noise stress floor established above—derived for stress spectra in Ref. [29] and modelled through the stress evolution in Ref. [30]—is exactly the explanation [28] themselves give—the spectrum of B^2 "can never be steeper than that of white noise"—and it accounts for the stress but not, by itself, for the gravitational waves: with $\Omega_{\rm T} \propto k^3$ the two response branches of Eq. (43) give $\Omega_{\rm GW} \propto k^3$ and $\Omega_{\rm GW} \propto k^1$ respectively, from the *same* stress. Which branch a mode sits in is a question about the source's temporal history, and that is where the remaining gap lies.

Crucially, [28] add that the causal k^3 is not absent but *transient*: a flat spectrum reaching arbitrarily large scales would be acausal, so at the very largest scales and earliest times the spectrum grows as k^3 before saturating to the flat k^1 plateau, and this transition "is much shorter than the time it takes for the GW spectrum to become stationary." We confirm this directly by reanalysing the public time-resolved spectra of their run `ini2` (Fig. 11; data from [28] via Zenodo). At source switch-on the infrared slope is the causal $\Omega_{\rm GW} \sim k^3$ (measured $k^{3.3}$ at $t = 1.00$), and within $\Delta t \sim 0.02$ it flattens to the saturated $\Omega_{\rm GW} \sim k^1$ shown in the paper's Fig. 1. The total GW energy then fluctuates about a constant mean—the "steady oscillatory state" of the freely ringing GW field—while the magnetic energy keeps decaying; and the magnetic peak $k_0(t)$ migrates to smaller k, the signature of the helical inverse transfer.

i. How fast the transient should have been. The transient admits a sharp quantitative test, and it fails it in an informative direction. For a source switched on at t_0 and *coherent* thereafter, the per-mode energy is $\dot h_k^2 + k^2 h_k^2 =$

$|\int_{t_0}^{t} e^{ikt'} S_k(t')dt'|^2$ exactly, whose oscillation-averaged value is $(2/k^2)\,B(k,\Delta t)$ with the build-up fraction

$$B(k,\Delta t) \;=\; 1-\mathrm{sinc}(k\Delta t) \;\xrightarrow[k\Delta t\ll 1]{}\; \tfrac{1}{6}(k\Delta t)^2, \qquad B \;\xrightarrow[k\Delta t\gg 1]{}\; 1, \tag{44}$$

so that $\Omega_{\rm GW}(k,t) = \Omega_{\rm sat}(k)B(k,\Delta t)$ with $\Omega_{\rm sat} \propto k\,\Pi(k)$. This is the oscillation-averaged interpolation between the two limits of Eq. (47), and it makes the transient concrete: the spectrum fills in from the ultraviolet downwards, the knee at $k \sim 1/\Delta t$ marching to lower k as the source acts, and if the source switches off the knee freezes where it stood. The causal k^3 is what lies *below* that knee at any instant. Measured against the `ini2` time series, however, the simulated spectrum saturates at every scale within $\Delta t \simeq 0.02$–0.06, whereas Eq. (44) would require $\Delta t \simeq 0.1$–0.4 over the same band of k. The source therefore saturates *faster than any coherent window can*, which is the signature of an effectively *impulsive* onset: a sudden switch-on is broadband in time, so every k lights up at once rather than in sequence. That is also the reason the peak lands on the spatial stress scale $\simeq 2k_0$ rather than on the sweeping scale Mk_0 (Secs. IV, IV C): an impulsive source has ample temporal power at $\omega = k$ for every k in the box, so nothing throttles the spatial maximum. The continuous, decorrelating emission that the stationary kernel describes is present but M-suppressed and subdominant. Computed with `Notebooks/time_dependent_gw_spectrum.py`, which also verifies Eq. (44) against a direct numerical solution of the driven oscillator to six digits.

j. Impulsive onset and sustained emission are not competing accounts. Two mechanisms have now been offered for the same simulated spectrum, and as stated they look like alternatives. The finite-duration branch of Eq. (47) requires a source that acts over a long window; the onset just inferred is effectively instantaneous. They are not alternatives, because they answer different questions. The build-up test measures the sharpness of the switch-*on*; which branch of Eq. (47) a mode sits in depends on the *length* of the emission window $T_{\rm em}$, and a source may be switched on abruptly and go on radiating for a long time afterwards. Read that way the two control different observables and do not compete. A sharp onset is broadband in time, so every mode in the box lights up at once and nothing throttles the spatial maximum: that is what fixes the peak at $\simeq 2k_0$ rather than at Mk_0. A long emission window pushes the break $k_{\rm br} \simeq 1/\delta t$ [9, 26] below the box fundamental: that is what places every mode the simulation contains on the k^1 side. The two statements concern opposite ends of the same window, and Sec. V A exhibits one finite lifetime delivering both at once—the linear band of Table IV and a peak that stays at $p \simeq 1.1$–2.2 throughout.

What this does *not* reconcile, and should not be presented as though it did, is the coherence requirement. The k^1 branch is reached only if the source stays correlated with itself across the window, and the build-up measurement of the preceding paragraph is evidence pointing the other way: it bounds how slowly a *coherent* source would have had to saturate, and the simulation saturates faster than that bound. The excess admits two readings—an onset sharper than the coherent model allows, or a source that is not coherent over $T_{\rm em}$ at all—and the time series quoted here does not separate them. That is the same open question raised in Sec. VI D and restated in Conclusion 3; no calculation in this paper closes it, and the measurement that would is the magnetic unequal-time correlator proposed in the next paragraph. [Referee C: Sec. V inferred an impulsive onset in one paragraph and appealed to sustained coherence in another, without ever setting the two beside each other]

k. What the simulation does and does not compute. The comparison is structurally asymmetric, in a way that bears directly on the apparatus of this paper. Every analytic treatment here begins from a *modelled* unequal-time correlator and convolves it with the Green function [Eq. (14)]; the simulation never forms one. Its gravitational-wave sector keeps h and $\dot h$ permanently in Fourier space and advances each mode with an *exact analytic propagator*—with the source held fixed across a step, $h(t+\Delta t) = (h - S/\omega^2)\cos\omega\Delta t + (\dot h/\omega)\sin\omega\Delta t + S/\omega^2$ and $\omega^2 = k^2 - a''/a$—sitting entirely outside the Runge–Kutta integrator that carries the MHD. There is therefore no spatial differencing of h and no Courant condition on the gravitational-wave sector, and the simulated infrared slope and peak cannot be artefacts of the wave integration; the Fourier transform the module performs is demanded by the transverse-traceless projection, which is genuinely nonlocal in real space, and not by retardation, which the propagator already *is*, mode by mode.

Two consequences follow. First, no unequal-time correlator is computed anywhere in that calculation—not as solver input, not in the single stored previous step (which supplies a finite-difference $\dot S$, a two-time *difference* rather than a two-time *product*), and not in the diagnostics, which are equal-time. There is thus nothing on the simulation side that can be set beside the Gaussian and power-law decorrelation models of Sec. II D: the two descriptions can only be confronted downstream, at the level of the GW spectrum, which is what Sec. IV C does. Second, and more usefully, the correlator is recoverable in post-processing. The projected stress amplitudes $S_+, S_\times$ live in the code's auxiliary arrays in Fourier space and are already exposed as output; dumping them at a cadence that resolves the eddy-turnover time at the spectral peak and forming the shell average of $\langle S_+(\mathbf{k},t_1)S_+^*(\mathbf{k},t_2) + S_\times(\mathbf{k},t_1)S_\times^*(\mathbf{k},t_2)\rangle$ would measure directly the one object that every analytic treatment in this paper is obliged to assume. We are not aware of such a measurement in the literature. Since the peak discrepancy of Sec. IV turns on precisely that assumption, it is the sharpest single test available, and we propose it here as the natural next step.

l. What the discrepancy points to. This reframes the tension considerably. The causal k^3 of our quasi-stationary and self-similar kernels is not contradicted by the simulation: it is the early-time transient, and the discrepancy with the simulated k^1 is a comparison of an *asymptotic* analytic slope with a *time-saturated* simulated one (the same axis mismatch anticipated for the finite-window band, Sec. V C). Two ingredients set the saturated slope: the white-noise ceiling of the quadratic stress, which is established, and a source that remains correlated with itself across the emission window, which is not—the inverse transfer visible in Fig. 11 is suggestive of it and the build-up test above tells against it. Neither is captured by a single-time hydrodynamic kernel. The honest statement is therefore weaker than a genuine failure of causality and stronger than a mere convention: our kernels give the correct transient infrared, and reproducing the stationary k^1 requires following the source through its saturation, a target for a dedicated time-dependent MHD calculation. It should be added that the two slopes are in any case established in disjoint bands of k, separated by the duration break $k_{\rm break} \simeq 1/\Delta t_s$ of Eq. (48), which for the published runs lies one to two decades below the largest mode the simulation box contains (Sec. V C). No simulation of this class can exhibit the causal asymptote, and none should be read as excluding it.

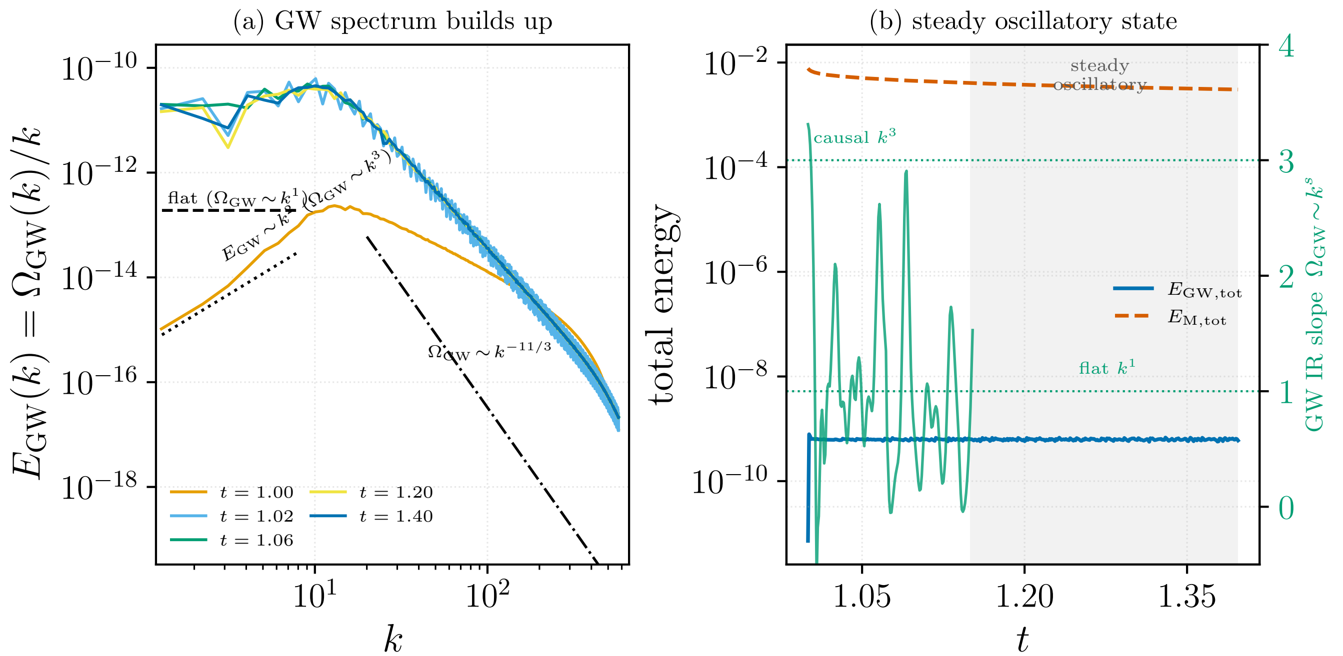


FIG. 11. Time-series reanalysis of the Roper Pol et al. [28] run `ini2` (public Pencil Code spectra, Zenodo 10.5281/zenodo.3692072), the run of their Fig. 1. *(a)* The GW energy spectrum $E_{\rm GW} = \Omega_{\rm GW}/k$ at successive times: at switch-on ($t \simeq 1.0$) the infrared follows the causal $E_{\rm GW} \sim k^2$ ($\Omega_{\rm GW} \sim k^3$) and within $\Delta t \sim 0.02$ saturates to the flat plateau ($\Omega_{\rm GW} \sim k^1$) of their Fig. 1. *(b)* Total energies versus time: the GW energy rises while the source acts, then fluctuates about a steady mean (grey band; the "steady oscillatory state"), while the magnetic energy keeps decaying; the right axis (green) shows the measured GW infrared slope crossing from the causal k^3 to the saturated k^1. Generated by `Notebooks/roperpol_timeseries.py`, which auto-downloads the data.

A. The infrared branch diagram

Everything above has been an argument about which branch of Eq. (47) a given band lies in, conducted without ever displaying the two branches together. They can be exhibited directly rather than inferred, and doing so settles the question the preceding paragraphs keep deferring: whether any temporal history at all produces k^1 over a band while leaving the $k \to 0$ asymptote alone. We evaluate the kernel with the triangular correlator $f(\tau) = (1 - |\tau|/\tau_c)\Theta(\tau_c - |\tau|)$—the autocorrelation of a source switched on for a window τ_c, and the admissible replacement for the top-hat—and measure the *local* logarithmic slope $d \ln \Omega_{\rm GW}/d \ln p$ across eight decades of p, so that the predicted break is straddled on both sides.

Because the stress is quadratic, the two-leg temporal factor is the transform of the *squared* tent, whose one-sided slope at zero lag is $f'(0^+) = -2/\tau_c$. The cusp theorem Eq. (37) then gives $T \to 4/(\tau_c\omega^2)$ against $T(0) = 2\tau_c/3$, so

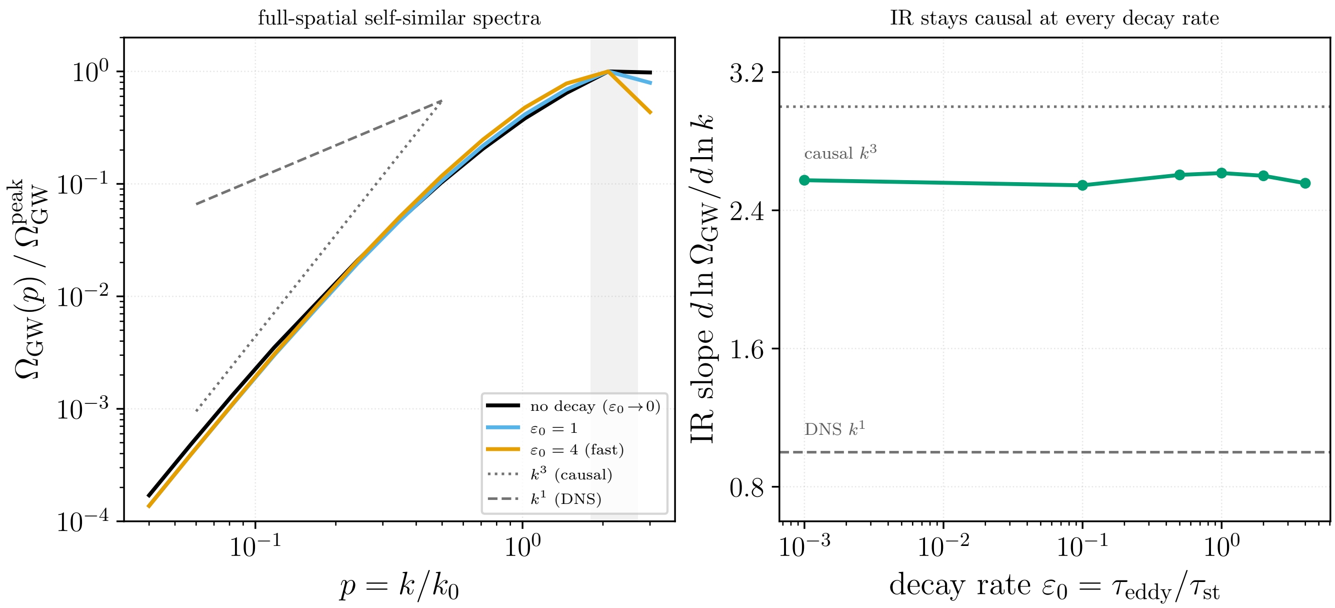


FIG. 12. The finite duration of a *decaying* source does not flatten the infrared. *Left:* GW spectra from the first-principles full-spatial self-similar kernel ($T_{\rm em} = 40\,\tau_{e0}$, $M = 1$) at three decay rates; the infrared slopes overlap on the causal k^3 guide (dotted) rather than the k^1 guide (dashed), and share the source-scale peak (shaded). *Right:* the infrared slope versus the decay rate $\varepsilon_0 = \tau_{\rm eddy}/\tau_{\rm st}$ stays at the causal value ($\simeq 2.5$–3) for every ε_0, never approaching the simulated k^1; the physically realised self-similar decay sits at $\varepsilon_0 \sim 1$. Generated by `Notebooks/fullspatial_selfsimilar_exact.py`.

treatment	temporal decorrelation	GW peak	infrared $\Omega_{\rm GW}$	high-k $\Omega_{\rm GW}$
Gogoberidze 2007 [2]	Gaussian, $\eta_k \propto k^{2/3}$	$\sim Mk_0$	k^3	steep ($\sim k^{-9/2}$)
Kahniashvili 2008 [3]	Gaussian + helical inverse cascade	$\to f_H$ (Hubble scale)	k^3	Kolmogorov input
this paper	Gaussian / power-law / self-similar decay	$1.49Mk_0$ / $2.4k_0$	k^3 (causal)	k^{-5} / $k^{-8/3}$
Roper Pol 2020 [28]	freely decaying	$2k_0$	k^1	$k^{-8/3}$

TABLE III. Temporal-decorrelation assumption, GW spectral peak, and spectral slopes for the three literature treatments and the kernels of this paper. The peak frequency spans orders of magnitude across the three works; the infrared slope is the discrepancy that is universal across *every* analytic treatment—the quasi-stationary estimates *and* the first-principles self-similar decaying kernel of this paper all give the causal k^3 (Fig. 12)—yet is contradicted by the simulation (k^1). Restoring the finite duration of a freely decaying source does *not* flatten the infrared: the growing eddy coherence is cancelled by the faster amplitude decay, so the hydrodynamic infrared stays causal for every decay rate. The rows nevertheless disagree about a band rather than about physics: once the stress is recognised as white noise below the peak—which is Ref. [28]'s own explanation, derived in Ref. [29]—the same stress feeds both branches of Eq. (47), and what separates the k^3 and k^1 entries is only which branch the tabulated band of k lies in. ~~The tension is therefore genuine, and Sec. V localizes it to two identifiable steps: on the analytic side the substitution $\Omega_{\rm T} \propto \Omega_{\rm M}$, which a quadratic source forbids, and on both sides the question of which branch of the finite-duration response the observed band of k lies in.~~ The infrared slope is not quoted explicitly by [3] but is the generic causal value of their aeroacoustic Gaussian-closure framework. Slopes are in the $\Omega_{\rm GW} = d\rho_{\rm GW}/d\ln k$ convention.

the two asymptotes cross at $\omega\tau_c = \sqrt{6}$, and the local slope passes through its midpoint where $1 + \cos u = 2\sin u/u$, i.e. at $u = \pi$ exactly. The break therefore sits at

$$k_{\rm break} = \pi/\tau_c, \tag{45}$$

a factor π above the naive $1/\tau_c$. Table IV bears this out: the measured coefficient converges to 3.07 for a hard global lifetime and 3.17 for the scale-dependent eddy lifetime, against $\pi = 3.1416$.

The infrared slope is $+3.000$ in every case, and the intermediate slope is $+0.974$ and $+0.995$ for $\hat\tau = 10^3$ and 10^4 ($+0.977$, $+0.996$ with the spatial factor divided out). Two caveats are worth stating plainly. The linear band is only *resolved* once $\hat\tau \gtrsim 10^3$: below that the break has not separated from the source-scale roll-over, and the fitted coefficient is contaminated by the peak. And the band is not a clean power law—the $\sin^2$ ringing of Eq. (47) makes the local slope oscillate between 0.90 and 1.25 about its mean. The peak itself stays at $p \simeq 1.1$–2.2 throughout, confirming that this is the same mechanism as the source-scale pinning of Sec. IV B: one finite lifetime produces both.

	$\hat{\tau}$	IR slope	band slope	$p_{\text{break}}\tau_c$	peak
global	10^2	+3.000	—	3.065	1.29
	10^3	+3.000	+0.974	3.072	1.28
	10^4	+3.000	+0.995	3.072	1.29
eddy	10^2	+3.000	—	3.150	1.62
	10^3	+3.000	+0.965	3.172	1.71
	10^4	+3.000	+0.996	3.175	1.63

TABLE IV. Infrared branch diagram for a hard finite lifetime, from `Notebooks/ir_branch_diagram.py`. "global" is $\tau_1 = \tau_2 = \tau_c$, a scale-independent lifetime; "eddy" is $\tau_i = \sqrt{x_i}/M$, the scale-dependent lifetime of Sec. IV B. The band slope is quoted only where the break is separated from the source scale by more than a decade. $p_{\text{break}}\tau_c$ is to be compared with $\pi = 3.1416$, Eq. (45).

This settles, in a controlled and reproducible way, what the retracted $\Omega_T \propto \Omega_M$ argument was reaching for: a temporal history *can* produce k^1, over a band, without touching the $k \to 0$ asymptote. [Task 2.1 — the figure both referees asked for]

B. The kernel at the simulation's own parameters

The branch diagram above is generic. The obvious next demand—and the one thing this paper has so far declined to do—is to run the same kernel at Ref. [28]'s own parameters and see what it gives over the band their box actually contains. Explaining a measurement away is not the same as reproducing it, and until this is done the manuscript has only done the former.

a. Their parameters, taken from their data. From the public `ini2` spectra, shell index times 100 gives the Hubble-unit wavenumber (the calibration is fixed independently at both ends of the range and agrees to 1.6%). That gives a source scale $k_0 = 673$, a box fundamental $k_{\text{box}} = 129$, hence $p_{\text{box}} = 0.19$: the box holds 0.66 decades below the peak and nothing more. The inertial range ends at $k_d/k_0 = 44.6$, i.e. $R = 158$, and $\Omega_{\text{M}} = 1.8 \times 10^{-3}$ gives $M = 0.043$–0.060. Two checks confirm the extraction: the GW peak comes out at $1.836\,k_0$ against the 1.84 of Sec. IV C, and the infrared slope measured from the raw shells, +1.295, reproduces the digitised +1.298 independently.

The one parameter their data do not give directly is τ_c, so we carry four closures through: the eddy time at the initial Alfvén speed ($\hat{\tau} = \tau_c k_0 = 8.0$), the eddy time at $M = \sqrt{2\Omega_{\text{M}}}$ and $\sqrt{\Omega_{\text{M}}}$ (16.6 and 23.4—this range contains the manuscript's own $\tau_c = 1/(k_0 u_0)$), and the magnetic-energy e-folding time (410.8), which bounds the lifetime from above.

b. What the kernel gives, and where it falls short. Table V collects the band-averaged slope over their own fit band $p = 0.193$–0.891. The prediction spans +0.57 to +1.59 and the measured +1.30 lies inside that span—but near its steep end, and the agreement should not be oversold. The single closure that reproduces the measurement, $\hat{\tau} = 8.0$, is also the closure for which the break $p_{\text{break}} = \pi/\tau_c = 0.39$ falls *inside* the fit band; there the local slope sweeps from +2.70 to 0.00 across the band, so +1.59 is a crossover average over a region that is not a power law at all. The three longer-lifetime closures—including the manuscript's own—sit on the asymptotic branch and give +0.70 to +0.77, which is 0.5–0.6 *shallower* than measured.

That residual is diagnosed, and it is not temporal. On the asymptotic branch the temporal factor contributes exactly -2, so the slope is $3 + d\ln\mathcal{G}/d\ln p - 2$ with $\mathcal{G}$ the spatial factor. The infinitely coherent control—no window at all, and exactly k^3 as $p \to 0$—reads +2.709 over this band, i.e. $d\ln\mathcal{G}/d\ln p = -0.291$, and $1 - 0.291 = 0.709$ reproduces the measured band slope to three decimals. The whole deficit is our spatial factor rolling over towards the peak inside the fit band, which is unavoidable when the band's upper end is within a factor two of k_0. The same fact shows up on the second diagnostic: our peak sits at 1.58–$1.69\,k_0$ against their $1.84\,k_0$. We roll over slightly too early, and the simulated spectrum stays straighter below its peak than ours does.

c. What this does establish. Two things, and neither needs a fitted parameter. First, *the causal +3 is not in the box*: the infinitely coherent control, which has no finite window and no temporal suppression whatever, already reads only +2.71 over the band the box contains. A measured infrared slope below 3 is therefore not evidence against causality even before any duration is invoked. Second, for every closure with $\hat{\tau} \gtrsim 16$ the break lands at $k_{\text{break}} = 5$–127 in Hubble units against $k_{\text{box}} = 129$—at or below the box floor, and with $M = \sqrt{2\Omega_{\text{M}}}$ within 1.2% of the first resolved shell. The box sits essentially *on* the break, which is the quantitative form of the claim Sec. V C makes from the published run parameters alone.

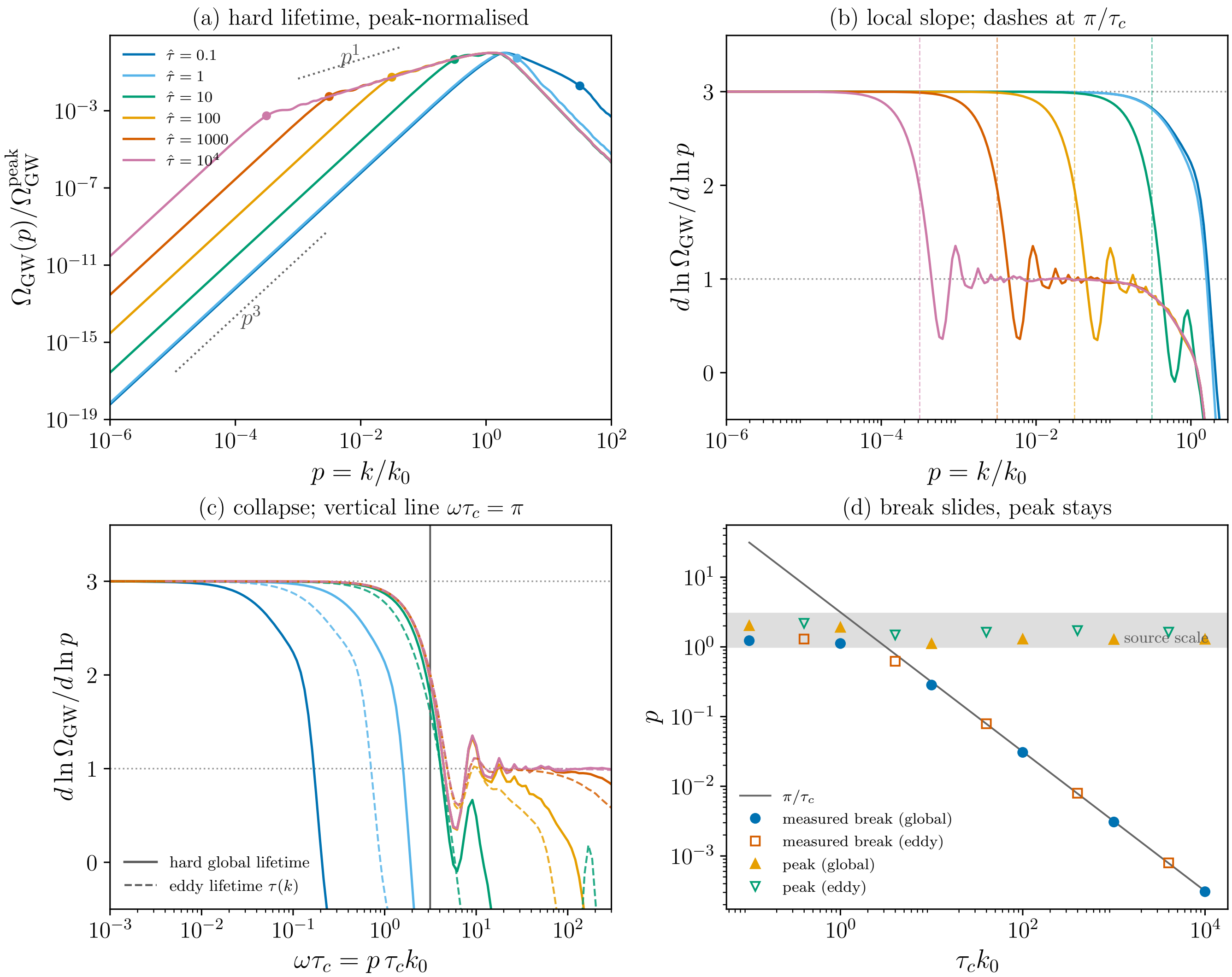


FIG. 13. Local logarithmic slope of $\Omega_{\rm GW}$ against $p = k/k_0$ for the triangular (finite-lifetime) correlator, at several $\hat\tau = \tau_c k_0$. The slope sits at 3 in the infrared and falls to 1 above the break Eq. (45); the residual oscillation about the linear band is the $\sin^2$ ringing of the finite window, not noise. The linear band is resolved only for $\hat\tau \gtrsim 10^3$.

Their raw time series says the same thing without reference to our kernel. The shells saturate at $t - t_0 = 0.0070$ down to 0.0020 for shells 1–6, i.e. $t_{\rm sat}k$ between 0.9 and 1.2: saturation goes as $1/k$, not at a common time, so every mode the box holds completes its oscillations while the source is still acting—the $k\Delta\eta \gg 1$ branch of Eq. (47). And the infrared slope is +3.50 at switch-on, +3.17 by $\Delta t = 0.002$ and +1.28 by $\Delta t = 0.02$, where it stays. This is the one piece of evidence in the section that bears on the coherence question of Sec. VI D in the *favourable* direction, and it should be weighed against the build-up bound above, which pulls the other way.

The honest summary is therefore: at the simulation's own parameters the kernel produces a linear band rather than a causal one, for the right reason and with the break in the right place, but it predicts $k^{0.7}$ where $k^{1.3}$ is measured, and the missing 0.6 is a peak-proximity effect in the spatial factor rather than anything about the temporal model.

[Task 2.5 — all three referees converged on this: the manuscript explained the simulations away rather than running its own kernel at their parameters]

C. Reproduction of the universal infrared theorem

The causal k^3 invoked repeatedly above is not a property of our kernels but a theorem. Cai, Pi and Sasaki [27] prove that $\Omega_{\rm GW} \propto k^3$ in the infrared for any source satisfying

lifetime model	$\hat{\tau} = \tau_c k_0$	$p_{\rm break}$	band slope	$p_{\rm peak}$
coherent (control)	∞	—	+2.709	2.010
global	8.0	0.393	+1.380	1.351
global	16.6	0.189	+0.570	1.405
global	23.4	0.134	+0.645	1.276
global	410.8	0.0076	+0.709	1.287
eddy	8.0	0.393	+1.591	1.691
eddy	16.6	0.189	+0.710	1.649
eddy	23.4	0.134	+0.696	1.651
eddy	410.8	0.0076	+0.771	1.581

TABLE V. Our kernel evaluated at the parameters of Ref. [28]'s run `ini2` ($k_0 = 673$, $k_{\rm box} = 129$, $R = 158$, $R_{\rm IR} = k_0/k_{\rm box} = 5.2$), with the triangular finite-lifetime correlator. The band slope is averaged over their own fit band $p = 0.193$–0.891, where the measurement is $+1.298$ (digitised) and $+1.295$ (raw shells); their GW peak is $1.836\,k_0$. "global" is a scale-independent lifetime, "eddy" the scale-dependent one. Results are stable to three decimals against R over 36–376 and $R_{\rm IR}$ over 5.2–100. Generated by **`Notebooks/kernel_at_rp20.py`**.

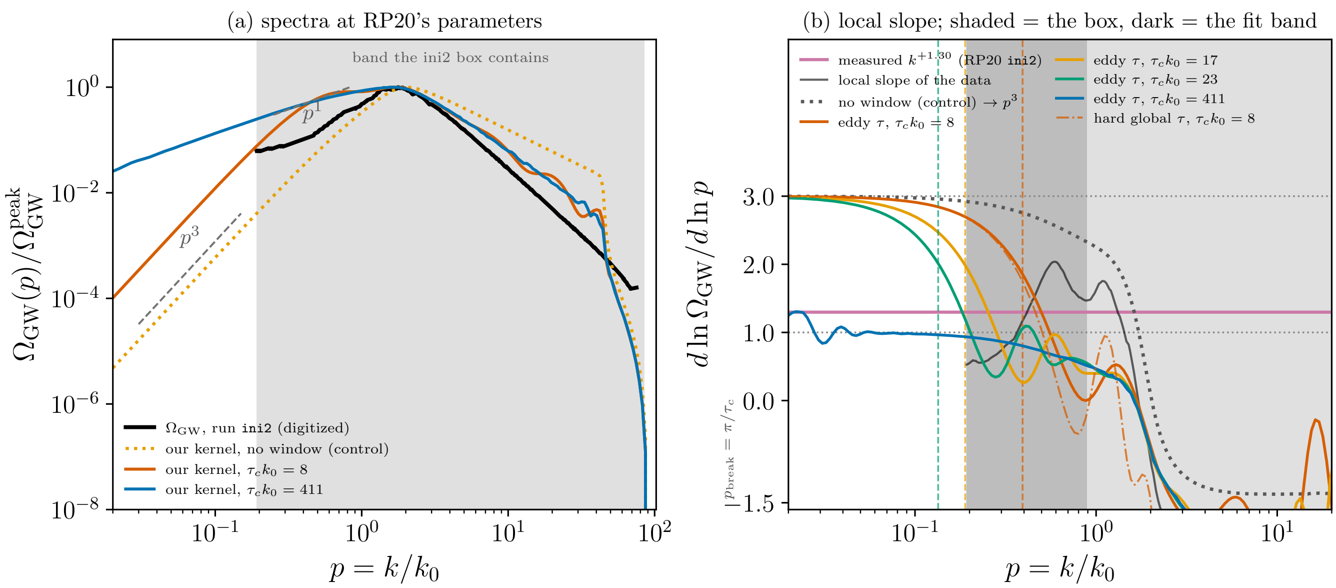


FIG. 14. The kernel of this paper at Ref. [28]'s own parameters. *(a)* The digitised **`ini2`** spectrum against our kernel at $\hat{\tau} = 8$ and 411 and against the no-window control, with the band the box contains shaded and p^1, p^3 guides. *(b)* Local logarithmic slope: the four lifetime closures, the control tending to $+3$, the smoothed local slope of the data, and the measured band average $+1.30$; vertical dashes mark $p_{\rm break} = \pi/\tau_c$ for each closure, and the dark shading is the fit band. The control already fails to reach $+3$ inside the box, which is the point of panel (b).

(I) $0 < \int d\ell\,[\text{source spectra}]^2 < \infty$ (finiteness);

(II) k lies below *every* scale associated with the source—its peak k_*, its spectral width Δk, *and* the inverse duration $\Delta t_s^{-1} = (a_* \Delta\eta_s)^{-1}$;

(III) the modes re-enter the horizon during radiation domination.

The exponent itself is pure counting: at horizon crossing of $1/k_*$ there are $N = (k_*/k)^3$ causally disconnected patches, so Poisson statistics give $\langle h_k h_k \rangle = N^{-1}|h_*|^2 = (k/k_*)^3 |h_*|^2$, and the k^3 follows from the dimensionality of space alone, independently of the later evolution.

This matters here because *both* of the exceptions they identify are realised by models in this paper, so the theorem is an external check rather than a restatement of our own kernels. We therefore reproduce it directly, from scratch and independently of Sec. C (Fig. 15).

a. Spectral width, and why a monochromatic source gives k^2. The anisotropic stress is the convolution of two source shells, which in one dimension carries an explicit pole,

$$\Pi(k) = \frac{2\pi}{k}\int dp\, p\, P(p)\int_{|k-p|}^{k+p} dq\, q\, P(q). \tag{46}$$

For a source of finite width the inner interval shrinks as $2k$ when $k \to 0$, cancelling the $1/k$ and leaving $\Pi(0)$ finite, so $\Omega_{\rm GW} \propto k^3$. For a δ-shell the inner integral does not shrink; the pole survives, and $\Omega_{\rm GW} \propto k^3 \cdot k^{-1} = k^2$. Numerically, a shell of log-width σ gives slope 3.000 for $k \ll \sigma$ and 2.000 in the unscreened band $\sigma \ll k \ll k_*$ (measured for $\sigma = 10^{-2}$ and 10^{-3}), while an exact δ-shell gives 2.000 over the whole infrared—a zero-width source has no lower scale, which is precisely the violation of condition (II). This is the same $1/p$ pole that appears in our monochromatic kernel as $\mathcal{K}_0(p)/p$, and it is why both panels of Fig. 23 show slope **2** rather than **3**: the monochromatic models are Cai–Pi–Sasaki's δ-function exception, not an anomaly of our temporal treatment.

b. Duration, and the intermediate k^1 band. Condition (II) also bounds k by the inverse duration. For a source coherent over a window $\Delta\eta$ the temporal factor is

$$\Big|\int_0^{\Delta\eta} d\eta\, e^{ik\eta}\Big|^2 = \frac{4\sin^2(k\Delta\eta/2)}{k^2} \;\longrightarrow\; \begin{cases} \Delta\eta^2, & k\Delta\eta \ll 1, \\ 2/k^2, & k\Delta\eta \gg 1 \text{ (oscillation-averaged)}, \end{cases} \tag{47}$$

so the spectrum measures slope $+2.999$ below $1/\Delta\eta$ and exactly $+1.000$ above it: the finite emission window removes two powers. The k^1 that appears at intermediate wavenumbers is therefore not a violation of causality but condition (II) being read outside its domain—k^3 is the asymptotic law for $k \ll 1/\Delta\eta$, and k^1 is the finite-window band above it. (Cai–Pi–Sasaki quote k^9 above $1/\Delta\eta$ for acoustic sources, where a coherent *oscillating* source resonates with the Green function; for an incoherent window such as ours the loss is two powers.)

c. Where the break sits, and why the simulation box cannot contain it. This locates the $k^1 \to k^3$ break precisely,

$$k_{\rm break} \;\simeq\; 1/\Delta t_s, \tag{48}$$

which is the order-of-magnitude form. It is refined by a factor π in Sec. V A, where the break is read off the *two-leg* temporal factor—the transform of the squared window rather than of the window—and located at $k_{\rm break} = \pi/\tau_c$, Eq. (45). Eq. (48) is used below only where a factor of a few does not matter, namely in comparing the break with the box fundamental; the placement against the observing bands in Sec. V D uses the refined form. [the manuscript quoted two different coefficients for the same break in two sections without ever relating them] And the same white-noise stress $\Omega_{\rm T} \propto k^3$ feeds *both* sides of it: what changes across $k_{\rm break}$ is only whether a mode completes oscillations while the source lives, i.e. which branch of Eq. (47) applies. It is worth inserting the numbers for the published runs. In units where the Hubble rate at generation is $\mathcal{H}_* = 1$, a source acting for a fraction of a Hubble time gives $k_{\rm break} \sim 1$; a source decaying instead on the outer-scale eddy-turnover time $1/(k_0 u_0)$, with $k_0 \sim 10^2$ and $u_0 \sim 10^{-1}$, gives $k_{\rm break} \sim 10$. Both lie one to two decades *below* the box fundamental $k_{\rm box} = 2\pi/L \sim 10^2$ that the simulations must adopt in order to place the source scale well inside the horizon. Every mode in such a box therefore sits in the $k\Delta\eta \gg 1$ branch, which is exactly why the measured spectra show a clean k^1 with no turnover at their largest scale.

The correct reading of that is not that the k^3 regime is under-resolved but that it is *absent from the box altogether*. Recovering it would require $k_{\rm box} \lesssim k_{\rm break}$ plus a decade below to fit a slope—a box 10^2–10^3 times larger at fixed small-scale physics, i.e. simulating a comoving Hubble volume *while* resolving the inertial range, a dynamic range of 10^4–10^6 per dimension. The universal k^3 of [27] must accordingly be imported analytically and never expected from a measurement, and a simulated infrared slope should never be quoted as contradicting it: the two statements are made in disjoint bands of k, separated by Eq. (48). (The break itself is not ours: Ref. [9] derives $k_{\rm br} = 1/\delta t_{\rm fin}$ and confirms it in simulations, and Ref. [26] obtains the same causal-to-linear transition in the sound-shell context. What we add is its numerical value *relative to the box fundamental* for the runs of Ref. [28], computed from the published run parameters, and the consequent observation that the causal band is absent from the box rather than merely under-resolved.)

D. The break in observed frequency

The break has so far been located relative to a simulation box. It is more useful to locate it relative to a detector, and the dimensional dictionary of Sec. IV A does that in one step. With $\hat\tau_c = \tau_c \mathcal{H}_*$ the source lifetime in Hubble units, Eq. (45) gives $\hat k_{\rm break} = \pi/\hat\tau_c$, and $f = (\hat k/2\pi) f_H$ then collapses the π against the 2π:

$$f_{\rm break} \;=\; \frac{f_H}{2\,\hat\tau_c}, \qquad f_H \simeq 1.6 \times 10^{-5}\,{\rm Hz}\left(\frac{g_*}{100}\right)^{1/6} \frac{T_*}{100\,{\rm GeV}}. \tag{49}$$

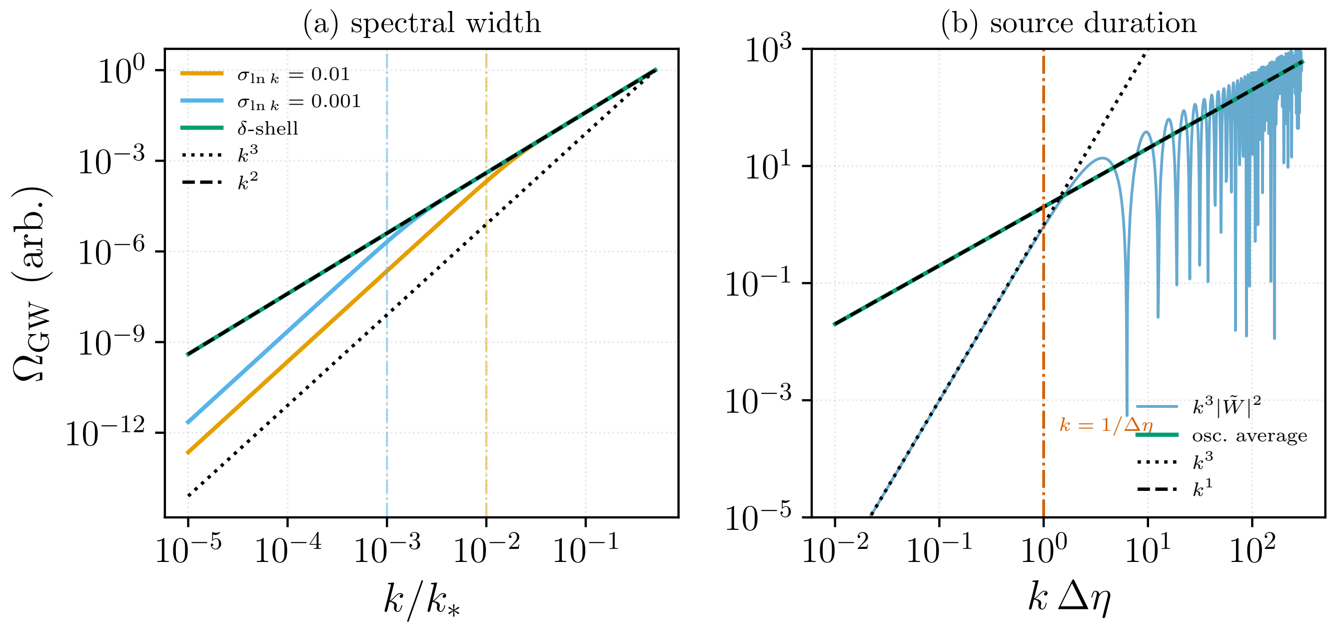


FIG. 15. Independent reproduction of the two exceptions to the universal k^3 infrared theorem of [27], computed from Eqs. (46) and (47) with no input from the kernels of this paper. *(a)* Shells of log-width $\sigma_{\ln k} = 10^{-2}$ and 10^{-3} (dash-dotted verticals mark $k = \sigma$) against an exact δ-shell: the finite-width curves follow k^3 for $k \ll \sigma$ and k^2 in the band $\sigma \ll k \ll k_*$, while the δ-shell follows k^2 throughout, having no lower scale. *(b)* Finite emission window $\Delta\eta$: the spectrum follows k^3 below $k = 1/\Delta\eta$ and, once the $\sin^2$ oscillations are averaged, k^1 above. Measured slopes are 3.000/2.000 and 2.999/1.000 respectively. Generated by `Notebooks/cai_pi_sasaki_reproduction.py`.

The eddy size $\gamma = l_0\mathcal{H}_*$ has cancelled: the break is set by how long the source lives, not by how large its eddies are, and it enters only when $\hat{\tau}_c$ is itself tied to k_0. Two closures bracket the possibilities. A lifetime that is a fraction ϵ of a Hubble time gives $\hat{\tau}_c = \epsilon$ and $f_{\rm break} = f_H/2\epsilon$, independent of γ outright; an outer-scale eddy turnover $\tau_c = 1/(k_0 u_0)$ gives $\hat{\tau}_c = \gamma/2\pi u_0$ and

$$f_{\rm break} = \frac{\pi u_0}{\gamma}\, f_H = \pi u_0 f_0, \qquad p_{\rm break} = \pi u_0, \qquad \hat{\tau} = \tau_c k_0 = 1/u_0. \tag{50}$$

Table VI evaluates both at the electroweak and QCD epochs ($f_H = 1.60 \times 10^{-5}$ and 1.75×10^{-8} Hz).

Three things follow, and the first is a caution about the second. Across the admitted parameter ranges $f_{\rm break}$ moves by two to three and a half decades, so no single value should be quoted; what is robust is the *ordering* of the break and the observing bands. Second, for an electroweak source the break lies below LISA's window in most of the parameter space, but not all of it. Recomputing the LISA energy-density sensitivity from the SciRD analytic noise model [45] gives a minimum of $\Omega_{\rm sens}$ at 2.52 mHz and a factor-of-two window 1.04–4.60 mHz (the strain minimum, and the origin of the often-quoted 3–10 mHz, is separately at 7.26 mHz; the f^3 weighting moves the Ω minimum down). Against that, the global-lifetime break runs 8 μHz–0.8 mHz and the eddy-lifetime break 5 μHz–15 mHz. Of the nine eddy-closure corners of Table VI, six put the break below the window—LISA then observes on the k^1 side, and the causal k^3 asymptote lies below the instrument, the same conclusion Sec. V C reaches for the simulation box. One corner, $\gamma \sim 10^{-2}$ with $u_0 \sim 0.3$, puts the break at 1.5 mHz, inside the window and at $p_{\rm break} = 0.94$, i.e. at the spectral knee where it is not separable from the peak. The remaining two, $\gamma \sim 10^{-3}$ with $u_0 \sim 0.1$ and 0.3, push it to 5.0 and 15.1 mHz—*above* the window, so that LISA would observe the causal side instead. Short-lived, small-scale sources are therefore the one case in which the k^3 asymptote is the observable rather than an inaccessible limit, and the ordering we emphasise elsewhere is reversed. ~~LISA therefore observes on the k^1 side of the break, and the causal k^3 asymptote lies below the instrument. The single corner that pushes the break into the LISA window, $\gamma \sim 10^{-3}$ with $u_0 \sim 0.3$, puts $p_{\rm break} = 0.94$ at the spectral knee itself.~~ [re-run of `k_break_frequency.py`: the grid gives 6 below / 1 inside / 2 above, and the corner named as "inside" is in fact above] Third, for a QCD source the break spans 5.5 nHz–17 μHz and so overlaps the pulsar-timing band, 1.6 nHz to a few $\times 10^{-7}$ Hz, but only in the long-lived corner; at mid-range parameters it sits above the band, so pulsar timing reads the causal tail rather than the break.

A resolution caveat carries over from Table IV, which resolves a clean linear band only for $\hat{\tau} \gtrsim 10^3$. Under the eddy closure $\hat{\tau} = 1/u_0 \le 10^2$ for any $u_0 \ge 0.01$, and $p_{\rm break} = \pi u_0$ is never more than about 1.5 decades below the peak, so the linear band is narrow and not cleanly resolved; a wide one needs the global closure with $\epsilon/\gamma \gtrsim 160$. None of this is a claim on the break itself, which is Ref. [9]'s $k_{\rm br} = 1/\delta t_{\rm fin}$; what is added here is the factor π of Eq. (45), which the quadratic source supplies, and the placement of the result against the two instruments. [Task 2.3. The LISA numbers use the standard SciRD analytic fit ($L = 2.5 \times 10^9$ m, optical-metrology and acceleration noise terms, written out in `Notebooks/k_break_frequency.py`), following Ref. [45]]

closure	epoch	parameter	$f_{\rm break}$/Hz	$p_{\rm break}$	$\hat{\tau}$
global	EW	$\epsilon = 0.01$	8.0×10^{-4}	0.5	6.3
	EW	$\epsilon = 1$	8.0×10^{-6}	5×10^{-3}	6.3×10^2
	QCD	$\epsilon = 0.01$	8.7×10^{-7}	0.5	6.3
	QCD	$\epsilon = 1$	8.7×10^{-9}	5×10^{-3}	6.3×10^2
eddy	EW	$u_0 = 0.01$	5.0×10^{-5}	0.031	10^2
	EW	$u_0 = 0.3$	1.5×10^{-3}	0.94	3.3
	QCD	$u_0 = 0.01$	5.5×10^{-8}	0.031	10^2
	QCD	$u_0 = 0.3$	1.6×10^{-6}	0.94	3.3

TABLE VI. The $k^3 \to k^1$ break in observed frequency, from Eqs. (49)–(50), at $\gamma = 10^{-2}$ for the rows where γ enters. Under the global closure $f_{\rm break}$ does not depend on γ at all and $p_{\rm break} = \gamma/2\epsilon$ does; under the eddy closure $p_{\rm break} = \pi u_0$ and $\hat{\tau} = 1/u_0$ depend on neither γ nor the epoch. Compare LISA's Ω-sensitivity window 1.04–4.60 mHz (minimum 2.52 mHz) and the pulsar-timing band 1.6 nHz–4×10^{-7} Hz. Computed by `Notebooks/k_break_frequency.py`, which also plots Fig. 16.

[Task 2.3 — Referees A and C both asked for the break in Hz against the sensitivity curves]

E. Which band sources the gravitational-wave infrared? A band-split experiment

The full fluid input of Eq. (25) carries two distinct bands: a causal Batchelor band $E \propto k^4$ below the stirring scale k_0 and the Kolmogorov range $E \propto k^{-5/3}$ above it. It is natural to ask which of them the gravitational-wave infrared inherits, and the answer is not obvious from Eq. (20), because the source is the *quadratic* stress $T_{ij} \sim v_i v_j$: the stress spectrum is the fluid spectrum convolved with itself, and a convolution has no reason to remember the slope of either factor. The hypothesis to be tested is the one already used qualitatively in Sec. C 2, that for $k \ll k_0$ the convolution is saturated by the antiparallel pairs $\boldsymbol{k}_1 \simeq -\boldsymbol{k}_2$ with $|\boldsymbol{k}_1| \simeq |\boldsymbol{k}_2|$ sitting at the top of whichever band is populated, so that the stress *floors at white noise* irrespective of how steep—or how absent—the fluid infrared is. If that is right, $\Omega_{\rm GW} \propto k^3$ is a statement about the analytic structure of a self-convolution and not about the input spectrum at all.

We test it by running the same stationary kernel three times, changing nothing kinematic or temporal and only masking the shape factor $S(k) = A(k)/A_{\rm Kol}(k)$ of Sec. III D band by band: (i) the Batchelor band alone, $S = 0$ for $k \geq k_0$; (ii) the inertial range alone, $S = 0$ for $k < k_0$, which is identically the original calculation of Sec. V C; and (iii) both, the full spectrum. The mask multiplies *both* convolution legs, $S(k_1)S(u)$, and is imposed by zeroing the integrand rather than by narrowing the integration domain, so that the triangle inequality $|k_1 - k| \leq u \leq k_1 + k$ and the spectral support $[k_{\rm IR}, k_d]$ remain exactly those of the unmasked kernel. Figure 17 shows the three inputs and the three resulting spectra at $M = 0.5$, $R = 10^4$, $R_{\rm IR} = 10^2$.

All three gravitational-wave spectra rise with the causal slope. Fitted over the decade $2 \times 10^{-3} \leq p \leq 2 \times 10^{-2}$, more than a decade and a half below every peak, the slopes are $p^{3.00}$ (Batchelor band alone), $p^{3.00}$ (inertial range alone) and $p^{3.02}$ (full spectrum); over the lower decade $10^{-3} \leq p \leq 10^{-2}$ they tighten to 3.001, 3.005 and 3.008. The inertial-range-only run is the sharpest test, and it is decisive: that input has *no infrared power whatsoever*—$E(k)$ is identically zero for $k < k_0$—yet its gravitational-wave spectrum is a clean p^3 across every decade below the peak. The white-noise floor is therefore generated internally, by the self-convolution of modes near k_0, and cannot be removed by removing the source's infrared. Two further checks confirm that it is the *upper* edge of each band that supplies the floor. First, the infrared amplitude of the Batchelor-only run is independent of the bandwidth: increasing $R_{\rm IR}$ from 10 to 300 changes $\Omega_{\rm GW}(p = 5 \times 10^{-3})$ by 0.4%, so the k^4 band's own long-wavelength modes contribute essentially nothing. Second, replacing Batchelor k^4 by Saffman k^2 leaves the slope at $p^{3.01}$ and moves the amplitude by a factor of only 2.7, although the input spectra differ by four decades at the bottom of the band. As a kinematic consistency check, the Batchelor-only spectrum terminates exactly at $p = 2$, the largest wavenumber two modes with $k < k_0$ can add up to.

The hypothesis therefore holds in its essential claim, with one quantitative correction to its sharper form. It is *not* true that the inertial band's lower edge dominates the full-spectrum infrared to the exclusion of the k^4 band: at $p = 3 \times 10^{-3}$ the two bands contribute 8.2×10^{-11} and 6.5×10^{-11} in the same units, a ratio of only 1.26, and the

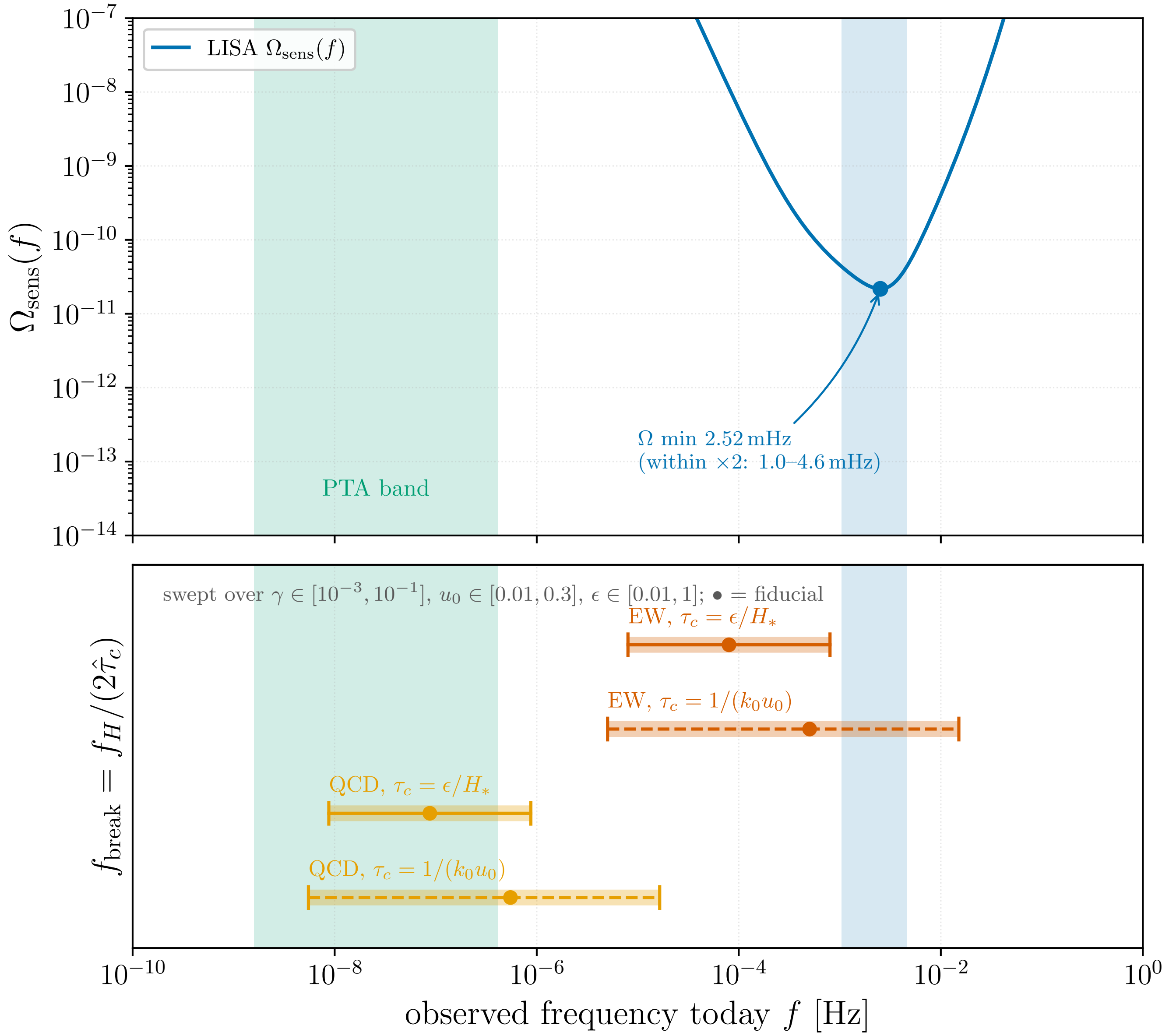


FIG. 16. Where the break falls relative to the observing bands. *Top:* the LISA energy-density sensitivity $\Omega_{\rm sens}(f)$ from the SciRD analytic noise model, with its minimum at 2.52 mHz and the factor-of-two window shaded, and the pulsar-timing band. *Bottom:* $f_{\rm break}$ swept over the parameter ranges of Table VI for the two epochs and the two lifetime closures. The bands are two to three decades wide, which is the point: the break's position relative to each instrument is robust where its value is not. Generated by `Notebooks/k_break_frequency.py`.

two bands are very nearly additive there (the full run exceeds the sum of the two single-band runs by 0.4%, i.e. the k_1-in-one-band, u-in-the-other cross term is negligible in the deep infrared). Both bands floor at white noise because both are bounded by k_0; they contribute comparably because both are dominated by the energy-containing modes on their respective sides of k_0. The cross term is not negligible near the peak, where at $p \simeq 0.5$ the full run exceeds the sum of the two single-band runs by 46%, which is why the full-spectrum peak ($p = 0.66$, amplitude 3.96×10^{-4}) is not the sum of the inertial-only peak ($p = 0.75$, 2.43×10^{-4}, i.e. the $1.49M$ sweeping scaling) and the Batchelor-only peak ($p = 0.47$, 5.48×10^{-5}). The practical consequence for Sec. V is a negative one and worth stating plainly: no choice of fluid infrared slope, and no widening of the infrared band, can bend the analytic k^3 towards the simulated k^1. The causal slope is a property of the quadratic source, and the discrepancy must be resolved through the source's *temporal history*—specifically its finite duration, Eq. (47) [9, 26]—and not on the spatial axis. Note that duration is not a choice of stationary UETC; on that point see the corollary of Appendix G.

a. The same two bands with the source fired at a single epoch. Replace the sweeping correlation by a source that acts at one instant, $\Pi(k;\tau_1,\tau_2) = P(k)\,\delta(\tau_1 - t_0)\,\delta(\tau_2 - t_0)$. Both δ's collapse in Eq. (7) and the cosine is evaluated at zero lag, so the temporal factor is identically unity *for any* t_0: the epoch at which the source fires cancels exactly and cannot appear in $\Omega_{\rm GW}$. Setting $t_0 = 1$, or any other value, changes nothing. What is left is the bare spatial convolution, and it is worth being explicit that this is *not* the $q \to 0$ limit of the sweeping kernel. The weight $(x+y)^{-1/2}$ in Eq. (23) is the $\sqrt{\pi/(A+B)}$ of the Gaussian ω_1 convolution, Eq. (B59)—it belongs to the temporal model, not the spatial one, and a δ-in-time source drops it too. The two coincide only where k_1 and u are pinned to a single shell, which is why Eq. (B105) may be read as a $q \to 0$ limit for the monochromatic source but not here.

The four curves separate the two axes cleanly. The infrared exponent is 3.00 (k^4 band, sweeping), 3.01 ($k^{-5/3}$ band, sweeping), 2.99 and 2.99 for the same two bands fired impulsively: neither the band nor the presence of a temporal model at all moves it, which is Eq. (B121) in its strongest form—the causal slope survives the deletion of the entire temporal sector. Everything else moves. The peak slides outward from $p = 0.47$ to 1.37 for the k^4 band and from 0.75 to 2.50 for the inertial band, because the sweeping factor was the only thing cutting the source off at $p \gtrsim M$; and the inertial band's ultraviolet tail flattens from $p^{-4.5}$ to $p^{-1.4}$, leaving only the fall-off of the spatial convolution itself. The k^4 band terminates at $p = 2$ in both cases: that edge is kinematic, Eq. (B123), and is the one feature no temporal model can move.

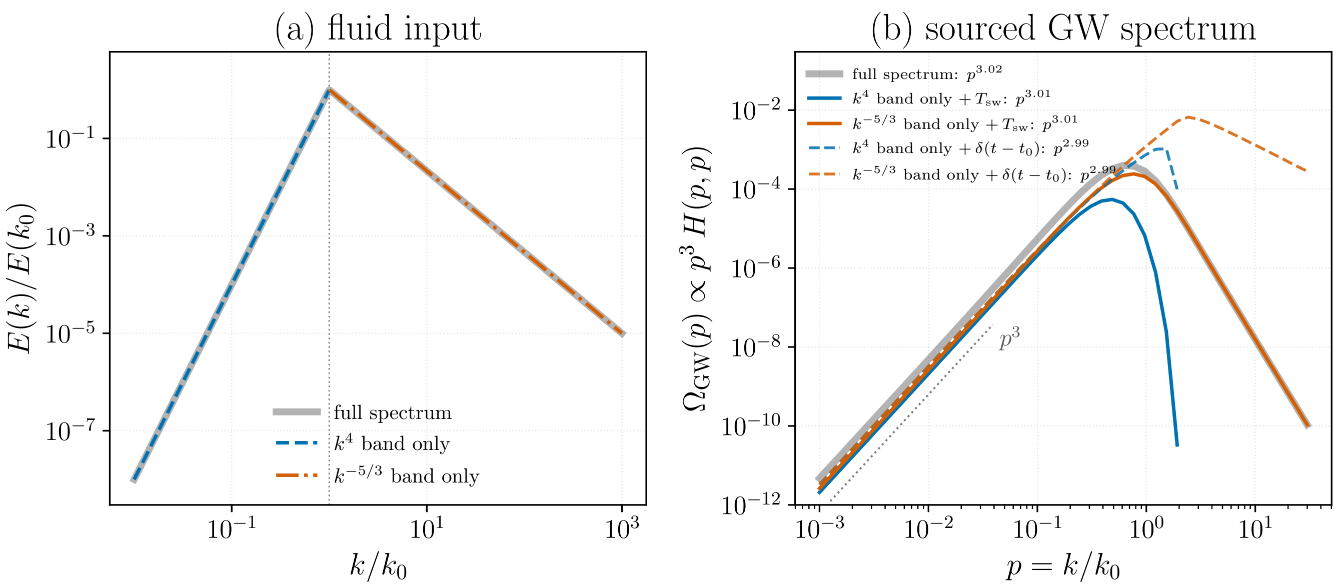


FIG. 17. Band-split test of the white-noise-floor hypothesis. *(a)* The three fluid inputs fed to the kernel: the full spectrum of Eq. (25) (grey), its Batchelor band $E \propto k^4$ alone (blue, dashed) and its Kolmogorov range $E \propto k^{-5/3}$ alone (orange, dash-dotted), obtained by masking the shape factor $S(k) = A(k)/A_{\rm Kol}(k)$ on one side of k_0 (dotted vertical line). *(b)* The gravitational-wave spectra $\Omega_{\rm GW}(p) \propto p^3 H_{ijij}(p,p;M,R)$ they source on the sound-cone diagonal $q = p$, with the infrared log–log slopes fitted over $2\times10^{-3} \le p \le 2\times10^{-2}$ quoted in the legend; the dotted guide is p^3. All are causal to better than 2% in the exponent, including the inertial-range-only run, whose input has no infrared power at all. The Batchelor-only spectrum ends at $p = 2$, the kinematic limit of a convolution of two modes with $k < k_0$. The two dashed curves repeat the single-band runs with the temporal factor removed entirely—evaluated at $q = 0$ rather than on the diagonal $q = p$, so that the Gaussian sweeping factor and its erfc in Eq. (23) are both unity and the source carries no decorrelation and no time dependence at all [the impulsive limit, Eq. (B105)]. The spatial integral is otherwise identical, so each solid/dashed pair isolates what the temporal model does and does not control. Parameters: $M = 0.5$, $R = (k_d/k_0)^{4/3} = 10^4$, $R_{\rm IR} = k_0/k_{\rm IR} = 10^2$, Batchelor infrared slope $s = 4$. Both panels are dimensionless and the vertical offset between them carries no physical meaning. Produced by `Notebooks/band_split_gw.py`.

b. Mach dependence of each band separately. Repeating both single-band runs over $M = 0.03$–3 separates the Mach dependence from the band decomposition, and the two turn out to be independent (Fig. 18). The infrared exponent is untouched: it stays within 2.99–3.06 of the causal value for every M and for both bands, so M joins the infrared slope s, the bandwidth $R_{\rm IR}$ and the temporal model on the list of things that cannot bend it. What M does control is what it controls for the full spectrum. The infrared amplitude follows the bare M^3 prefactor of Eq. (23)—compensating by M^3 flattens it to 16.6% (k^4 band) and 14.0% ($k^{-5/3}$ band) across two decades of M—and the peak follows the sweeping law, at $p_{\rm peak} \simeq 1.5\,M$ for the inertial band, which is the $1.49\,M$ of Eq. (31) recovered from that band alone, and at a slightly lower $p_{\rm peak} \simeq 1.1\,M$ for the Batchelor band.

The two bands part company only where kinematics intervenes. Above $M \simeq 1$ both peaks fall below the sweeping law, and the k^4 band saturates hardest, reaching only $p_{\rm peak} = 1.33$ at $M = 3$ where $1.1\,M$ would predict 3.3: its support is bounded above by k_0, so its convolution cannot place power beyond $p = 2$ at all [Eq. (B123)], and the sweeping scale runs into that ceiling. The inertial band, whose support extends to k_d, saturates later and less. Note finally that with the temporal factor removed the Mach dependence disappears entirely except for the prefactor: at $q = 0$ the exponential and the erfc of Eq. (23) are both unity and no M survives in the integrand, so the no-correlation spectra are M^3 times a single universal shape. Every M-dependent feature in Fig. 18 is therefore a property of the sweeping model, not of the source spectrum.

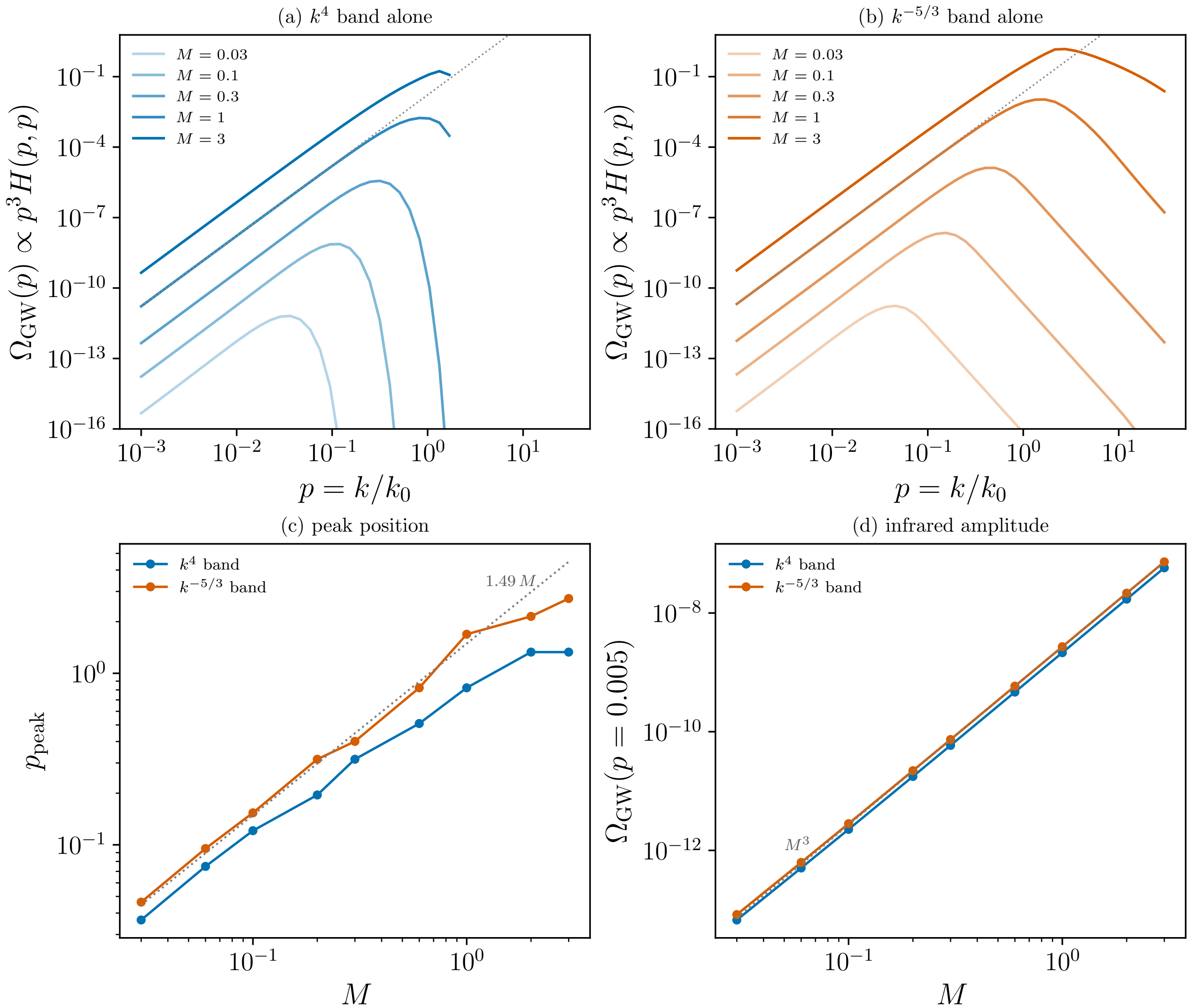


FIG. 18. Mach dependence of the band-split spectra, at $R = 10^4$, $R_{\rm IR} = 10^2$, Kraichnan sweeping on the diagonal $q = p$. *(a),(b)* $\Omega_{\rm GW}(p)$ from the Batchelor k^4 band alone and from the Kolmogorov $k^{-5/3}$ band alone, at $M = 0.03, 0.1, 0.3, 1, 3$ (increasing shade); dotted line p^3. The infrared exponent is M-independent for both bands; the amplitude and the peak are not. *(c)* Peak position against M, with the $1.49\,M$ sweeping law of Eq. (31) dotted. The inertial band reproduces it; the Batchelor band tracks a slightly lower $1.1\,M$ and saturates above $M \simeq 1$, where the sweeping scale meets the kinematic ceiling $p \leq 2$ that bounds a convolution of two modes with $k < k_0$. *(d)* Infrared amplitude at $p = 5 \times 10^{-3}$ against M, with the M^3 prefactor dotted; both bands follow it to 16.6% and 14.0% over two decades of M. Produced by `Notebooks/band_split_mach.py`.

VI. TEMPORAL MODELS FOR DECAYING TURBULENCE: TIMESCALE REGIMES AND COSMOLOGICAL RELEVANCE

The spatial axis is closed: no choice of S moves the asymptotic infrared (Sec. V E). Everything that distinguishes one predicted spectrum from another above the infrared therefore lives on the temporal axis. This section sets out, as a method rather than as a single model, how the temporal factor of a freely decaying, statistically homogeneous and isotropic source is constructed; classifies the possibilities by *which physical timescale controls the decorrelation*; and asks which regime each early-Universe process actually realises. The four models $T_{\rm sw}, T_{\rm dec}, T_{\rm imp}, T_{\rm burst}$ of Sec. II D are four entries in the resulting catalogue, not an exhaustive list.

A. Two-time structure of a decaying source

A decaying source is not stationary, so its UETC is not a function of the lag alone. Write the two times as a mean and a lag, $\bar{t} = \frac{1}{2}(t_1 + t_2)$, $\tau = t_1 - t_2$, and split the correlator into an *amplitude* and a *decorrelation*,

$$\Pi(k; t_1, t_2) = \sqrt{P(k, t_1)\, P(k, t_2)}\; f(k; \bar{t}, \tau), \qquad f(k; \bar{t}, 0) = 1. \tag{51}$$

The two factors answer different questions: P says how much stress there is at each instant, f says how long it remembers itself. Only f enters the temporal factor; P sets the amplitude and the emission window.

For self-similar decay the amplitude is carried by a single length and a single velocity,

$$P(k, t) = u^4(t) L^3(t)\, \mathfrak{P}(kL(t))\,, \quad L \propto (1 + t/\tau_{\rm st})^{n_L}, \quad u \propto (1 + t/\tau_{\rm st})^{-n_u}, \tag{52}$$

with $\tau_{\rm st}$ the decay time of the envelope. The exponents n_L, n_u are not free: they are fixed by whichever integral invariant the decay conserves — Loitsyansky [36], Saffman [46], or, for non-helical MHD, the Hosking integral [47, 48], which supersedes the earlier $q = 1/2$ magnetic class and is the invariant behind the resistively controlled decay of Ref. [49]. The split Eq. (51) and the locally stationary approximation that follows are standard in this literature [19, 23, 40]; we restate them only to fix notation. The kernels of Table I all assume the fast and slow times separate,

$$\varepsilon_0(k) \equiv \frac{\tau_c(k)}{\tau_{\rm st}} \ll 1 \implies f(k; \bar{t}, \tau) \simeq f(k; \tau), \tag{53}$$

so that the source is *locally* stationary and the whole $\bar{t}$ dependence sits in the amplitude. Equation (53) is comfortably satisfied deep in the inertial range, where $\tau_c \propto k^{-2/3}$ falls while $\tau_{\rm st}$ is fixed, and is *marginal at the outer scale*, where $\varepsilon_0 = O(1)$: the energy-containing eddies decorrelate on the same timescale as the decay itself. This is the one place where the quasi-stationary construction is not controlled, and it is also the scale that sets the peak.

Given f, the temporal factor is

$$T(k, \omega) = 2 \int_0^\infty d\tau\, \cos(\omega\tau)\, f(k; \tau), \tag{54}$$

evaluated on the light cone $\omega = k$, and combined across the two legs of the convolution by the product rule of Eq. (35). The single structural fact that organises everything below is the cusp theorem Eq. (37): the large-ω behaviour of Eq. (54) is fixed by $f'(0^+)$ alone, so a correlator that is smooth at zero lag is suppressed faster than any power, while any correlator with a kink there gives $T \to -2f'(0^+)/\omega^2$.

B. The competing timescales

Which τ_c appears in Eq. (54) is a physical question, not a modelling convention. Table VII collects the candidates.

C. Catalogue of temporal models

Table VIII lists each decorrelation model together with the one number that fixes its ultraviolet behaviour. The pattern is uniform: the models divide into those with $f'(0^+) = 0$, which cut off super-polynomially, and those with a cusp, which all give the *same* ω^{-2} tail and differ only in its coefficient.

Timescale	Definition	k-scaling	Physical origin
$\tau_{\rm sw}(k)$	$1/(u_{\rm rms}k)$	k^{-1}	Advection of a k-eddy past the observer by the energy-containing flow [20–22]
$\tau_{\rm eddy}(k)$	$1/(k\,u_k) = \varepsilon^{-1/3}k^{-2/3}$	$k^{-2/3}$	Local nonlinear turnover (Kolmogorov) [37, 38]
$\tau_A(k)$	$1/(k\,v_A)$	k^{-1}	Alfvén crossing; MHD counterpart of sweeping [33, 50, 51]
$\tau_\nu(k)$	$1/(\nu k^2)$	k^{-2}	Diffusive momentum transport (short mean free path)
$\tau_\eta(k)$	$1/(\eta k^2)$	k^{-2}	Resistive diffusion
$\tau_{\rm drag}$	$1/\alpha_{\rm fs}$	k^0	Free-streaming drag (long mean free path); *not* diffusive [31, 32, 52]
$\tau_{\rm st}$	L/u	k^0	Self-similar decay of the envelope, Eq. (52)
$T_{\rm em}$	Duration of active emission	k^0	Driving lifetime, or a Hubble time, whichever is shorter
$1/\omega$	$1/k$ on the light cone	k^{-1}	The period the gravitational wave is sensitive to

TABLE VII. Candidate decorrelation timescales for a homogeneous isotropic source. The k-scaling column is the discriminator: two mechanisms with the same value at the outer scale but different exponents predict different ultraviolet spectra. Note in particular that sweeping (k^{-1}) and turnover ($k^{-2/3}$) are *not* the same rate. The kernel of Sec. III follows Ref. [2] in using $\eta_k = \varepsilon^{1/3}k^{2/3}/\sqrt{2\pi}$, which is the turnover rate, under the name "sweeping"; the distinction is nominal only if one never compares ultraviolet slopes. Ref. [24] makes the same objection and adopts the Eulerian k^{-1} time for this reason.

Model	$f(\tau)$	τ_c	$f'(0^+)$	$T(\omega)$	Status / where
Impulsive $T_{\rm imp}$	$\tau_0\,\delta(\tau)$	0	—	τ_0 (const)	Sec. D
Short-correlated (white)	any, with $\omega\tau_c \ll 1$	$\tau_c(k)$	—	$2\tau_c(k)$	Limit of every row below
Sweeping $T_{\rm sw}$	$\exp(-\frac{\pi}{4}\eta_k^2\tau^2)$	$1/\eta_k$	0	$\frac{2}{\eta_k}e^{-\omega^2/\pi\eta_k^2}$	Gaussian cutoff; Sec. III
Exponential (Lagrangian)	$e^{-\lvert\tau\rvert/\tau_e}$	τ_e	$-1/\tau_e$	$\frac{2\tau_e}{1+\omega^2\tau_e^2} \to \frac{2}{\tau_e\omega^2}$	Cusped; not used above
Power law $T_{\rm dec}$	$(1+\lvert\tau\rvert/\tau_k)^{-2/3}$	τ_k	$-\frac{2}{3\tau_k}$	$\to \frac{4}{3\tau_k\omega^2}$	BK2016; Sec. IV B
Finite lifetime $T_{\rm burst}$	$(1-\lvert\tau\rvert/\tau_c)\Theta(\tau_c-\lvert\tau\rvert)$	τ_c	$-1/\tau_c$	$\to \frac{2}{\tau_c\omega^2}$	Autocorrelation of a box; Sec. E 2
Alfvénic	$e^{-\lvert\tau\rvert/\tau_A}$, $\tau_A = 1/(kv_A)$	$1/(kv_A)$	$-kv_A$	$\to \frac{2kv_A}{\omega^2}$	Cusped, but $\tau_c \propto k^{-1}$
Hard cutoff (top hat)	$\Theta(\tau_c-\lvert\tau\rvert)$	τ_c	0	$2\sin(\omega\tau_c)/\omega$	**Inadmissible**: changes sign
Resistive envelope	unchanged; acts on P, not f	—	—	—	$\tau_{\rm st} \simeq {\rm Lu}^{1/4}\tau_A$ [49]

TABLE VIII. Temporal models for a decaying homogeneous isotropic source. The $f'(0^+)$ column determines the ultraviolet by Eq. (37) and nothing else does: every cusped model gives ω^{-2}, and the three distinct-looking decaying models differ only in the coefficient. The top-hat row is included because it is the intuitive way to say "the source switches off", and it is not a correlation function: its transform changes sign, in violation of Bochner's theorem [37], giving negative GW energy over roughly half the spectrum; positive-definiteness of the UETC was imposed as a modelling constraint by Ref. [23]. Its admissible replacement is the triangle in the row above, which is the autocorrelation of the switched-on box. The last row is not a decorrelation model at all — resistivity controls the decay of the *amplitude* P, so it changes the emission window and the normalisation while leaving $T(\omega)$ untouched.

D. Regime classification

Two dimensionless ratios sort the catalogue. The first is the number of source coherence times per gravitational-wave period, evaluated on the light cone,

$$\mathcal{N}_\omega(k) \equiv \omega\tau_c(k)\big|_{\omega=k} = k\,\tau_c(k), \tag{55}$$

and the second is the number of independent samples the emission window contains, $\mathcal{N}_{\rm em}(k) \equiv T_{\rm em}/\tau_c(k)$. Then:

- $\mathcal{N}_\omega \ll 1$ (*incoherent*). The source forgets itself many times per wave period; $T \to 2\tau_c(k)$ and all memory of the shape of f is lost. Only the k-dependence of τ_c survives, so Table VII — not Table VIII — is what matters in this regime.

- $\mathcal{N}_\omega \sim 1$ (*transition*). The peak of $\Omega_{\rm GW}$ sits here. Both axes contribute, which is why the peak is the one feature that is not cleanly attributable to either.
- $\mathcal{N}_\omega \gg 1$ (*coherent*). The wave resolves the internal dynamics of the source and the cusp theorem takes over: smooth f is exponentially suppressed, cusped f leaves the universal ω^{-2}.

For $k \lesssim 1/T_{\rm em}$ the quasi-stationary construction fails outright: the source cannot be resolved as stationary over a window shorter than the wave period, and the window itself, not f, shapes the response. Two cautions belong here. First, this is a *band* effect confined to $k \lesssim 1/T_{\rm em}$; the strict $k \to 0$ limit remains the causal k^3 of Sec. V C for any admissible source, as it must [27]. Second, whether the energy-containing modes actually stay coherent over $T_{\rm em}$ — which is what would populate such an intermediate band — is a dynamical question we have *not* answered; imposing it as a free parameter is circular, and no calculation in this paper establishes it.

E. Relevance to early-Universe processes

Table IX maps the catalogue onto the processes that are actually invoked as GW sources. The column that does the work is the third: once the controlling timescale is identified, the row of Table VIII and hence the ultraviolet behaviour follow.

Process	Epoch	Controlling timescale	Temporal model	GW consequence and caveat
Preheating / parametric resonance [53]	end of inflation	$T_{\rm em} \ll \tau_{\rm st}$; explosive, broadband	$T_{\rm imp}$, $T_{\rm burst}$	Source-scale peak; ultraviolet inherited from the spatial spectrum
Bubble collisions, first-order PT [4, 54, 55]	EW, QCD	single crossing time R_*/v_w	$T_{\rm burst}$	Cusp supplied by the finite window; ω^{-2}. *In tension with* the envelope-approximation f^{-1} of Ref. [54]
Sound waves [17, 26, 56]	EW	long-lived, but only where shocks form late [57]	coherent, $\mathcal{N}_\omega \gg 1$	Longest-lived source; coherence is the whole amplitude story
Freely decaying HD turbulence [23, 39, 58]	post-PT	$\tau_{\rm eddy}$, with $\varepsilon_0 = O(1)$	$T_{\rm dec}$	Peak pinned at $p \simeq 2.4$; Eq. (53) marginal. Ref. [23] instead measures Gaussian sweeping
Decaying non-helical MHD [47–49]	radiation era	τ_A for f; ${\rm Lu}^{1/4}\tau_A$ for P, ${\rm Lu} = v_A L/\eta$	Alfvénic + resistive envelope	$\tau_c \propto k^{-1}$, not $k^{-2/3}$: a different ultraviolet
Helical MHD, inverse transfer [3, 59, 60]	radiation era	τ_A at a growing $L(t)$	$T_{\rm dec}$ with drifting k_0	Peak migrates with the source scale; coherence at k_0 *unresolved*
Chiral plasma instability [44, 61]	$T \gtrsim 80$ TeV	driven, $\tau_{\rm st} \to \infty$	$T_{\rm sw}$ (quasi-stationary)	The one genuinely stationary case; sweeping peak $1.49M$
PT magnetic fields [62]	EW, QCD	τ_A; single injection	$T_{\rm burst} \to T_{\rm dec}$	Burst then self-similar decay; two regimes in one history
Free-streaming epochs [31, 32]	ν, γ decoupling	$\tau_{\rm drag} \propto k^0$	modifies $\tau_c(k)$	Damping rate loses its k-dependence; cutoff is not Kolmogorov

TABLE IX. Early-Universe processes and the temporal regime each realises. Two entries are worth isolating. The chiral instability is the only process in the list that is genuinely *driven* and therefore the only one for which the stationary sweeping kernel of Sec. III is the appropriate model rather than an approximation. Conversely, decaying MHD is the case most often computed with that kernel and the one where it is least justified: the Alfvénic correlation time scales as k^{-1} rather than the $k^{-2/3}$ built into η_k, so the ultraviolet slope of a magnetically sourced spectrum should not be read off the hydrodynamic kernel. Propagation-side damping of the waves themselves [63] is a separate effect and is not included in any row.

Three conclusions follow from the tables that are not visible from any single model. None is a new calculation; the tables are an organising device, and their value is in what the juxtaposition makes obvious. First, the ultraviolet slope is not a prediction of "turbulence" but of a choice between two exponents, $\tau_c \propto k^{-2/3}$ (hydrodynamic turnover) and $\tau_c \propto k^{-1}$ (sweeping, Alfvénic); most of the literature adopts the first even when modelling magnetic sources [2, 3, 62]. The objection is not new—Ref. [24] raises it and adopts the Eulerian k^{-1} time for this reason—but it has not been acted on since. Second, all cusped models collapse onto one prediction, so effort spent discriminating among decaying correlators is misdirected — the question that matters is only whether the correlator is differentiable at zero lag. Third, resistivity and viscosity enter this axis through the *amplitude*, not the decorrelation, and therefore cannot affect the infrared at all; that is the subject of Sec. VII.

VII. OUTLOOK: THE DISSIPATION MICROPHYSICS

Everything above concerns the first two inputs of the pipeline, the spatial spectrum and the unequal-time correlator. The results are largely negative for the first and sharp for the second: no spatial choice moves the infrared, and the ultraviolet and the peak are controlled by a single feature of the temporal correlator. That leaves the third input, and it is the one the field has treated most crudely.

a. What is currently assumed. Every analytic treatment of GWs from turbulence known to us terminates the cascade with a Navier–Stokes viscosity: a single kinematic ν, a Kolmogorov dissipation wavenumber $k_d = (\varepsilon/\nu^3)^{1/4}$, and a cutoff whose shape is then argued to be immaterial because the inertial range is long. Ref. [24] states the position explicitly, imposing $E(k > k_d) = 0$ and noting that "the precise modeling of the cutoff" should not matter. Ref. [23] goes further and drops the dissipation range altogether, with a reason: the bulk of the GW signal is produced faster than the turbulence decays, so the dynamics at the viscous scale has no time to matter. That argument is sound as far as it goes—it is the same separation of timescales that makes the GW spectrum a near-instantaneous photograph of the source—but it addresses the *cutoff wavenumber* and the *decay of the amplitude*, not the contribution of viscosity to the correlator itself.

On the simulation side the assumption is even more explicit. Ref. [28] fixes $\nu = \eta$, i.e. magnetic Prandtl number $\mathrm{Pm} = 1$, and chooses its value "as small as possible, but still large enough" for the grid; the hydrodynamic simulations of Ref. [23] contain no physical viscosity at all, only the numerical viscosity of the lattice.

b. What the plasma actually does. Neither assumption describes the early Universe. Two features of the real dissipation physics are well established in the primordial-magnetic-field literature and absent from the GW literature.

First, momentum transport is not always diffusive. When the mean free path of the dominant carrier is short compared with the scale of interest the damping is the familiar νk^2; when it is long the carrier free-streams and exerts a *drag*, with a different wavenumber dependence and a different functional form of the cutoff. Both regimes occur, and the system alternates between them as species decouple; the damping scales in each were worked out for primordial magnetic fields long ago [31, 32] and are reviewed in Refs. [14, 15]. The dissipation range that terminates the source spectrum is therefore not a Kolmogorov range at the epochs of interest.

Second, the magnetic Prandtl number is enormous, not unity. The kinematic viscosity is set by weakly coupled free-streamers and the resistivity by collisional processes, and the resulting Pm exceeds unity by many orders of magnitude through the radiation era [14, 32]. The consequence is standard in the dynamo literature but has no counterpart here: above the kinetic viscous scale the velocity field is dead while the magnetic field continues to cascade, producing a broad *subviscous* magnetic range [33, 64–66]. Since the GW source is the quadratic stress B_iB_j and not the velocity, that range radiates. It is present in neither the analytic calculations, which stop at a single k_d, nor the *gravitational-wave* simulations, which are all at $\mathrm{Pm} = 1$; high-Pm MHD simulations exist —Ref. [66] reaches $\mathrm{Pm} \simeq 2500$ in three dimensions, while Ref. [49] spans $\mathrm{Pm} \leq 5$ in 3D and $\mathrm{Pm} \leq 100$ in 2D—but have never been coupled to a GW calculation. Free-streaming, separately, is explicitly set aside in the field's own review of the relevant magnetohydrodynamics [67], which notes that its effect on MHD has been studied elsewhere; that review does not treat the magnetic Prandtl number at all. ~~The omission is acknowledged in the field's own review of the relevant magnetohydrodynamics.~~ [the review is declaring its scope, not conceding a gap, and has no Pm content]

c. Why this is now the interesting input. The axis-separation result makes the consequence predictable in kind if not in size. Dissipation is a spatial modification of the source, so by Sec. V E it cannot alter the infrared slope—a subviscous magnetic tail will radiate its own k^3 like every other band. What it can change is the amplitude and, through the ultraviolet support of the convolution, the peak and the high-frequency tail. A subviscous range spanning several decades in k is, on dimensional grounds, a large amount of stress that the current calculations discard, and its contribution is directly computable with the kernel of Sec. III: it amounts to replacing the shape factor $\mathcal{S}(k)$ of Eq. (25) by one that continues past k_d. We stress that we have not performed that calculation, and one caution applies. In the Kulsrud–Anderson picture [64] the subviscous spectrum $M(k) \propto k^{3/2} K_0(k/k_\eta)$ places the bulk of the magnetic energy at the *resistive* scale [66]—far above k_d—so the additional stress enters the convolution in the deep ultraviolet, where the gravitational-wave response is smallest, rather than near the peak. That picture is moreover a velocity-driven dynamo regime, not the magnetically dominated decay relevant here, so the extrapolation is not automatic. "Several decades of k" is a statement about bandwidth, not about energy, and the two must not be conflated until the integral is done. ~~the subviscous magnetic energy is concentrated near the resistive scale and is a modest fraction of the total, so the contribution could be small;~~ [corrected: the bulk of the magnetic energy IS subviscous, so the old caution was self-defeating]

There is also a temporal consequence, and it interacts with the cusp result of Sec. IV B. A viscously damped mode decorrelates as $e^{-\nu k^2|\tau|}$, whose slope at zero lag is $-\nu k^2$, not zero. Viscosity therefore *supplies* the kink that Eq. (37) requires, even for a flow whose sweeping correlator is exactly the Gaussian measured in Ref. [23]. The coefficient follows the two-leg rule of Sec. V A: the source is quadratic, so the temporal factor is the transform of the *squared* correlator, and $f_2'(0^+) = 2f_1(0)f_1'(0^+) = -2\nu k^2$. The cusp theorem then gives $T \to 4\nu k^2/\omega^2$, which on the cone $\omega = k$ is the constant 4ν. ~~Evaluated on the sound cone the resulting temporal factor tends to the constant 2ν,~~ [2ν is the one-leg value; the same factor two that Sec. V A gets right for the tent was missed here. Verified numerically, `Notebooks/viscous_cusp_floor.py`] In the dimensionless variables of this paper the result is sharper than that. With $k_d = (\varepsilon/\nu^3)^{1/4}$ and $R = (k_d/k_0)^{4/3}$ one has $\nu = \varepsilon^{1/3}k_0^{-4/3}R^{-1}$, and $u_0 = (\varepsilon/k_0)^{1/3} = M$, so the dimensionless viscous rate is Mp^2/R and

$$\hat{T} \longrightarrow \frac{4M}{R}, \tag{56}$$

a constant *independent of* p. For $M \lesssim 0.1$ we reproduce it to better than 1%; the agreement degrades to a few per cent by $M = 0.5$, the residual being the next cusp term $\propto \omega^{-4}$ carried by the sweeping rate. ~~We reproduce it to better than 0.5% at every p tested and for $M = 0.05$–0.5, $R = 10^3$–10^4.~~

The consequence is not the one the paragraph above anticipates, and it runs in the uncomfortable direction. On the cone the two-leg sweeping factor is $\propto \exp(-p^{2/3}/M^2)$, which for subsonic flow is already negligible by $p \sim M$; a p-independent floor therefore does not sit *beneath* that cutoff so much as *replace* it. Solving $\exp(-p^{2/3}/M^2)\times$prefactor $= 4M/R$ puts the crossover above the sweeping peak $\xi_* M$ for $M \gtrsim 0.3$—where the picture of a low floor is right—but *below* it for $M \lesssim 0.1$, which is the regime cosmological flows occupy: at $M = 0.05$ the crossover is at $p \simeq 0.008$–0.011 against a peak at 0.074. Taken at face value that would mean the stationary sweeping peak does not survive the introduction of a physical viscosity at deeply subsonic Mach number, since above the crossover the spectrum reverts to p^3 times the bare spatial convolution, scaled by $4M/R$.

We stop short of asserting that, because Eq. (56) is a statement about the temporal factor alone. In the full kernel the two convolution legs carry different wavenumbers, each with its own damping and its own weight, and whether the crossover survives that averaging is exactly the calculation this section says has not been performed. What is established here is narrower and still worth having: the coefficient is 4ν and not 2ν; the floor is $4M/R$ and so is suppressed by the Reynolds number rather than by the viscosity alone; and the expectation that it is *small* is unsafe for the Mach numbers that matter. Quantifying it in the full kernel is the natural next calculation, and it is now a well-posed one. ~~so the prediction is a low but strictly power-law floor beneath the Gaussian sweeping cutoff, with amplitude linear in the viscosity. Whether that floor is ever large enough to matter is a quantitative question we have not settled, and it is the natural next calculation.~~ [Task 2.2, partially closed: the coefficient and the parametric size are now computed and the "low floor" framing is shown to be unsafe below $M \simeq 0.1$; the full-kernel evaluation is still outstanding]

We therefore regard the pipeline as closed on its spatial axis, well constrained on its temporal axis, and open on its dissipative one.

VIII. SUMMARY AND CONCLUSIONS

1. The turbulence-to-GW pipeline factorises *asymptotically*. The strict $k \to 0$ infrared slope is set by the spatial source model alone—this is the universal theorem of Ref. [27], not a result of ours—and the ultraviolet asymptote by the temporal model alone, a statement whose aeroacoustic form predates the cosmological literature [68, 69]. Every intermediate feature, the peak and the break alike, responds to both. The factorisation is *not* a statement about the observable band: a source of finite duration moves the infrared by two powers [9, 26], and duration belongs to neither axis (Sec. II D, Appendix H).

2. The causal infrared $\Omega_{\rm GW} \propto k^3$ is a property of the quadratic source, not of the fluid spectrum. That the stress self-convolution floors at white noise regardless of how blue the input is was established by Refs. [18, 25] and, for stress spectra specifically, by Ref. [29]; our contribution is a sharper test of it. The band-split experiment of Sec. V E takes the limit those papers do not, deleting the infrared band outright: the Kolmogorov range alone, an input with *identically zero* infrared power, still radiates $p^{3.00}$, and it continues to do so with the temporal sector removed entirely.

3. Consequently the k^3-versus-k^1 tension cannot be resolved on the spatial axis. ~~It localises to one assumption, $\Omega_T \propto \Omega_M$, which the self-convolution of the stress forbids.~~ [removed, same reason as in the abstract] It is not in fact a disagreement about physics. The two slopes are the two branches of one finite-duration response, in disjoint bands separated by $k_{\rm br} \simeq 1/\delta t$: k^3 is the asymptotic law and k^1 the finite-duration regime [9, 26]. The break lies below most observationally accessible bands *and* below the fundamental mode of every published simulation

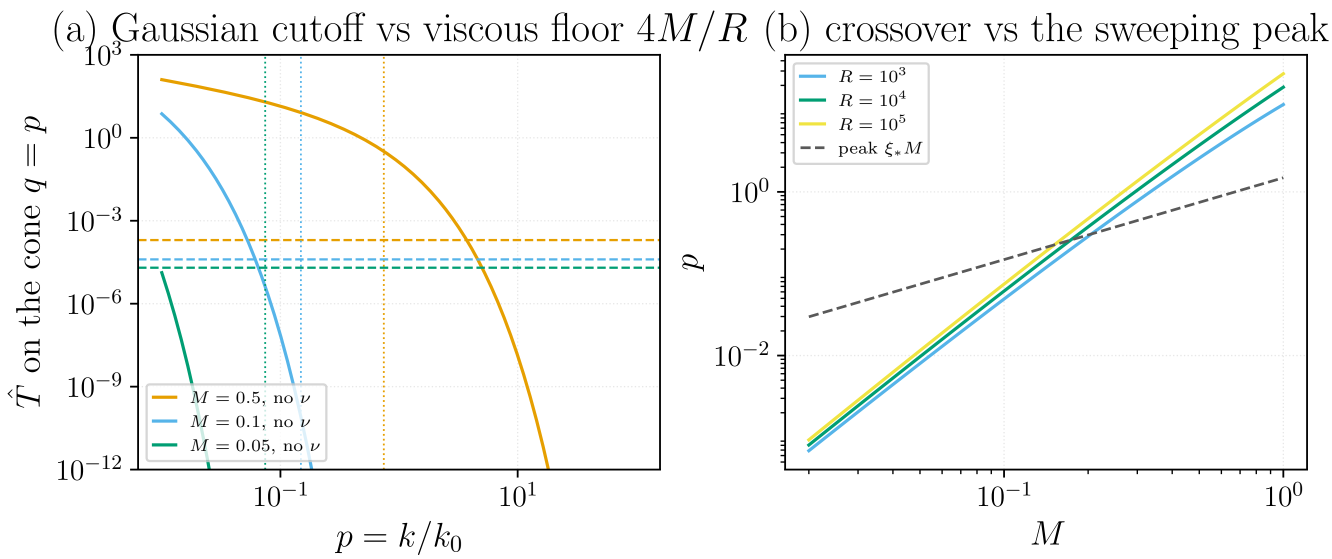


FIG. 19. The viscous cusp. *(a)* The two-leg sweeping factor on the cone $q = p$ (solid) against the viscous floor $4M/R$ of Eq. (56) (dashed) at $R = 10^4$; dotted verticals mark the sweeping peak $\xi_* M$. The Gaussian is super-exponential in $p^{2/3}/M^2$, so the constant floor overtakes it, and the more subsonic the flow the earlier. *(b)* The crossover against M for three Reynolds numbers, with the peak $\xi_* M$ dashed: the floor emerges above the peak for $M \gtrsim 0.3$ and below it for $M \lesssim 0.1$. Temporal factor only — the full two-wavevector kernel is not evaluated here. Generated by `Notebooks/viscous_cusp_floor.py`.

box—though not universally: for a short-lived, small-scale electroweak source it rises above the LISA window (Sec. V D)— so no simulation of this class can exhibit the causal asymptote and none should be read as excluding it (Sec. V). Both halves of that statement are now quantified. In observed frequency $f_{\rm break} = f_H/2\hat{\tau}_c$ (Sec. V D): for an electroweak source it runs 5 μHz–15 mHz against LISA's energy-density window 1.0–4.6 mHz, so LISA observes on the k^1 side in every corner of the parameter space but one, and that one puts the break at the spectral knee where it is not separable from the peak. The value itself moves by two to three decades with the source lifetime and eddy size and should not be quoted as a number; the ordering of break and band is what is robust. Run at the simulation's own parameters (Sec. V B) the kernel produces a linear band rather than a causal one, with the break at or below the box floor, and the infinitely coherent control already fails to reach +3 anywhere the box can see—so a measured slope below 3 is not evidence against causality. It nevertheless predicts $k^{0.7}$ where $k^{1.3}$ is measured; the missing 0.6 traces to our spatial factor rolling over towards the peak inside a fit band only 0.66 decades wide, not to the temporal model. The manuscript therefore reproduces the shape of the simulated infrared but not yet its slope. What remains genuinely open is whether real MHD sources stay coherent long enough to populate the k^1 band; we have not established that, and assuming it is circular.

4. The displacement of the GW peak to the source scale is controlled by a single number, the slope of the source correlator at zero lag. That a cusp gives an ω^{-2} tail is the standard Abelian theorem for Fourier integrals, and its consequences for GW spectra—including for the *peak location*—were already drawn by Ref. [25], which states that the differentiability of the source envelope moves the peak from the inverse duration to the inverse correlation scale. What is new here is the object to which the argument is applied: their regularity sits at the switch-on and switch-off endpoints of a deterministic, separable envelope $g(t)$, whereas ours sits at zero lag of a *stationary, decorrelating* UETC. The consequence is that all cusped decorrelation models collapse onto a single prediction, so the specific decorrelation law does not matter (Sec. IV B). [Referee-verification found Caprini+2009 already links regularity to the peak] The question that would overturn the source-scale peak is therefore whether the correlator of decaying turbulence is differentiable at zero lag, not whether it follows any particular power law. Ref. [23] measures a Gaussian hydrodynamic UETC, for which it is; the corresponding *magnetic* measurement has not been made, and this conclusion is conditional on it. The nearest available evidence does not favour us: Eulerian space–time correlations measured in forced MHD turbulence [70] are consistent with sweeping by the energy-containing motions, which would give $f'(0^+) = 0$ and remove the cusp. That measurement is of a forced, hydrodynamically swept field rather than of the anisotropic stress of a decaying, magnetically dominated one, so it does not settle the case—but it is the outcome to beat.

5. The dissipation input is the one still open, and it is the only claim here for which we could find no prior

art: the early Universe is neither Navier–Stokes nor Pm = 1, and the resulting subviscous magnetic range radiates because the source is B_iB_j rather than the velocity (Sec. VII). The field's own review of the relevant magnetohydrodynamics [67] sets free-streaming aside explicitly and does not treat the magnetic Prandtl number at all—a declaration of scope, which is weaker than a concession but still locates the gap. ~~The omission is acknowledged in the field's own review of the relevant magnetohydrodynamics.~~ [same over-claim as in the abstract; Sec. VII was corrected in the first pass, this conclusion and the abstract were not] We identify the gap here but do not close it; the size of the discarded contribution is not yet computed, and until it is, this is a programme rather than a result.

Appendix A: Table of symbols

Symbols used in this paper. Glyphs flagged **[MULTI]** carry more than one meaning depending on the case and are disambiguated in place.

Gravitational-wave field and source

Symbol	Definition	Introduced
h_{ij}	Transverse–traceless metric (strain) perturbation	(1)
$h_c(p)$	Characteristic strain, $h_c^2(p) = p\,H(p,p)$	(8)
Π_{ij}^{TT}	TT part of the anisotropic (Reynolds) stress source	(3)
$\Lambda_{ij,lm}$	TT projection operator, $P_{il}P_{jm} - \frac{1}{2}P_{ij}P_{lm}$	(3)
P_{ij}	Transverse projector, $\delta_{ij} - \hat{k}_i\hat{k}_j$	(3)
T_{lm}	Reynolds stress, $T_{lm} = w\,v_l v_m$	(3)
w	Enthalpy density, $w = \bar{\rho} + \bar{p}$	(3)
v_i	Turbulent (incompressible) velocity field	(3)
ρ_{GW}	GW energy density (Isaacson), $\langle \dot{h}_{ij}\dot{h}_{ij}\rangle/32\pi G$	(B14)
$\Pi(k;\tau_1,\tau_2)$	Unequal-time correlator (UETC) of the TT stress	(6)
$\widetilde{\Pi}(k,\omega)$	Temporal transform of the UETC, $= H_{ijij}$ for a stationary source	(B37)
G	Newton's gravitational constant	(2)
c	Speed of light; set to 1 only when evaluating numerically	(21)
$\Omega_{\mathrm{GW}}(p)$	GW energy spectrum, $d\rho_{\mathrm{GW}}/d\ln k$ (dimensionless)	(8)
Ω_{M}	Magnetic (source) energy fraction; $M \simeq \sqrt{\Omega_{\mathrm{M}}}$	(24)
Ω_K	Kinetic-energy fraction of the source	Sec. II C
$H_{ijij}(\mathbf{k},\omega)$	Contracted source spatio-temporal correlator (the GW kernel)	(12)
$H[S,T](p,q;M,R)$	The kernel labelled by its (spatial, temporal) model pair	(10)
$A_{ij}^{\mathrm{TT}}(\mathbf{k})$	Spatial amplitude of an impulsive burst	(D1)
$P(k)$	Spatial power of the burst amplitude	(D7)
T_{src}	Source duration (statistically stationary case)	(B37)
T_{em}	Finite emission window of the self-similar source	(43)

Model grid (Sec. II D)

Symbol	Definition	Introduced
$S_{\rm K41}$	Kolmogorov inertial range, $E(k) \propto k^{-5/3}$ on $k_0 < k < k_d$	Sec. III
$S_{\delta k}$	Monochromatic spatial source, $E(k) = E_0\delta(k - k_0)$	(C1)
$S_{\delta r}$	Real-space white noise, $R_{ij} \propto \delta^{(3)}(\mathbf{r})$, i.e. $E(k) \propto k^2$	(C10)
$S_{\rm full}$	Batchelor/Saffman infrared band joined to the Kolmogorov range	(25)
$T_{\rm sw}$	Kraichnan sweeping: Gaussian in the time difference	(11)
$T_{\rm dec}$	BK2016 decaying turbulence: power law $(1 + \lvert\tau\rvert/\tau_k)^{-2/3}$	(B69)
$T_{\rm imp}$	Impulsive, δ-correlated in time	(D1)
$T_{\rm burst}$	Source of finite duration Δt	(E9)

Background / cosmology

Symbol	Definition	Introduced
$a(\tau)$	FLRW scale factor	(1)
τ [MULTI]	Conformal time. **[also:** time lag $\tau = \tau_1 - \tau_2$ in UETCs/decorrelation]	(1)
$\Delta\tau$	Time lag $\tau_1 - \tau_2$ in the master integral	(7)
H_*, $\mathcal{H}_*$	Hubble rate at GW production (physical, conformal)	(32)
f_H	Hubble frequency today, $f_H = H_*(a_*/a_0)$	(32)
f, f_0	Observed GW frequency; stirring-scale frequency, $f = f_0\, p$	(33)
γ	Largest-eddy size in Hubble radii, $\gamma \equiv l_0 H_*$; $\hat{k}_0 = 2\pi/\gamma$	(33)
$\hat{\omega}$, $\hat{k}$	Frequency/wavenumber in Hubble units, $\hat{\omega} = \omega/H_*$	(32)

Spatial turbulence spectrum

Symbol	Definition	Introduced
$F_{ij}(\mathbf{k}, \tau)$	Spectral (Fourier) two-point velocity correlator	(11)
$A(k)$	Spectral amplitude, $A(k) = E(k)/4\pi k^2$	(11)
$E(k)$, E_k	Kinetic energy spectrum; $E_k = C_k \varepsilon^{2/3} k^{-5/3}$ (Kolmogorov)	(17)
E_0	Total energy of the monochromatic source, $\int_0^\infty E(k)\, dk = E_0$	(C1)
v_0^2	Velocity variance of the white-noise source	(C10)
C_k	Kolmogorov constant	(11)
ε	Turbulent energy dissipation (cascade) rate; $\varepsilon = c^3 M^3 k_0$	(11)
$\mathcal{K}(k, k_1, u)$	Geometric (tensor-contraction) kernel	(16)
$\widetilde{\mathcal{K}}$	$\mathcal{K}$ in dimensionless arguments (homogeneous of degree zero)	(19)
$\mathcal{K}_0(p)$	Monochromatic geometric kernel, $\frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}$	(C3)
Θ	Heaviside step (triangle-inequality constraint $p \le 2$)	(C2)
$\mathcal{S}_{ij}(\mathbf{r})$	Spatial profile of the real-space burst source	(B110)
ℓ	Spatial coherence length of $\Sigma(r)$	(E10)

Real-space lane (App. B 1)

Symbol	Definition	Introduced
$G_{\rm ret}$	Retarded Green function, $\delta(t-t'-\lvert\mathbf{x}-\mathbf{x}'\rvert)/4\pi\lvert\mathbf{x}-\mathbf{x}'\rvert$	(B3)
$\hat{n}$	Local propagation direction, $\hat{n}=(\mathbf{x}-\mathbf{x}')/\lvert\mathbf{x}-\mathbf{x}'\rvert$; the TT projection is taken along it	(B5)
$R_{ij}(\mathbf{r},\tau)$	Real-space two-point velocity correlator, $\langle v_i(\mathbf{y},t)v_j(\mathbf{y}+\mathbf{r},t+\tau)\rangle$	(B6)
R_L, R_N	Longitudinal / transverse correlation functions; $R_N=R_L+\frac{r}{2}\partial_r R_L$ (incompressible)	(B7)
$R_{ijij}(\mathbf{r},\tau)$	Contracted real-space source correlator, $R_{ii}R_{jj}+\frac{1}{3}R_{ij}R_{ij}=\frac{14}{3}R_N^2+4R_NR_L+\frac{4}{3}R_L^2$	(B10)
$\Sigma(r,\tau)$	Real-space TT-stress correlator, $\Sigma=w^2R_{ijij}$	(B13)
$\widehat{\Sigma}(r,\omega)$	Temporal transform of Σ at fixed separation r (space stays real)	(B21)
$\widetilde{\Sigma}(k,\omega)$	Spatial transform of $\widehat{\Sigma}$; $\widetilde{\Sigma}(k,k)=T_{\rm src}\widetilde{\Pi}(k,k)$	(B27)
$L(\mathbf{x},\tau)$	Unequal-time GW correlator, $\langle\dot{h}_{ij}(\mathbf{x},t)\dot{h}_{ij}(\mathbf{x},t+\tau)\rangle/32\pi G$	(B14)
$I(\mathbf{x},\omega)$	Spectral energy density, $(2\pi)^{-1}\int d\tau\, e^{i\omega\tau}L$; $d\rho_{\rm GW}/d\ln\omega=\omega I$	(B17)

Stress spectrum and slope bookkeeping (Sec. V)

Symbol	Definition	Introduced
$n[X]$	Logarithmic slope of a spectrum, $n[X]=d\ln X/d\ln k$	(41)
$\Omega_{\rm T}$, $E_{\rm T}$	Anisotropic-stress energy fraction and energy spectrum; $n[\Omega_{\rm GW}]=n[\Omega_{\rm T}]-2$	(41)
$P_{\rm T}$, $P_{\rm M}$	Stress and magnetic *power* spectra; $P_{\rm T}$ is $P_{\rm M}$ self-convolved	(41)
$B(k,\Delta t)$	Build-up fraction $1-\mathrm{sinc}(k\Delta t)$: how much of $\Omega_{\rm sat}$ has accumulated	(44)
$\Omega_{\rm sat}(k)$	Saturated (fully built-up) GW spectrum	(44)
S_+, $S_\times$	TT-projected stress polarisation amplitudes held in the simulation's k-space aux arrays	Sec. V

Wavenumbers

Symbol	Definition	Introduced
k	GW / Fourier wavenumber magnitude	(12)
k_0, k_*	Outer (integral / stirring) scale wavenumber	(20)
k_d	Dissipation-scale wavenumber, $k_d/k_0=R^{3/4}$	(21)
k_1	First source wavevector magnitude	(13)
u	Second source wavevector magnitude, $u=\lvert\mathbf{k}-\mathbf{k}_1\rvert$	(B43)
μ	Cosine of the source angle, $\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}_1$	(B50)
$k_{\rm peak}$, $p_{\rm peak}$	Wavenumber of the GW spectral peak	Sec. IV
$k_{\rm break}$	Duration break, $k_{\rm break}\simeq 1/\Delta t_s$, separating the k^1 and k^3 branches	(48)
$k_{\rm box}$	Simulation box fundamental, $k_{\rm box}=2\pi/L$	(48)
$\omega_{\rm peak}$	Peak frequency of the real-space burst spectrum	(E10)

Temporal / decorrelation model

Symbol	Definition	Introduced
$f(\eta_k, \tau)$	Temporal decorrelation function; Kraichnan $\exp(-\frac{\pi}{4}\eta_k^2\tau^2)$	(11)
$\tilde{f}(\eta_k, \omega)$	Frequency transform of f; Kraichnan $(2/\eta_k)\exp(-\omega^2/\pi\eta_k^2)$	(17)
η_k	Kraichnan eddy/sweeping rate, $\frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3}$	(11)
η_0	Sweeping rate at the outer scale, $\eta_0 \equiv \eta_{k_0} = Mk_0/\sqrt{2\pi}$	(C5)
$g(k, \omega)$	Combined spectral $\times$ temporal factor, $A(k)\tilde{f}(\eta_k, \omega)$	(17)
τ_k	Sweeping/eddy decorrelation time, $1/\eta_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$	(B69)
τ_1, τ_2 [MULTI]	The two legs' decorrelation times, $\sqrt{x}/M$, $\sqrt{y}/M$. **[also:** the two integration times τ_1, τ_2 of the master formula (7)]	(B69)
$\tau_*(t)$	Feynman-combined time, $1/\tau_* = t/\tau_1 + (1-t)/\tau_2$	(B96)
τ_s, t_0	Epoch of the impulsive burst	(D1)
Δt	Duration of a finite burst; $g \to \delta$ as $\Delta t \to 0$	(E9)
$g(t)$, $\tilde{g}(\omega)$ [MULTI]	Burst temporal profile and its transform. **[distinct from** $g(k, \omega)$ **and** $\hat{g}$]	(B112)
$\hat{g}(\bar{q})$	Decaying-turbulence temporal kernel, $e^{-i\bar{q}}(-i\bar{q})^{-1/3}\Gamma(\frac{1}{3}, -i\bar{q})$	(B70)
$\mathcal{C}(\bar{q})$	Transform of the *squared* decorrelation; $\mathcal{C}(0) = 6$, tail $8/3\bar{q}^2$	(C7)
$I_\omega(\omega)$	Temporal convolution for the monochromatic source	(C4)
J, $\tilde{J}$	Temporal convolution retained inside the white-noise spatial integral	(C11)

Dimensionless variables (note the heavy reuse of p, q, s, Ω, x, Φ)

Symbol	Definition	Introduced
p	Dimensionless wavenumber, $p = k/k_0$	(8)
q [MULTI]	Dimensionless frequency $q = \omega/(c\,k_0)$; on-shell $q = p$. **[also:** $q = k_1/k$ in (19)]	(8)
$\bar{q}$	Rescaled frequency of the decay kernel, $\bar{q} = \omega\tau_1 = \sqrt{2\pi}\,q/M$	(C6)
q_1, ω_1	Convolution frequency (integration variable)	(14)
s	$s = u/k$	(19)
Ω [MULTI]	Dimensionless frequency $\omega/(\varepsilon^{1/3}k^{2/3})$. **[distinct from** $\Omega_{\rm GW}, \Omega_{\rm M}, \Omega_K$]	(19)
Ω_1	$\omega_1/(\varepsilon^{1/3}k^{2/3})$	(19)
Φ [MULTI]	Gaussian envelope of the scaling form (B57). **[also:** $\Phi_{\rm eq}$ = equal-time velocity spectral correlator]	(19)
$\mathfrak{H}(\Omega)$	Dimensionless scaling function of the K41 kernel	(19)
x [MULTI]	Substitution variable $(k_1/k_0)^{-4/3}$. **[also:** comoving coordinate x^i in (1)]	(21)
y [MULTI]	Substitution variable $(u/k_0)^{-4/3}$. **[also:** $y = u/k_0$ in the white-noise model (C11)]	(21)
z	$z = k_1/k_0$ (white-noise model)	(C11)
t [MULTI]	Feynman parameter on $[0, 1]$. **[also:** cosmic/simulation time]	(C15)
M	Outer-scale turbulent Mach number, $M = u_0/c = (\varepsilon/k_0)^{1/3}/c$	(21)
u_0	Outer-scale eddy velocity; $u_0 = Mc$, luminal ceiling $u_0 = c$	(21)
R	Reynolds-like inertial-range width, $k_d/k_0 = R^{3/4}$	(21)
ξ	Sweeping collapse variable, $\xi \equiv p/M = k/(Mk_0)$	(28)
$\Psi(p, p/M)$	M-independent rescaled spatial integral, $\Omega_{\rm GW} = M^4p^3\Psi$	(38)
$y_{\rm min}, y_{\rm max}$	Triangle-inequality bounds of the y integration	(22)

Decaying-turbulence (self-similar) parameters

Symbol	Definition	Introduced
$\tau_c(k)$	Source correlation time at wavenumber k, $1/(\varepsilon^{1/3}k^{2/3})$	Sec. V
$\tau_{\rm st}$	Self-similar decay time, $\tau_{\rm st} \sim L/u$	Sec. V
$\tau_{\rm eddy}$	Outer-scale eddy turnover time	Sec. V
ε_0	Dimensionless decay rate, $\varepsilon_0 = \tau_{\rm eddy}/\tau_{\rm st} = O(1)$	Sec. V
$\Phi_{\rm eq}(k,t)$	Equal-time velocity spectral correlator, $\propto (1+t/\tau_{\rm st})^{-34/21}$	Sec. V
$\tau_{\rm sw}(k)$	Sweeping time, $1/(u_{\rm rms}k)$; scales as k^{-1}, unlike $\tau_{\rm eddy} \propto k^{-2/3}$	Table VII
$\tau_A(k)$	Alfvén crossing time, $1/(kv_A)$	Table VII
v_A	Alfvén speed	Table VII
$\tau_\nu(k)$	Viscous time, $1/(\nu k^2)$	Table VII
$\tau_\eta(k)$	Resistive time, $1/(\eta k^2)$	Table VII
$\tau_{\rm drag}$	Free-streaming drag time, $1/\alpha_{\rm fs}$; independent of k	Table VII
$\alpha_{\rm fs}$	Free-streaming drag rate	Table VII
τ_e	Correlation time of the exponential (Lagrangian) model	Table VIII
τ_0	Normalisation of the impulsive correlator, $f = \tau_0\delta(\tau)$	Table VIII
Lu	Lundquist number, $\mathrm{Lu} = v_AL/\eta$; sets $\tau_{\rm st} \simeq \mathrm{Lu}^{1/4}\tau_A$ [49]	Table VIII
$P(k,t)$	Equal-time source power in the amplitude–decorrelation split	(51)
$\mathfrak{P}$	Self-similar scaling function of the source power	(52)
$L(t)$, $u(t)$	Self-similar length and velocity of the decaying envelope	(52)
n_L, n_u	Self-similar growth/decay exponents of L and u	(52)
$\bar{t}$	Mean time $\frac{1}{2}(t_1+t_2)$ of the two-time correlator	(51)
$\mathcal{N}_\omega$	Coherence number, $\omega\tau_c$ on the light cone $= k\tau_c(k)$	(55)
$\mathcal{N}_{\rm em}$	Sampling number, $T_{\rm em}/\tau_c(k)$	Sec. VI D

Appendix B: Consolidated derivation chain

1. Framework in real space

$$ds^2 = a^2(\tau)\big[-d\tau^2 + (\delta_{ij}+h_{ij})dx^i dx^j\big], \qquad \partial_i h_{ij} = h_{ii} = 0, \qquad a \to 1 \tag{B1}$$

$$\Box\, h_{ij}(\mathbf{x},t) = 16\pi G\,\Pi^{\rm TT}_{ij}(\mathbf{x},t), \qquad \Box = \partial_t^2 - \nabla^2 \tag{B2}$$

$$\Box\, G_{\rm ret}(\mathbf{x}-\mathbf{x}', t-t') = -\delta^{(3)}(\mathbf{x}-\mathbf{x}')\,\delta(t-t'), \qquad G_{\rm ret} = \frac{\delta\big(t-t'-|\mathbf{x}-\mathbf{x}'|\big)}{4\pi|\mathbf{x}-\mathbf{x}'|} \tag{B3}$$

$$h_{ij}(\mathbf{x},t) = 4G\int d^3x'\,\frac{\Pi^{\rm TT}_{ij}\big(\mathbf{x}', t-|\mathbf{x}-\mathbf{x}'|\big)}{|\mathbf{x}-\mathbf{x}'|}, \qquad \dot{h}_{ij}(\mathbf{x},t) = 4G\int d^3x'\,\frac{\dot{\Pi}^{\rm TT}_{ij}\big(\mathbf{x}', t-|\mathbf{x}-\mathbf{x}'|\big)}{|\mathbf{x}-\mathbf{x}'|} \tag{B4}$$

$$\hat{n} = \frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|}, \qquad P_{ij}(\hat{n}) = \delta_{ij} - \hat{n}_i\hat{n}_j, \qquad \Lambda_{ij,lm}(\hat{n}) = P_{il}P_{jm} - \tfrac{1}{2}P_{ij}P_{lm}, \qquad \Pi^{\rm TT}_{ij} = \Lambda_{ij,lm}(\hat{n})\,T_{lm}, \qquad T_{lm} = w\,v_l v_m \tag{B5}$$

$$R_{ij}(\mathbf{r},\tau) \equiv \big\langle v_i(\mathbf{y},t)\,v_j(\mathbf{y}+\mathbf{r},t+\tau)\big\rangle \qquad \text{(homogeneous: independent of } \mathbf{y}) \tag{B6}$$

$$R_{ij}(\mathbf{r},\tau) = R_N(r,\tau)\,\delta_{ij} + \big[R_L(r,\tau) - R_N(r,\tau)\big]\,\hat r_i \hat r_j, \qquad \hat r = \mathbf{r}/r, \quad r = |\mathbf{r}| \tag{B7}$$

$$\partial_i R_{ij} = 0 \implies R_N = R_L + \frac{r}{2}\frac{\partial R_L}{\partial r}, \qquad R_L(0,0) = R_N(0,0) = \tfrac{1}{3}\langle v^2\rangle, \qquad R_{ii}(0,0) = \langle v^2\rangle \tag{B8}$$

$$\big\langle v_i v_j v_k v_l\big\rangle = \langle v_i v_j\rangle\langle v_k v_l\rangle + \langle v_i v_k\rangle\langle v_j v_l\rangle + \langle v_i v_l\rangle\langle v_j v_k\rangle \tag{B9}$$

$$\Lambda_{ij,lm}\Lambda_{ij,l'm'} \otimes (\text{B9}) \implies R_{ijij}(\mathbf{r},\tau) \equiv R_{ii}(\mathbf{r},\tau)R_{jj}(\mathbf{r},\tau) + \tfrac{1}{3}R_{ij}(\mathbf{r},\tau)R_{ij}(\mathbf{r},\tau) \tag{B10}$$

$$\begin{aligned} R_{ii} = 2R_N + R_L, \qquad R_{ij}R_{ij} = 2R_N^2 + R_L^2 \\ R_{ijij}(r,\tau) = (2R_N + R_L)^2 + \tfrac{1}{3}\big(2R_N^2 + R_L^2\big) = \tfrac{14}{3}R_N^2 + 4R_N R_L + \tfrac{4}{3}R_L^2 \end{aligned} \tag{B11}$$

$$r \to 0,\ \tau \to 0: \qquad R_{ijij} \to \tfrac{10}{9}\langle v^2\rangle^2 \tag{B12}$$

$$\Sigma(r,\tau) \equiv w^2\, R_{ijij}(r,\tau) = \big\langle \Pi^{\rm TT}_{ij}(\mathbf{y},t)\,\Pi^{\rm TT}_{ij}(\mathbf{y}+\mathbf{r},t+\tau)\big\rangle \tag{B13}$$

$$\rho_{\rm GW}(\mathbf{x}) = \frac{\big\langle \dot h_{ij}\dot h_{ij}\big\rangle}{32\pi G}, \qquad L(\mathbf{x},\tau) \equiv \frac{1}{32\pi G}\big\langle \dot h_{ij}(\mathbf{x},t)\,\dot h_{ij}(\mathbf{x},t+\tau)\big\rangle, \qquad \rho_{\rm GW}(\mathbf{x}) = L(\mathbf{x},0) \tag{B14}$$

$$\frac{(4G)^2}{32\pi G} = \frac{G}{2\pi}: \qquad L(\mathbf{x},\tau) = \frac{G}{2\pi}\iint \frac{d^3x_1\,d^3x_2}{|\mathbf{x}-\mathbf{x}_1|\,|\mathbf{x}-\mathbf{x}_2|}\,\big\langle \dot\Pi^{\rm TT}_{ij}(\mathbf{x}_1,t_1)\,\dot\Pi^{\rm TT}_{ij}(\mathbf{x}_2,t_2)\big\rangle, \qquad \begin{aligned} t_1 &= t - |\mathbf{x}-\mathbf{x}_1| \\ t_2 &= t + \tau - |\mathbf{x}-\mathbf{x}_2| \end{aligned} \tag{B15}$$

$$\big\langle \dot\Pi^{\rm TT}_{ij}(\mathbf{x}_1,t_1)\dot\Pi^{\rm TT}_{ij}(\mathbf{x}_2,t_2)\big\rangle = -\partial_\tau^2\,\big\langle \Pi^{\rm TT}_{ij}(\mathbf{x}_1,t_1)\Pi^{\rm TT}_{ij}(\mathbf{x}_2,t_2)\big\rangle = -\partial_\tau^2\,\Sigma\big(|\mathbf{x}_1-\mathbf{x}_2|,\ t_2-t_1\big) \tag{B16}$$

$$I(\mathbf{x},\omega) \equiv \frac{1}{2\pi}\int d\tau\, e^{i\omega\tau} L(\mathbf{x},\tau), \qquad \rho_{\rm GW}(\mathbf{x}) = \int d\omega\, I(\mathbf{x},\omega), \qquad \frac{d\rho_{\rm GW}}{d\ln\omega} = \omega\, I(\mathbf{x},\omega) \tag{B17}$$

$$\int d\tau\, e^{i\omega\tau}\big(-\partial_\tau^2\big)(\cdot) = \omega^2\int d\tau\, e^{i\omega\tau}(\cdot) \tag{B18}$$

$$I(\mathbf{x},\omega) = \frac{G\,\omega^2}{(2\pi)^2}\iint \frac{d^3x_1\,d^3x_2}{|\mathbf{x}-\mathbf{x}_1|\,|\mathbf{x}-\mathbf{x}_2|}\int d\tau\, e^{i\omega\tau}\,\Sigma\big(|\mathbf{x}_1-\mathbf{x}_2|,\ t_2-t_1\big) \tag{B19}$$

$$\mathbf{x} = 0, \quad x_a = |\mathbf{x}_a|, \quad \mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2, \quad r = |\mathbf{r}|: \qquad t_2 - t_1 = \tau - (x_2 - x_1) \tag{B20}$$

$$\widehat\Sigma(r,\omega) \equiv \int ds\, e^{i\omega s}\,\Sigma(r,s) \implies \int d\tau\, e^{i\omega\tau}\Sigma\big(r,\tau-(x_2-x_1)\big) = e^{i\omega(x_2-x_1)}\,\widehat\Sigma(r,\omega) \tag{B21}$$

$$x_2 - x_1 \longrightarrow -\hat n\cdot\mathbf{r}, \qquad \big\langle e^{-i\omega\,\hat n\cdot\mathbf{r}}\big\rangle_{\hat n} = \frac{\sin(\omega r)}{\omega r} \tag{B22}$$

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega} = 8G\,\omega^2\int_0^\infty dr\ r\ \sin(\omega r)\,\widehat\Sigma(r,\omega)} \tag{B23}$$

$$\Sigma(r,s) = \Sigma(r)\,\delta(s) \implies \widehat\Sigma(r,\omega) = \Sigma(r) \implies (\text{B23}) = (\text{B111}) \tag{B24}$$

$$\Sigma(r,s) = \Sigma(r)\,g_{\rm c}(s), \quad \tilde g_{\rm c} = |\tilde g|^2 \implies (\text{B23}) = (\text{B113}) \tag{B25}$$

$$\omega r \ll 1:\ \sin(\omega r) \to \omega r \implies \frac{d\rho_{\rm GW}}{d\ln\omega} \to 8G\,\omega^3\int_0^\infty dr\, r^2\,\widehat\Sigma(r,0)\ \propto\ \omega^3 \tag{B26}$$

2. From the real-space lane to the spectral kernel

$$\widetilde{\Sigma}(k,\omega) \equiv \int d^3r\, e^{-i\mathbf{k}\cdot\mathbf{r}}\, \widehat{\Sigma}(r,\omega), \qquad \int d\Omega_{\hat{r}}\, e^{-ikr\cos\theta} = \frac{4\pi\sin(kr)}{kr} \implies \widetilde{\Sigma}(k,\omega) = \frac{4\pi}{k}\int_0^\infty dr\, r\sin(kr)\, \widehat{\Sigma}(r,\omega) \tag{B27}$$

$$\int_0^\infty dr\, r\sin(\omega r)\, \widehat{\Sigma}(r,\omega) = \frac{\omega}{4\pi}\, \widetilde{\Sigma}(\omega,\omega) \tag{B28}$$

$$\text{(B23)} \implies \frac{d\rho_{\mathrm{GW}}}{d\ln\omega} = 8G\omega^2 \cdot \frac{\omega}{4\pi}\widetilde{\Sigma}(\omega,\omega) = \frac{2G}{\pi}\,\omega^3\, \widetilde{\Sigma}(\omega,\omega) \tag{B29}$$

$$\widetilde{\Sigma}(k,\omega) = w^2\, \widetilde{R_{ijij}}(k,\omega), \qquad H_{ijij}(\mathbf{k},\omega) \equiv \frac{1}{(2\pi)^4}\int d\tau\, d^3\boldsymbol{\xi}\; e^{i\omega\tau - i\mathbf{k}\cdot\boldsymbol{\xi}}\, R_{ijij}(\boldsymbol{\xi},\tau) \tag{B30}$$

$$\frac{d\rho_{\mathrm{GW}}}{d\ln k} \propto k^3 H_{ijij}(\mathbf{k},\omega)\big|_{\omega=k} \implies \Omega_{\mathrm{GW}}(p) \propto p^3 H(p,p), \qquad p = \frac{k}{k_0}, \quad q = \frac{\omega}{c\,k_0}, \qquad h_c^2(p) = p\, H(p,p) \tag{B31}$$

Equivalent spectral route (independent check of the coefficient).

$$h''_{ij}(\mathbf{k},\tau) + k^2 h_{ij}(\mathbf{k},\tau) = 16\pi G\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau) \tag{B32}$$

$$h_{ij}(\mathbf{k},\tau) = \frac{16\pi G}{k}\int_{\tau_i}^{\tau} d\tau'\, \sin[k(\tau-\tau')]\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau'), \qquad \dot{h}_{ij}(\mathbf{k},\tau) = 16\pi G\int_{\tau_i}^{\tau} d\tau'\, \cos[k(\tau-\tau')]\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau') \tag{B33}$$

$$\big\langle \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau_1)\Pi^{\mathrm{TT}*}_{ij}(\mathbf{k}',\tau_2)\big\rangle = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{k}')\, \Pi(k;\tau_1,\tau_2) \tag{B34}$$

$$\cos[k(\tau-\tau_1)]\cos[k(\tau-\tau_2)] \longrightarrow \tfrac{1}{2}\cos[k(\tau_1-\tau_2)] \tag{B35}$$

$$\frac{1}{32\pi G}(16\pi G)^2 \cdot \frac{1}{2}\cdot\frac{k^3}{2\pi^2} = \frac{2Gk^3}{\pi} \implies \frac{d\rho_{\mathrm{GW}}}{d\ln k} \simeq \frac{2G\,k^3}{\pi}\int d\tau_1\int d\tau_2\, \cos[k(\tau_1-\tau_2)]\, \Pi(k;\tau_1,\tau_2) \tag{B36}$$

$$\Pi(k;\tau_1,\tau_2) = \Pi(k;\tau_1-\tau_2) \implies \int d\tau_1\int d\tau_2 = T_{\mathrm{src}}\int d(\Delta\tau), \qquad \widetilde{\Pi}(k,\omega) = \int d(\Delta\tau)\, e^{i\omega\Delta\tau}\Pi(k;\Delta\tau) \tag{B37}$$

$$\frac{2Gk^3}{\pi}T_{\mathrm{src}}\widetilde{\Pi}(k,k) \;=\; \frac{2G}{\pi}\omega^3\widetilde{\Sigma}(\omega,\omega)\Big|_{\omega=k} \implies \widetilde{\Sigma}(k,k) = T_{\mathrm{src}}\, \widetilde{\Pi}(k,k) \tag{B38}$$

3. Spatial reduction

$$R_{ijij}(\mathbf{r},\tau) \overset{\text{(B10)}}{=} R_{ii}R_{jj} + \tfrac{1}{3}R_{ij}R_{ij} \overset{\text{(B11)}}{=} \tfrac{14}{3}R_N^2 + 4R_NR_L + \tfrac{4}{3}R_L^2, \qquad H_{ijij} \overset{\text{(B30)}}{=} \mathcal{F}_{\boldsymbol{\xi},\tau}\big[R_{ijij}\big] \tag{B39}$$

$$R_{ij}(\mathbf{r},\tau) = \int\frac{d^3k}{(2\pi)^3}e^{i\mathbf{k}\cdot\mathbf{r}}F_{ij}(\mathbf{k},\tau), \qquad f(\eta_k,\tau) = \int\frac{d\omega}{2\pi}e^{-i\omega\tau}\tilde{f}(\eta_k,\omega) \tag{B40}$$

$$F_{ij}(\mathbf{k},\tau) = A(k)\, P_{ij}(\hat{k})\, f(\eta_k,\tau) \tag{B41}$$

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^4}\int d\tau\, d^3\mathbf{r}\; e^{i\omega\tau - i\mathbf{k}\cdot\mathbf{r}}\Big[R_{ii}(\mathbf{r},\tau)R_{jj}(\mathbf{r},\tau) + \tfrac{1}{3}R_{ij}(\mathbf{r},\tau)R_{ij}(\mathbf{r},\tau)\Big] \tag{B42}$$

$$\mathbf{u} \equiv \mathbf{k} - \mathbf{k}_1, \quad u \equiv |\mathbf{u}| \tag{B43}$$

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^7}\int d\tau\, e^{i\omega\tau} d^3k_1\Big[F_{ii}(\mathbf{k}_1,\tau)F_{jj}(\mathbf{u},\tau) + \tfrac{1}{3}F_{ij}(\mathbf{k}_1,\tau)F_{ij}(\mathbf{u},\tau)\Big] \tag{B44}$$

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1\Big[\tilde{F}_{ii}(\mathbf{k}_1,\omega_1)\tilde{F}_{jj}(\mathbf{u},\omega-\omega_1) + \tfrac{1}{3}\tilde{F}_{ij}(\mathbf{k}_1,\omega_1)\tilde{F}_{ij}(\mathbf{u},\omega-\omega_1)\Big] \tag{B45}$$

$$P_{ii}(\hat{k}) = 2, \qquad P_{ij}(\hat{k}_1)P_{ij}(\hat{u}) = 1 + (\hat{k}_1\cdot\hat{u})^2 \tag{B46}$$

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1\, A(k_1)A(u)\,\tilde{f}(\eta_{k_1},\omega_1)\tilde{f}(\eta_u,\omega-\omega_1)\Big[4 + \tfrac{1}{3}\big(1 + (\hat{k}_1\cdot\hat{u})^2\big)\Big] \tag{B47}$$

$$\hat{k}_1\cdot\hat{u} = \frac{k^2 - k_1^2 - u^2}{2k_1 u} \tag{B48}$$

$$\begin{aligned} 4 + \tfrac{1}{3} + \tfrac{1}{3}(\hat{k}_1\cdot\hat{u})^2 &= \frac{13}{3} + \frac{(k^2-k_1^2-u^2)^2}{12k_1^2u^2} \\ &= \frac{13}{3} + \frac{1}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \\ &\equiv \mathcal{K}(k,k_1,u) = \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \end{aligned} \tag{B49}$$

$$d^3k_1 = k_1^2\, dk_1\, d\phi\, d\mu, \quad \mu = \hat{k}\cdot\hat{k}_1, \quad \int d\phi = 2\pi, \quad u = \sqrt{k^2 + k_1^2 - 2kk_1\mu}, \quad u\, du = k\, k_1\, d\mu, \quad u \in [|k-k_1|,\, k+k_1] \tag{B50}$$

$$g(k,\omega) \equiv A(k)\,\tilde{f}(\eta_k,\omega) \tag{B51}$$

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^7}\int_0^\infty \frac{k_1\, dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\, du \int_{-\infty}^{\infty} d\omega_1\; g(k_1,\omega_1)\, g(u,\omega-\omega_1)\, \mathcal{K}(k,k_1,u) \tag{B52}$$

4. $S_{\rm K41} \times T_{\rm sw}$

$$A(k) = \frac{C_k\epsilon^{2/3}k^{-11/3}}{4\pi}, \qquad E_k = C_k\epsilon^{2/3}k^{-5/3}, \qquad k \in [k_0, k_d] \tag{B53}$$

$$f(\eta_k,\tau) = e^{-\frac{\pi}{4}\eta_k^2\tau^2}, \qquad \tilde{f}(\eta_k,\omega) = \frac{2}{\eta_k}e^{-\omega^2/\pi\eta_k^2}, \qquad \eta_k = \frac{1}{\sqrt{2\pi}}\epsilon^{1/3}k^{2/3}, \qquad \eta_k^2 = \frac{\epsilon^{2/3}k^{4/3}}{2\pi} \tag{B54}$$

$$g(k,\omega) = \frac{2E_k}{4\pi k^2\eta_k}\exp\!\left(-\frac{\omega^2}{\pi\eta_k^2}\right) \tag{B55}$$

$$q=\frac{k_1}{k},\qquad s=\frac{u}{k},\qquad \Omega=\frac{\omega}{\epsilon^{1/3}k^{2/3}},\qquad \Omega_1=\frac{\omega_1}{\epsilon^{1/3}k^{2/3}},\qquad g(k,\omega)=\frac{C_k\epsilon^{1/3}k^{-13/3}}{\sqrt{2\pi}}e^{-2\Omega^2} \tag{B56}$$

$$\Phi(q,\Omega_1;s,\Omega-\Omega_1)=q^{-13/3}s^{-13/3}\exp\bigl[-2\Omega_1^2q^{-4/3}-2(\Omega-\Omega_1)^2s^{-4/3}\bigr] \tag{B57}$$

$$H_{ijij}(\mathbf{k},\omega)=\frac{C_k^2\epsilon}{256\pi^8}\,k^{-5}\,\mathfrak{H}(\Omega),\qquad \mathfrak{H}(\Omega)=\int_0^\infty dq\,q\int_{-\infty}^{\infty}d\Omega_1\int_{|1-q|}^{1+q}ds\,s\;\Phi\,\widetilde{\mathcal{K}}(q,s) \tag{B58}$$

$$a=\frac{1}{\pi\eta_{k_1}^2},\qquad b=\frac{1}{\pi\eta_u^2},\qquad \int_{-\infty}^{\infty}d\omega_1\,e^{-a\omega_1^2-b(\omega-\omega_1)^2}=\sqrt{\frac{\pi}{a+b}}\,e^{-ab\omega^2/(a+b)} \tag{B59}$$

$$\frac{ab}{a+b}=\frac{1}{\pi(\eta_{k_1}^2+\eta_u^2)},\qquad \eta_{k_1}^2+\eta_u^2=\frac{\epsilon^{2/3}\bigl(k_1^{4/3}+u^{4/3}\bigr)}{2\pi}\implies e^{-ab\omega^2/(a+b)}=\exp\left(-\frac{2\epsilon^{-2/3}\omega^2}{k_1^{4/3}+u^{4/3}}\right) \tag{B60}$$

$$\int_0^\infty dy\,e^{-Ay^2-B(y-x)^2}=\sqrt{\frac{\pi}{4(A+B)}}\,e^{-ABx^2/(A+B)}\,\mathrm{erfc}\left[-\frac{Bx}{\sqrt{A+B}}\right] \tag{B61}$$

Note (corrected 2026-08-14). The factor $u^{-4/3}$ in the erfc argument is required: without it the argument carries $L^{-4/3}$ rather than being dimensionless. It follows from the Gaussian–erfc identity Eq. (B61) with $A=1/(\pi\eta_{k_1}^2)$, $B=1/(\pi\eta_u^2)$, $x=\omega$. The main-text Eq. (20) and the code (`core.py:integrand_y`, `_fullspectrum_kernel.py`, `band_split_gw.py`) have been brought into line. The numerical effect is $\lesssim 10\%$ on the diagonal and moves no slope: the band-split infrared slopes are +2.996, +2.998, +3.007 after the correction (stable to four decimals against grid refinement). An independent check fixes the sign of the error: the closed-form aeroacoustic limit $H(p\to 0,q)$, derived separately and never edited, is reproduced by the corrected kernel to 1.00000 at every q tested, whereas the uncorrected kernel disagreed with its own limit by up to 8%. [Referee C: the same equation was printed in two different forms]

$$\begin{aligned}H_{ijij}(k,\omega)=\;&\frac{\varepsilon}{24\pi(2\pi)^{3/2}k}\int_{k_0}^{k_d}dk_1\,k_1^{-10/3}\int_{\max(|k_1-k|,k_0)}^{\min(k_1+k,k_d)}du\,u^{-10/3}\,\bigl(k_1^{-4/3}+u^{-4/3}\bigr)^{-1/2}\\&\times\left[27-\tfrac{k^2}{k_1^2}-\tfrac{k^2}{u^2}+\tfrac{k^4}{2k_1^2u^2}+\tfrac{k_1^2}{2u^2}+\tfrac{u^2}{2k_1^2}\right]\\&\times\exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3}+u^{4/3}}\right)\mathrm{erfc}\left(-\frac{\sqrt{2}\,\varepsilon^{-1/3}\omega\,u^{-4/3}}{(k_1^{-4/3}+u^{-4/3})^{1/2}}\right)\end{aligned} \tag{B62}$$

$$M=\frac{1}{c}\left(\frac{\varepsilon}{k_0}\right)^{1/3},\qquad \frac{k_d}{k_0}=R^{3/4},\qquad \varepsilon=c^3M^3k_0,\qquad x=\left(\frac{k_1}{k_0}\right)^{-4/3},\qquad y=\left(\frac{u}{k_0}\right)^{-4/3} \tag{B63}$$

$$dk_1\,k_1^{-10/3}=-\tfrac{3}{4}k_0^{-7/3}x^{3/4}dx,\qquad \bigl(k_1^{-4/3}+u^{-4/3}\bigr)^{-1/2}=k_0^{2/3}(x+y)^{-1/2} \tag{B64}$$

$$y_{\max}(x)=\min\bigl(|x^{-3/4}-p|^{-4/3},\,1\bigr),\qquad y_{\min}(x)=\max\bigl((x^{-3/4}+p)^{-4/3},\,R^{-1}\bigr) \tag{B65}$$

$$\begin{aligned}H[S_{\mathrm{K41}},T_{\mathrm{sw}}](p,q;M,R)=\;&\frac{3M^3k_0^{-4}}{256(2\pi)^{3/2}\,p}\int_{R^{-1}}^{1}dx\,x^{3/4}\int_{y_{\min}}^{y_{\max}}dy\,y^{3/4}\,(x+y)^{-1/2}\\&\times\left[27-p^2\bigl(x^{3/2}+y^{3/2}\bigr)+\tfrac{p^4}{2}x^{3/2}y^{3/2}+\tfrac{1}{2}\bigl(x^{-3/2}y^{3/2}+y^{-3/2}x^{3/2}\bigr)\right]\\&\times\exp\left(-\frac{2xy}{x+y}\frac{q^2}{M^2}\right)\mathrm{erfc}\left(-\frac{\sqrt{2}\,q\,y}{M\sqrt{x+y}}\right)\end{aligned} \tag{B66}$$

$$p\ll M:\quad \exp(\cdot)\to 1,\ \mathrm{erfc}(\cdot)\to\mathrm{const}\implies \Omega_{\mathrm{GW}}\to C\,M^3p^3,\qquad h_c\sim M^{3/2}\sqrt{p} \tag{B67}$$

$$p_{\mathrm{peak}}=1.49\,M \tag{B68}$$

5. $S_{\rm K41} \times T_{\rm dec}$

$$f(\tau) = \Big(1 + \frac{|\tau|}{\tau_k}\Big)^{-2/3}, \qquad \tau_k = \frac{1}{\eta_k} = \frac{\sqrt{2\pi}}{M k_0^{1/3} k^{2/3}}, \qquad \tau_1 = \frac{\sqrt{x}}{M}, \quad \tau_2 = \frac{\sqrt{y}}{M} \tag{B69}$$

$$\hat{g}(\bar{q}) = \int_0^\infty d\sigma\, e^{i\bar{q}\sigma}(1+\sigma)^{-2/3} = e^{-i\bar{q}}(-i\bar{q})^{-1/3}\,\Gamma\big(\tfrac{1}{3}, -i\bar{q}\big) \tag{B70}$$

$$\int d\omega_1\, \hat{g}(\omega_1\tau_1)\hat{g}\big((\omega-\omega_1)\tau_2\big) = \frac{2\pi}{\tau_1\tau_2}\int_0^\infty d\tau\, \cos(\omega\tau)\Big(1+\frac{\tau}{\tau_1}\Big)^{-2/3}\Big(1+\frac{\tau}{\tau_2}\Big)^{-2/3} \tag{B71}$$

$$H[S_{\rm K41}, T_{\rm dec}](p,q;M,R): \quad \text{Eq. (B66) with } \exp(\cdot)\,\mathrm{erfc}(\cdot) \;\longrightarrow\; \text{Eq. (B71)} \tag{B72}$$

$$\tau \to \tau/M \implies \text{Eq. (B71)} = M\,F(\omega/M), \qquad \Omega_{\rm GW}(p;M) = M^4\,p^3\,\Psi\big(p,\,p/M\big) \tag{B73}$$

$$p/M \to 0: \quad \Omega_{\rm GW} \propto M^4, \qquad p_{\rm peak}^{\rm decay} \simeq 2.4 \tag{B74}$$

6. $S_{\delta k}$

$$E(k) = E_0\,\delta(k-k_0), \qquad A(k) = \frac{E_0\,\delta(k-k_0)}{4\pi k^2}, \qquad \tilde{F}_{ij}(\mathbf{k},\omega) = \frac{E_0\delta(k-k_0)}{4\pi k^2}P_{ij}(\hat{k})\tilde{f}(\omega) \tag{B75}$$

$$2\pi\int_{-1}^{1} d\mu\, A(u) = \frac{E_0}{2k\,k_0^2}\,\Theta(2k_0 - k) \tag{B76}$$

$$\begin{aligned}\mathcal{K}_0(p) \equiv \mathcal{K}(k,k_0,k_0) &= \frac{27}{6} + \frac{k^4}{12k_0^4} + \frac{k_0^2}{12k_0^2} + \frac{k_0^2}{12k_0^2} - \frac{k^2}{6k_0^2} - \frac{k^2}{6k_0^2} \\ &= \frac{27+1+1}{6} + \frac{p^4}{12} - \frac{p^2}{3} = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}\end{aligned} \tag{B77}$$

$$H_{ijij}(\mathbf{k},\omega) = \frac{E_0^2\,\Theta(2k_0-k)}{8\pi(2\pi)^8\,k\,k_0^2}\,\mathcal{K}_0(p)\,I_\omega(\omega), \qquad I_\omega(\omega) \equiv \int d\omega_1\,\tilde{f}(\omega_1)\tilde{f}(\omega-\omega_1) \tag{B78}$$

$$\eta_0 \equiv \eta_{k_0} = \frac{Mk_0}{\sqrt{2\pi}}, \qquad \tau_1 = \frac{1}{\eta_0} = \frac{\sqrt{2\pi}}{Mk_0}, \qquad q = \frac{\omega}{k_0}, \qquad \bar{q} \equiv \omega\tau_1 = \frac{\sqrt{2\pi}\,q}{M} \tag{B79}$$

$$\tilde{f}(\omega) = \frac{2}{\eta_0}e^{-\omega^2/\pi\eta_0^2} \implies I_\omega(\omega) = \frac{2\sqrt{2}\,\pi}{\eta_0}e^{-\omega^2/(2\pi\eta_0^2)}, \qquad \frac{\omega^2}{2\pi\eta_0^2} = \frac{q^2}{M^2} \tag{B80}$$

$$\boxed{H[S_{\delta k}, T_{\rm sw}](p,q;M) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\,e^{-q^2/M^2}} \tag{B81}$$

$$\tilde{f}(\omega) = \hat{g}(\omega\tau_1) \implies I_\omega = \frac{1}{\tau_1}\int dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\hat{g}(\bar{q}-q_1)\big] \tag{B82}$$

$$\boxed{H[S_{\delta k},T_{\rm dec}](p,q;M)=\frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\int_{-\infty}^{\infty}dq_1\,\mathrm{Re}\big[\hat g(q_1)\hat g(\bar q-q_1)\big]} \tag{B83}$$

$$\mathcal{C}(\bar q)\equiv\int_{-\infty}^{\infty}d\tau\,\cos(\bar q\tau)\big[(1+|\tau|)^{-2/3}\big]^2=2\,\mathrm{Re}\Big[e^{-i\bar q}(-i\bar q)^{1/3}\,\Gamma\big(-\tfrac{1}{3},-i\bar q\big)\Big] \tag{B84}$$

$$h=f\Theta,\quad f(s)=(1+s)^{-2/3}\implies(\hat g*\hat g)(\bar q)=2\pi\,\mathcal{F}[h^2](\bar q)\implies\int_{-\infty}^{\infty}dq_1\,\mathrm{Re}\big[\hat g(q_1)\hat g(\bar q-q_1)\big]=\pi\,\mathcal{C}(\bar q) \tag{B85}$$

$$\mathcal{C}(0)=6,\qquad\mathcal{C}(\bar q)\xrightarrow[\bar q\to 0]{}6-3\sqrt{3}\,\Gamma(\tfrac{2}{3})\,\bar q^{\,1/3},\qquad\mathcal{C}(\bar q)\xrightarrow[\bar q\to\infty]{}\frac{8}{3\bar q^{\,2}} \tag{B86}$$

$$T(\omega)=2\int_0^{\infty}d\tau\,\cos(\omega\tau)f(\tau)\;\longrightarrow\;-\frac{2f'(0^+)}{\omega^2},\qquad f=(1+|\tau|)^{-4/3}:\;f'(0^+)=-\tfrac{4}{3} \tag{B87}$$

7. $S_{\delta r}$

$$R_{ij}(\mathbf{r},\tau)=v_0^2\,\delta^{(3)}(\mathbf{r})\,P_{ij}\,T(\eta_k,\tau),\qquad\tilde F_{ij}(\mathbf{k},\omega)=v_0^2\,P_{ij}(\hat k)\,\widetilde T(\eta_k,\omega),\qquad A(k)=v_0^2 \tag{B88}$$

$$z=\frac{k_1}{k_0},\quad y=\frac{u}{k_0},\quad p=\frac{k}{k_0},\quad q=\frac{\omega}{k_0},\quad\frac{k_d}{k_0}=R^{3/4},\qquad\eta_k=\frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3} \tag{B89}$$

$$H_{ijij}(\mathbf{k},\omega)=\frac{v_0^4k_0^2}{(2\pi)^7}\int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})}y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,J(\omega;k_1,u),\quad J\equiv\int d\omega_1\,\widetilde T(\eta_{k_1},\omega_1)\widetilde T(\eta_u,\omega-\omega_1) \tag{B90}$$

$$\eta_{k_1}^2+\eta_u^2=\frac{M^2}{2\pi}k_0^{2/3}\big(k_1^{4/3}+u^{4/3}\big)\implies J=\frac{4\pi\sqrt{2\pi}}{Mk_0^{1/3}\sqrt{k_1^{4/3}+u^{4/3}}}\exp\left(-\frac{2\omega^2}{M^2k_0^{2/3}(k_1^{4/3}+u^{4/3})}\right) \tag{B91}$$

$$\boxed{H[S_{\delta r},T_{\rm sw}](p,q;M,R)=\int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})}y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,\big(z^{4/3}+y^{4/3}\big)^{-1/2}e^{-2q^2/[M^2(z^{4/3}+y^{4/3})]}} \tag{B92}$$

$$\tau_k=\frac{1}{\eta_k}=\frac{\sqrt{2\pi}}{Mk_0^{1/3}k^{2/3}},\qquad\tilde J(q;z,y,M)=\int d\omega_1\,\mathrm{Re}\big[\hat g(\omega_1\tau_{k_1})\hat g\big((\omega-\omega_1)\tau_u\big)\big] \tag{B93}$$

$$\boxed{H[S_{\delta r},T_{\rm dec}](p,q;M,R)=\int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})}y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,\tilde J(q;z,y,M)} \tag{B94}$$

$$A^{-2/3}B^{-2/3}=\frac{\Gamma(4/3)}{\Gamma(2/3)^2}\int_0^1\frac{dt}{[t(1-t)]^{1/3}}\big[tA+(1-t)B\big]^{-4/3} \tag{B95}$$

$$A=1+\frac{\tau}{\tau_1},\quad B=1+\frac{\tau}{\tau_2},\qquad\frac{1}{\tau_*(t)}=\frac{t}{\tau_1}+\frac{1-t}{\tau_2} \tag{B96}$$

$$\tilde J=\frac{\pi}{\tau_1\tau_2}\frac{\Gamma(4/3)}{\Gamma(2/3)^2}\int_0^1\frac{dt}{[t(1-t)]^{1/3}}\,\tau_*(t)\,\mathcal{C}\big(\omega\,\tau_*(t)\big) \tag{B97}$$

$$\tau_1=\tau_2:\quad\frac{\Gamma(4/3)}{\Gamma(2/3)^2}B\big(\tfrac{2}{3},\tfrac{2}{3}\big)=1\implies\text{Eq. (B97)}\to\text{Eq. (B85)} \tag{B98}$$

8. $T_{\rm imp}$, $T_{\rm burst}$

$$\Pi^{\rm TT}_{ij}(\mathbf{k},\tau)=A^{\rm TT}_{ij}(\mathbf{k})\,\delta(\tau-\tau_s),\qquad \big\langle A^{\rm TT}_{ij}(\mathbf{k})A^{\rm TT*}_{ij}(\mathbf{k}')\big\rangle=(2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{k}')P(k) \tag{B99}$$

$$\Pi(k;\tau_1,\tau_2)=P(k)\,\delta(\tau_1-\tau_s)\,\delta(\tau_2-\tau_s) \tag{B100}$$

$$\begin{aligned}\frac{d\rho_{\rm GW}}{d\ln k}&=\frac{2Gk^3}{\pi}\int d\tau_1\int d\tau_2\,\cos[k(\tau_1-\tau_2)]\,P(k)\delta(\tau_1-\tau_s)\delta(\tau_2-\tau_s)\\&=\frac{2Gk^3}{\pi}P(k)\cos(0)=\frac{2Gk^3}{\pi}P(k)\end{aligned} \tag{B101}$$

$$P(k)=\frac{1}{(2\pi)^7}\int d^3k_1\,A(k_1)A(u)\,\mathcal{K}(k,k_1,u)=\frac{1}{(2\pi)^6}\int_0^\infty\frac{k_1dk_1}{k}\int_{|k-k_1|}^{k+k_1}u\,du\,A(k_1)A(u)\,\mathcal{K}(k,k_1,u) \tag{B102}$$

$$\int_0^\infty\frac{k_1dk_1}{k}\int_{|k-k_1|}^{k+k_1}u\,du\;\frac{E_0\delta(k_1-k_0)}{4\pi k_1^2}\,\frac{E_0\delta(u-k_0)}{4\pi u^2}=\frac{E_0^2}{16\pi^2}\frac{1}{k_0^2}\frac{1}{k}\,\Theta(2k_0-k) \tag{B103}$$

$$P(k)=\frac{E_0^2\,\mathcal{K}_0(p)}{(2\pi)^6\cdot 16\pi^2\,k\,k_0^2}\,\Theta(2-p) \tag{B104}$$

$$\boxed{H[S_{\delta k},T_{\rm imp}](p)=\frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)=\lim_{q\to0}H[S_{\delta k},T_{\rm sw}](p,q;M)} \tag{B105}$$

$$\Box h_{ij}=16\pi G\,S_{ij},\quad \Box=\partial_t^2-\nabla^2,\qquad \dot h_{ij}(\mathbf{r},t)=4G\int d^3r'\,\frac{\dot S_{ij}(\mathbf{r}',t-|\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|} \tag{B106}$$

$$L(\mathbf{r},\tau)\equiv\frac{1}{32\pi G}\big\langle\dot h_{ij}(\mathbf{r},t)\dot h_{ij}(\mathbf{r},t+\tau)\big\rangle,\qquad I(\mathbf{r},\omega)\equiv\frac{1}{2\pi}\int d\tau\,e^{i\omega\tau}L(\mathbf{r},\tau),\qquad \frac{d\rho_{\rm GW}}{d\ln\omega}=\omega\,I(\mathbf{r},\omega) \tag{B107}$$

$$\frac{(4G)^2}{32\pi G}=\frac{G}{2\pi}:\qquad L(\mathbf{r},\tau)=\frac{G}{2\pi}\int\frac{d^3r_1d^3r_2}{|\mathbf{r}-\mathbf{r}_1||\mathbf{r}-\mathbf{r}_2|}\big\langle\dot S_{ij}(\mathbf{r}_1,t')\dot S_{ij}(\mathbf{r}_2,t'')\big\rangle,\qquad \begin{aligned}t'&=t-|\mathbf{r}-\mathbf{r}_1|\\t''&=t+\tau-|\mathbf{r}-\mathbf{r}_2|\end{aligned} \tag{B108}$$

$$\big\langle\dot S_{ij}(\mathbf{r}_1,t')\dot S_{ij}(\mathbf{r}_2,t'')\big\rangle=-\partial_\tau^2\big\langle S_{ij}(\mathbf{r}_1,t')S_{ij}(\mathbf{r}_2,t'')\big\rangle\implies I(\mathbf{r},\omega)=\frac{G\omega^2}{(2\pi)^2}\int\frac{d^3r_1d^3r_2}{|\mathbf{r}-\mathbf{r}_1||\mathbf{r}-\mathbf{r}_2|}\int d\tau\,e^{i\omega\tau}\big\langle S_{ij}S_{ij}\big\rangle \tag{B109}$$

$$S_{ij}=\mathcal{S}_{ij}(\mathbf{r})\delta(t-t_0),\qquad \Sigma(\mathbf{r}_1,\mathbf{r}_2)=\big\langle\mathcal{S}_{ij}(\mathbf{r}_1)\mathcal{S}_{ij}(\mathbf{r}_2)\big\rangle,\qquad \big\langle e^{i\boldsymbol{\omega}\cdot\mathbf{r}}\big\rangle=\frac{\sin(\omega r)}{\omega r} \tag{B110}$$

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega}=8G\,\omega^2\int_0^\infty dr\,r\,\sin(\omega r)\,\Sigma(r)} \tag{B111}$$

$$S_{ij}(\mathbf{r},t)=\mathcal{S}_{ij}(\mathbf{r})\,g(t-t_0),\qquad \tilde g(\omega)=\int dt\,g(t)e^{i\omega t} \tag{B112}$$

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega}=8G\,\omega^2\,|\tilde g(\omega)|^2\int_0^\infty dr\,r\,\sin(\omega r)\,\Sigma(r)} \tag{B113}$$

$$\boxed{\omega_{\rm peak}\simeq\min\Big(\frac{1}{\ell},\,\frac{1}{\Delta t}\Big)} \tag{B114}$$

9. Band-split source

$$E(k)=E_0\times\begin{cases}(k/k_0)^s, & k_{\rm IR}\le k<k_0,\\ (k/k_0)^{-5/3}, & k_0\le k\le k_d,\\ 0, & \text{otherwise},\end{cases}\qquad s=4,\qquad \frac{k_0}{k_{\rm IR}}=R_{\rm IR},\qquad \frac{k_d}{k_0}=R^{3/4} \tag{B115}$$

$$A(k)=\frac{E(k)}{4\pi k^2},\qquad A_{\rm Kol}(k)\propto k^{-11/3}\implies \mathcal{S}(k)\equiv\frac{A(k)}{A_{\rm Kol}(k)}=\Big(\frac{k}{k_0}\Big)^{s+5/3}\Theta(k_0-k)+\Theta(k-k_0) \tag{B116}$$

$$\mathcal{S}_{\rm IR}(k)=\mathcal{S}(k)\,\Theta(k_0-k),\qquad \mathcal{S}_{\rm in}(k)=\mathcal{S}(k)\,\Theta(k-k_0),\qquad \mathcal{S}=\mathcal{S}_{\rm IR}+\mathcal{S}_{\rm in} \tag{B117}$$

$$k_1=k_0x^{-3/4},\qquad u=k_0y^{-3/4},\qquad H[\mathcal{S}_b]:\quad \text{Eq. (B66) integrand}\longrightarrow \text{integrand}\times\mathcal{S}_b(k_1)\,\mathcal{S}_b(u) \tag{B118}$$

$$\mathcal{S}_b(k_1)\mathcal{S}_b(u)=0 \text{ by zeroing the integrand, not by narrowing } [y_{\min},y_{\max}] \text{ of Eq. (B65)} \tag{B119}$$

$$\mathcal{S}=\mathcal{S}_{\rm IR}+\mathcal{S}_{\rm in}\implies H[\mathcal{S}]=H[\mathcal{S}_{\rm IR}]+H[\mathcal{S}_{\rm in}]+2H_\times,\qquad H_\times:\ \mathcal{S}_{\rm IR}(k_1)\mathcal{S}_{\rm in}(u) \tag{B120}$$

$$p\to0:\quad u\to k_1\implies P_T(0)=\int d^3q\,P_M(q)^2\ \neq\ f\big(s\big),\ \neq f\big(R_{\rm IR}\big) \tag{B121}$$

$$\implies H[\mathcal{S}_b](p\to0)\to\text{const}\implies \Omega_{\rm GW}\propto p^3\qquad \text{for } b\in\{\text{IR, in, IR}+\text{in}\} \tag{B122}$$

$$\operatorname{supp}\mathcal{S}_{\rm IR}\subset[k_{\rm IR},k_0)\implies \operatorname{supp}H[\mathcal{S}_{\rm IR}]\subset[0,2k_0)\implies\Theta(2-p) \tag{B123}$$

$$M=0.5,\ R=10^4,\ R_{\rm IR}=10^2,\qquad n[\Omega_{\rm GW}]\big|_{2\times10^{-3}\le p\le2\times10^{-2}}=\begin{cases}3.00, & b=\text{IR},\\ 3.01, & b=\text{in},\\ 3.02, & b=\text{IR}+\text{in},\end{cases} \tag{B124}$$

$$\frac{H[\mathcal{S}_{\rm in}]}{H[\mathcal{S}_{\rm IR}]}\bigg|_{p=3\times10^{-3}}=1.26,\qquad \frac{H[\mathcal{S}]}{H[\mathcal{S}_{\rm IR}]+H[\mathcal{S}_{\rm in}]}=\begin{cases}1.004, & p=3\times10^{-3},\\ 1.46, & p\simeq0.5,\end{cases} \tag{B125}$$

$$R_{\rm IR}:10\to300\implies \Delta\Omega_{\rm GW}\big|_{p=5\times10^{-3}}=0.4\%;\qquad s:4\to2\implies n[\Omega_{\rm GW}]=3.01,\quad \Delta\Omega_{\rm GW}=\times2.7 \tag{B126}$$

Appendix C: Monochromatic and White-Noise Spectra

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function) distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k-space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. III: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω, (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u=|\mathbf{k}-\mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

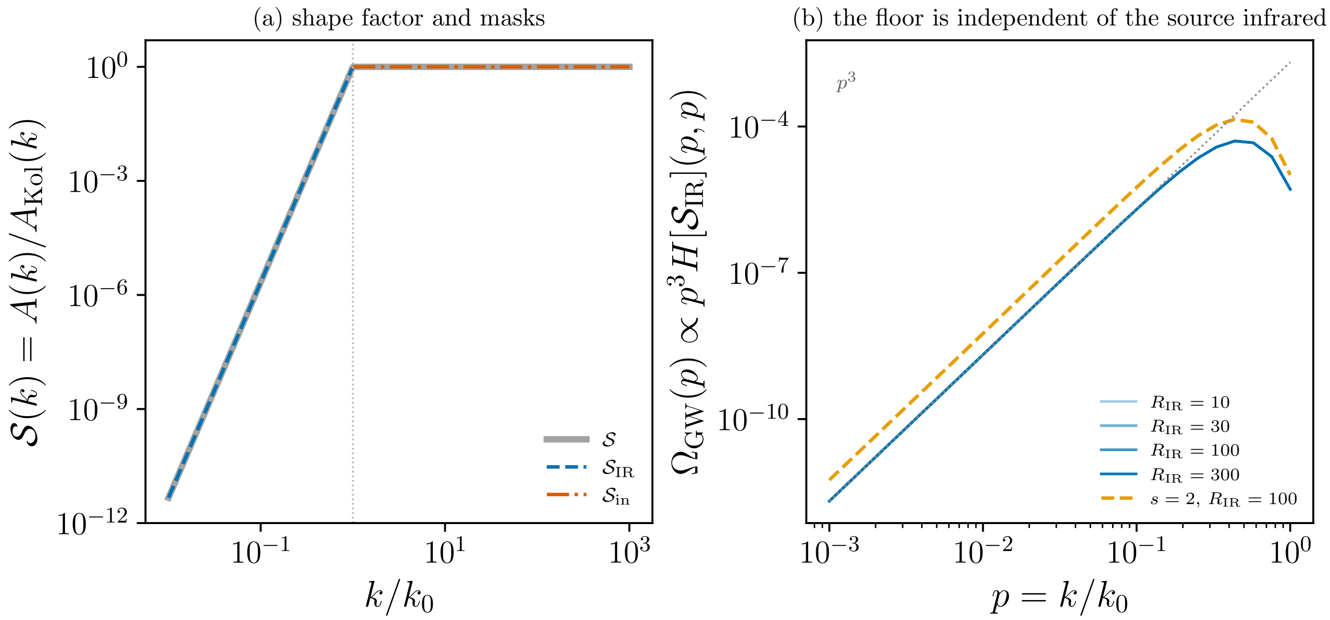


FIG. 20. The two ingredients of the band-split chain. *(a)* The shape factor $\mathcal{S}$ of Eq. (B116) and its masks $\mathcal{S}_{\rm IR}, \mathcal{S}_{\rm in}$ of Eq. (B117). The masks have disjoint support and sum to $\mathcal{S}$, so each traces it exactly on its own side of k_0 (dotted vertical); this is what makes the decomposition Eq. (B120) exact rather than approximate. *(b)* Eq. (B121) made visible: the infrared band alone, $\Omega_{\rm GW} \propto p^3 H[\mathcal{S}_{\rm IR}](p,p)$, at four infrared bandwidths $R_{\rm IR} = 10, 30, 100, 300$ and at two infrared slopes $s = 4$ (Batchelor) and $s = 2$ (Saffman), with $M = 0.5$, $R = 10^4$. The four bandwidth curves are drawn separately but are not distinguishable: they agree to 0.31% at $p = 5 \times 10^{-3}$ despite spanning 1.5 decades of $R_{\rm IR}$, because the floor $P_T(0) = \int d^3q\, P_M(q)^2$ is fixed by the modes at the top of the band and is blind to how far the band extends. Changing the infrared slope itself moves the amplitude by only 2.71 and leaves the exponent at 3.005 against 3.001, although the two inputs differ by four decades at the bottom of the band. Dotted guide: p^3. Produced by `Notebooks/band_split_masks.py`.

1. Monochromatic energy spectrum $E(k) = E_0\delta(k - k_0)$

Set $E(k) = E_0\, \delta(k - k_0)$ so that $A(k) = E_0\, \delta(k - k_0)/(4\pi k^2)$ and the spectral tensor reads

$$\tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0\, \delta(k - k_0)}{4\pi k^2}\, P_{ij}(\hat{k})\, \tilde{f}(\omega), \qquad \int_0^\infty E(k)\, dk = E_0, \tag{C1}$$

with $\tilde{f}(\omega)$ to be specified by the decorrelation model.

Starting from Eq. (15) in spherical coordinates, the radial integral collapses $k_1 \to k_0$ via $\delta(k_1 - k_0)$, fixing $u = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$. Trading μ for u (azimuth contributes 2π) and using $\delta(u - k_0)$ from $A(u)$ yields the closed-form spatial reduction

$$2\pi \int_{-1}^{1} d\mu\, A(u) = \frac{E_0}{2kk_0^2}\, \Theta(2k_0 - k), \tag{C2}$$

with the Heaviside factor enforcing the triangle inequality $p \equiv k/k_0 \le 2$. The geometric kernel at $k_1 = u = k_0$ evaluates to

$$\mathcal{K}_0(p) \equiv \mathcal{K}(k, k_0, k_0) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}. \tag{C3}$$

Normalisation (corrected 2026-08-14). The radial collapse contributes $\int_0^\infty k_1^2 dk_1\, A(k_1) = E_0/(4\pi)$, which was previously applied as $\times E_0$; the displayed prefactor below was therefore 4π too large. A second, independent factor of 2 was lost when substituting into Eq. (C5), making that equation 8π too large. Both are now corrected. The result $\int d^3k_1\, A(k_1)A(u)\,\mathcal{K} = E_0^2\mathcal{K}_0(p)\Theta(2k_0 - k)/(8\pi k k_0^2)$ agrees with the independent collapse of Eq. (D10), which was right all along—the two displays had disagreed with each other. The boxed $\mathfrak{H}$ is a shape and carries no part of this normalisation, so no slope, peak, edge, Mach scaling, figure or line of code changes; only the *absolute* amplitude of the monochromatic branch, which matters as soon as a detector forecast is attempted. [Referee A wants a LISA/PTA forecast, which needs absolute amplitudes]

Substituting Eqs. (C2)–(C3) in Eq. (15),

$$H_{ijij}(\mathbf{k},\omega) = \frac{E_0^2\,\Theta(2k_0-k)}{8\pi(2\pi)^8\,kk_0^2}\,\mathcal{K}_0(p)\,I_\omega(\omega), \qquad I_\omega(\omega) \equiv \int d\omega_1\,\tilde{f}(\omega_1)\tilde{f}(\omega-\omega_1). \tag{C4}$$

All spatial structure has been resolved in closed form; only the temporal convolution I_ω remains. Because both wavevectors are pinned to k_0 by the two δ-functions, there is no leftover wavenumber integration: the only spatial constraint is the M-independent triangle inequality $\Theta(2-p)$. The Mach number therefore cannot reshape any spatial integration *bounds* here; it enters solely through the temporal decorrelation rate at the fixed scale k_0, which we tie to M via $\eta_{k_0} = \frac{M}{\sqrt{2\pi}}k_0$ (cf. Eq. (21)). This contrasts with the white-noise case of Sec. C 2, where M does enter the surviving wavenumber integral.

a. *Kraichnan decorrelation*

For $\tilde{f}(\omega) = (2/\eta_0)e^{-\omega^2/\pi\eta_0^2}$ with $\eta_0 \equiv \eta_{k_0} = Mk_0/\sqrt{2\pi}$, the Gaussian convolution gives $I_\omega(\omega) = (2\sqrt{2}\pi/\eta_0)e^{-\omega^2/(2\pi\eta_0^2)}$. With $q \equiv \omega/k_0$ the exponent becomes $\omega^2/(2\pi\eta_0^2) = q^2/M^2$, so Eq. (C4) separates into a dimensional prefactor and the boxed dimensionless kernel

$$H_{ijij}(\mathbf{k},\omega) = \frac{\sqrt{2}\,E_0^2}{8\pi(2\pi)^7\,kk_0^2\,\eta_0}\,\mathcal{K}_0(p)\,\Theta(2-p)\,e^{-q^2/M^2}, \qquad \boxed{H[S_{\delta k},T_{\rm sw}](p,q;M) \;\equiv\; \mathfrak{H}^{(\delta k)}_{\rm Kraichnan}(p,q;M) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\,e^{-q^2/M^2}.} \tag{C5}$$

There is no remaining integral. The p-shape is fixed; M acts only as the Gaussian width in q, so the GW spectrum peaks at $q \sim M$.

b. *Decay model*

For decaying turbulence, $\tilde{f}(\omega) = \hat{g}(\omega\tau_1)$ with $\hat{g}(\bar{q}) = e^{-i\bar{q}}(-i\bar{q})^{-1/3}\Gamma(\frac{1}{3},-i\bar{q})$ and $\tau_1 = 1/\eta_0 = \sqrt{2\pi}/(Mk_0)$ the decay time at scale k_0. With $q = \omega/k_0$ the natural temporal argument is $\bar{q} \equiv \omega\tau_1 = \sqrt{2\pi}\,q/M$; changing variables in Eq. (C4), the temporal factor becomes $I_\omega = (1/\tau_1)\int dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\hat{g}(\bar{q}-q_1)\big]$. The real part is essential and not cosmetic: the convolution is genuinely complex (its imaginary part exceeds its real part already at $\bar{q} \simeq 1$), and it is the real part that carries the spectrum. The dimensionless result is

$$\boxed{H[S_{\delta k},T_{\rm dec}](p,q;M) \;\equiv\; \mathfrak{H}^{(\delta k)}_{\rm decay}(p,q;M) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\int_{-\infty}^{\infty} dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\,\hat{g}(\bar{q}-q_1)\big], \qquad \bar{q} = \frac{\sqrt{2\pi}\,q}{M}.} \tag{C6}$$

The spatial part is fully resolved by the two δ-functions; only a 1-D frequency integral remains, and M enters it solely through the rescaled frequency $\bar{q}$.

a. The frequency convolution in closed form. Contrary to what the singular appearance of $\hat{g}$ suggests, this convolution *does* have a closed form, and evaluating it as a written q_1 integral is both unnecessary and numerically treacherous. The GW source is quadratic in the velocity, so under quasi-normal closure the object the spectrum needs is the transform of the *squared* two-time correlation,

$$\mathcal{C}(\bar{q}) \;\equiv\; \int_{-\infty}^{\infty} d\tau\,\cos(\bar{q}\tau)\,\big[(1+|\tau|)^{-2/3}\big]^2 \;=\; 2\,\mathrm{Re}\Big[e^{-i\bar{q}}(-i\bar{q})^{1/3}\,\Gamma\big(-\tfrac{1}{3},-i\bar{q}\big)\Big], \tag{C7}$$

which is the *same* closed form as $\hat{g}$ with the power $2/3 \to 4/3$ — squaring the correlation simply shifts the exponent. Writing $h = f\Theta$ for the one-sided correlation $f(s) = (1+s)^{-2/3}$, the convolution theorem gives $(\hat{g}*\hat{g})(\bar{q}) = 2\pi\,\mathcal{F}[h^2](\bar{q})$, whose real part halves into the one-sided cosine integral. Hence the exact identity

$$\int_{-\infty}^{\infty} dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\hat{g}(\bar{q}-q_1)\big] \;=\; \pi\,\mathcal{C}(\bar{q}). \tag{C8}$$

Two consequences are worth stating explicitly. First, the one-sided/two-sided bookkeeping that distinguishes $\hat{g}$ from the two-sided kernel enters only as the constant π: it carries *no* shape information and therefore cannot bend a spectral slope. Second, the limits of $\mathcal{C}$ are elementary,

$$\mathcal{C}(0)=6,\qquad \mathcal{C}(\bar q)\xrightarrow[\bar q\to 0]{} 6-3\sqrt{3}\,\Gamma(\tfrac{2}{3})\,\bar q^{\,1/3},\qquad \mathcal{C}(\bar q)\xrightarrow[\bar q\to\infty]{}\frac{8}{3\bar q^{\,2}},\tag{C9}$$

the last following from the cusp of f^2 at $\tau=0$. This $\bar q^{-2}$ ultraviolet tail is the physically decisive one: it is a *heavy power law*, in contrast to the Gaussian cutoff of the stationary Kraichnan model, and it is what allows the decaying GW spectrum to extend past the sweeping cutoff so that its peak is pinned at the source scale (Sec. III E).

2. Real-space delta-function correlations

Take the spatial part of R_{ij} to be a $\delta^{(3)}(\mathbf{r})$, with velocity variance v_0^2 and temporal decorrelation $T(\eta_k,\tau)$:

$$R_{ij}(\mathbf{r},\tau)=v_0^2\,\delta^{(3)}(\mathbf{r})\,P_{ij}\,T(\eta_k,\tau),\qquad \tilde F_{ij}(\mathbf{k},\omega)=v_0^2\,P_{ij}(\hat k)\,\widetilde T(\eta_k,\omega),\tag{C10}$$

so $A(k)=v_0^2$ is k-independent (i.e. $E(k)\propto k^2$, ultraviolet-dominated); the integrals must be cut off between k_0 and k_d.

a. Physical relevance: an uncorrelated magnetic field. This model is not merely a mathematical limit. It is the natural description of a *spatially uncorrelated* source—a magnetic field with no correlation beyond a microscopic scale, $\langle B_i(\mathbf{r}_1)B_j(\mathbf{r}_2)\rangle\propto\delta^{(3)}(\mathbf{r}_1-\mathbf{r}_2)$, which is exactly the kind of field left by an uncorrelated causal process at a phase transition. A natural question, and one the user of such a model should confront, is how a source that is uncorrelated at large scales can radiate a *correlated* gravitational-wave signal there at all. The answer is that the GW source is not B but the *quadratic* stress $T_{ij}\sim B_iB_j$, whose two-point function is a *convolution* of two field spectra [Eqs. (13)–(14)]. The convolution of two structureless (white) spectra is not white: as $k\to 0$ its value tends to the finite constant $\int d^3q\,P_B(q)^2$, so the stress—and hence the GW spectrum—is smooth and analytic in the infrared even though the field is not. Large-scale correlation of the *signal* is manufactured by the quadratic coupling, and its correlation length is set by the convolution, not by the (vanishing) correlation length of B. This is the same $1/p$ pole and shell–shell convolution that governs the universal infrared theorem (Eq. (46)), and it is why a causal (Batchelor $E_M\propto k^4$) magnetic field yields a GW spectrum whose stress cannot be steeper than white noise—the mechanism [28] invoke for the flat plateau of their Fig. 1. Read in the opposite direction, the same convolution is what forbids the analytic substitution $\Omega_{\rm T}\propto\Omega_{\rm M}$ that Sec. V identifies as the single faulty step behind the infrared discrepancy: this subsection is the $k\to 0$ limit of that argument, taken with the steepest possible field.

Crucially, we adopt the same k-dependent sweeping rate as in Sec. III, $\eta_k=\frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3}=\frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, so that $\widetilde T$ depends on the wavenumber and the temporal factor no longer decouples from the spatial integral. With $A(k_1)=A(u)=v_0^2$ in Eq. (13) and dimensionless variables $z=k_1/k_0$, $y=u/k_0$, $p=k/k_0$, $q=\omega/k_0$, $R^{3/4}=k_d/k_0$,

$$H_{ijij}(\mathbf{k},\omega)=\frac{v_0^4k_0^2}{(2\pi)^7}\int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})}y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,J(\omega;k_1,u),\qquad J(\omega;k_1,u)\equiv\int d\omega_1\,\widetilde T(\eta_{k_1},\omega_1)\widetilde T(\eta_u,\omega-\omega_1),\tag{C11}$$

where $\widetilde{\mathcal{K}}(p,z,y)=\mathcal{K}(k,k_1,u)|_{k_1=zk_0,u=yk_0}$ and the frequency factor J now carries the k_1,u dependence inherited from η_{k_1},η_u.

a. Kraichnan decorrelation

With $\widetilde T(\eta_k,\omega)=(2/\eta_k)e^{-\omega^2/\pi\eta_k^2}$ the convolution combines two Gaussians of *different* widths η_{k_1},η_u; using $\int d\omega_1\,e^{-a\omega_1^2}e^{-b(\omega-\omega_1)^2}=\sqrt{\pi/(a+b)}\,e^{-ab\omega^2/(a+b)}$ with $a=1/\pi\eta_{k_1}^2$, $b=1/\pi\eta_u^2$, and $\eta_{k_1}^2+\eta_u^2=\frac{M^2}{2\pi}k_0^{2/3}(k_1^{4/3}+u^{4/3})$, the frequency factor becomes

$$J(\omega;k_1,u)=\frac{4\pi\sqrt{2\pi}}{Mk_0^{1/3}\sqrt{k_1^{4/3}+u^{4/3}}}\,\exp\Big(-\frac{2\,\omega^2}{M^2k_0^{2/3}(k_1^{4/3}+u^{4/3})}\Big).\tag{C12}$$

Inserting this into Eq. (C11) and absorbing the constants into the dimensional prefactor gives the boxed dimensionless kernel

$$\boxed{H[S_{\delta r},T_{\rm sw}](p,q;M,R)\;\equiv\;\mathfrak{H}^{(\delta r)}_{\rm Kraichnan}(p,q;M,R)=\int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})}y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,(z^{4/3}+y^{4/3})^{-1/2}\,e^{-2q^2/[M^2(z^{4/3}+y^{4/3})]}.}\tag{C13}$$

The exponential now acts as an M-dependent effective cutoff on the wavenumber integration: contributions require $z^{4/3} + y^{4/3} \gtrsim 2q^2/M^2$, so as M grows the effective wavenumber support widens and the GW spectrum peak shifts. This is precisely the structure of the Gogoberidze exponent $\exp[-2\varepsilon^{-2/3}\omega^2/(k_1^{4/3}+u^{4/3})]$ in Eq. (20); the k-dependent sweeping rate, rather than a fixed η_1, is what makes M control the integration support.

b. Decay model

For decaying turbulence $\widetilde{T}(\eta_k, \omega) = \hat{g}(\omega\tau_k)$ with the same $\hat{g}(q) = e^{-iq}(-iq)^{-1/3}\Gamma(\frac{1}{3}, -iq)$ as Sec. C 1 b, but now with the k-dependent decay time $\tau_k = 1/\eta_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$. The temporal factor is the convolution $J(\omega; k_1, u) = \int d\omega_1\, \mathrm{Re}\big[\hat{g}(\omega_1\tau_{k_1})\hat{g}((\omega-\omega_1)\tau_u)\big]$ which, because τ_{k_1}, τ_u depend on the wavenumbers, must remain *inside* the spatial integral. Writing $\tilde{J}(q; z, y, M)$ for this convolution with the per-mode dimensionless frequencies fixed by $\tau_{k_1}, \tau_u \propto 1/M$, the boxed result is

$$\boxed{H[S_{\delta r}, T_{\mathrm{dec}}](p, q; M, R) \;\equiv\; \mathfrak{H}^{(\delta r)}_{\mathrm{decay}}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z\,dz}{p} \int_{\max(|p-z|,1)}^{\min(p+z, R^{3/4})} y\,dy\, \widetilde{\mathcal{K}}(p, z, y)\, \tilde{J}(q; z, y, M).} \tag{C14}$$

The decay rates scale as $\eta_k \propto Mk_0^{1/3}k^{2/3}$, so M again sets the effective wavenumber support. Unlike the fixed-η_0 case, the three integrals no longer factorize: the k-dependence of the decay rate couples the frequency convolution to the wavenumber integration, mirroring the behavior of Eq. (20).

a. Reduction of the two-scale factor. The two-scale convolution $\tilde{J}$ admits no elementary closed form, but it reduces exactly to a one-dimensional integral over the single-scale form Eq. (C7), and it is evaluated that way throughout. By the convolution theorem it is first the transform of the time-domain *product*, Eq. (35); a Feynman parameter then collapses the product of two $-2/3$ powers onto a single $-4/3$ power,

$$A^{-2/3}B^{-2/3} = \frac{\Gamma(4/3)}{\Gamma(2/3)^2} \int_0^1 \frac{dt}{[t(1-t)]^{1/3}} \big[tA + (1-t)B\big]^{-4/3}, \tag{C15}$$

so that, with $A = 1 + \tau/\tau_1$, $B = 1 + \tau/\tau_2$ and $1/\tau_*(t) = t/\tau_1 + (1-t)/\tau_2$,

$$\int_{-\infty}^{\infty} d\omega_1\, \mathrm{Re}\big[\hat{g}(\omega_1\tau_1)\hat{g}((\omega-\omega_1)\tau_2)\big] = \frac{\pi}{\tau_1\tau_2} \frac{\Gamma(4/3)}{\Gamma(2/3)^2} \int_0^1 \frac{dt}{[t(1-t)]^{1/3}}\, \tau_*(t)\, \mathcal{C}\big(\omega\,\tau_*(t)\big). \tag{C16}$$

The t-integrand is smooth and non-oscillatory ($\mathcal{C}$ is positive and monotonically decreasing), and its endpoint singularities are exactly the Gauss–Jacobi weight, so a fixed 48-node rule is converged to ten digits; at $\tau_1 = \tau_2$ it collapses back to Eq. (C8), since $[\Gamma(4/3)/\Gamma(2/3)^2]\, B(2/3, 2/3) = 1$. This matters numerically as well as formally: evaluated as a written oscillatory integral the amplitude varies on τ_1, τ_2 while the cosine varies on $1/\omega$, and at small ω those scales are separated by many decades, so standard oscillatory quadratures either fail outright or return silently incorrect values. Eq. (C16) contains no oscillatory quadrature at all.

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k-k_0)$	Kraichnan	Eq. (C5): closed form; M in frequency scale only
$E_0\delta(k-k_0)$	Decay	Eq. (C6): 1D frequency integral; M in $\bar{q}$
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (C13): 2D wavevector integral; M-cutoff in bounds
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (C14): 2D wavevector $\times$ nested 1D frequency; M-cutoff
$E_0\delta(k-k_0)$	Impulsive $\delta(\tau-\tau_s)$	Eq. (D13): closed form; no M or q dependence

Appendix D: Impulsive (δ-in-Time) Source

We now consider a source that fires exactly once: the turbulent stress is active only at a single conformal time $\tau = \tau_s$ but remains monochromatic in wavenumber, $E(k) = E_0\,\delta(k-k_0)$. This idealisation approximates a combustion front, a rapid phase transition, or any process whose active lifetime is negligible compared with the GW oscillation period $2\pi/k$. Because the temporal part of the UETC is an exact double δ-function, the time integrals in the master formula collapse *without approximation*, yielding a GW spectrum that is purely spatial in character and entirely free of Mach-number or decorrelation-time parameters.

1. Source model and UETC

The stress tensor of the impulsive source factorises as

$$\Pi_{ij}^{\rm TT}(\mathbf{k},\tau) = A_{ij}^{\rm TT}(\mathbf{k})\,\delta(\tau-\tau_s), \tag{D1}$$

where $A_{ij}^{\rm TT}(\mathbf{k})$ carries the spatial amplitude of the burst and is independent of time. The two-point function of the TT stress defines the UETC via

$$\big\langle \Pi_{ij}^{\rm TT}(\mathbf{k},\tau_1)\,\Pi_{ij}^{\rm TT\,*}(\mathbf{k}',\tau_2)\big\rangle = (2\pi)^3\,\delta^{(3)}(\mathbf{k}-\mathbf{k}')\,\Pi(k;\tau_1,\tau_2). \tag{D2}$$

Inserting the ansatz (D1) and defining the spatial power $P(k)$ through $\langle A_{ij}^{\rm TT}(\mathbf{k})\,A_{ij}^{\rm TT\,*}(\mathbf{k}')\rangle = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{k}')P(k)$, the UETC becomes

$$\Pi(k;\tau_1,\tau_2) = P(k)\,\delta(\tau_1-\tau_s)\,\delta(\tau_2-\tau_s). \tag{D3}$$

No decorrelation model (η_k, $\hat{g}$, or any finite coherence time) is needed; the UETC is exact and requires no further closure assumption.

2. Temporal integral and master formula

The Isaacson GW energy density $\rho_{\rm GW} = \langle \dot{h}_{ij}\dot{h}_{ij}\rangle/(32\pi G)$, together with the retarded solution

$$\dot{h}_{ij}(\mathbf{k},\tau) = 16\pi G\int_{\tau_i}^{\tau} d\tau'\,\cos[k(\tau-\tau')]\,\Pi_{ij}^{\rm TT}(\mathbf{k},\tau'), \tag{D4}$$

and time-averaging the fast-oscillation product $\cos[k(\tau-\tau_1)]\cos[k(\tau-\tau_2)] \to \frac{1}{2}\cos[k(\tau_1-\tau_2)]$, gives the master formula

$$\frac{d\rho_{\rm GW}}{d\ln k} \simeq \frac{2G\,k^3}{\pi}\int d\tau_1\int d\tau_2\,\cos[k(\tau_1-\tau_2)]\,\Pi(k;\tau_1,\tau_2). \tag{D5}$$

Substituting the impulsive UETC (D3),

$$\begin{aligned}\frac{d\rho_{\rm GW}}{d\ln k} &= \frac{2G\,k^3}{\pi}\int d\tau_1\int d\tau_2\,\cos[k(\tau_1-\tau_2)]\;P(k)\,\delta(\tau_1-\tau_s)\,\delta(\tau_2-\tau_s)\\ &= \frac{2G\,k^3}{\pi}\,P(k)\,\cos[k(\tau_s-\tau_s)] \;=\; \frac{2G\,k^3}{\pi}\,P(k)\,\cos(0) \;=\; \frac{2G\,k^3}{\pi}\,P(k).\end{aligned} \tag{D6}$$

The temporal factor is identically unity regardless of k, k_0, or τ_s: both δ-functions collapse to set $\tau_1=\tau_2=\tau_s$, so the cosine argument vanishes. No approximation is involved. The GW spectrum is therefore determined entirely by the instantaneous spatial power $P(k)$ of the burst, with no temporal modulation and no on-shell frequency integral.

3. Spatial reduction for monochromatic $E(k) = E_0\,\delta(k-k_0)$

We now compute $P(k)$ for the monochromatic energy spectrum. Adopting the isotropic, incompressible velocity model $F_{ij}(\mathbf{k}) = A(k)P_{ij}(\hat{k})$ with $P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i\hat{k}_j$, the spatial power of the TT-projected quadratic source reads

$$P(k) = \frac{1}{(2\pi)^7}\int d^3k_1\,A(k_1)\,A(u)\,\mathcal{K}(k,k_1,u), \qquad \mathbf{u}\equiv\mathbf{k}-\mathbf{k}_1,\quad u\equiv|\mathbf{u}|, \tag{D7}$$

with the geometric kernel

$$\mathcal{K}(k,k_1,u) = \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2}. \tag{D8}$$

For the monochromatic spectrum $E(k) = E_0\,\delta(k-k_0)$, the amplitude is $A(k) = E_0\,\delta(k-k_0)/(4\pi k^2)$. Going to spherical coordinates $d^3k_1 = k_1^2\,dk_1\,d\phi\,d\mu$ with $\mu = \hat{k}\cdot\hat{k}_1$, the azimuthal integral gives 2π; trading μ for $u = \sqrt{k^2+k_1^2-2kk_1\mu}$ (so $u\,du = kk_1\,d\mu$),

$$P(k) = \frac{1}{(2\pi)^6}\int_0^\infty \frac{k_1\,dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\,du\;A(k_1)\,A(u)\,\mathcal{K}(k,k_1,u). \tag{D9}$$

Inserting $A(k_1) = E_0\delta(k_1-k_0)/(4\pi k_1^2)$, the k_1 integral collapses to $k_1 = k_0$; inserting $A(u) = E_0\delta(u-k_0)/(4\pi u^2)$, the u integral collapses to $u = k_0$, which is inside the range $[|k-k_0|, k+k_0]$ only when $k \le 2k_0$, i.e. $p \le 2$. The angular residue from $\delta(u-k_0)$ is $1/(du/d\mu)|_{u=k_0} = u/(kk_1)|_{u=k_0} = k_0/(kk_0) = 1/k$, so the two-dimensional integration reduces to

$$\int_0^\infty \frac{k_1\,dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\,du\;\frac{E_0\,\delta(k_1-k_0)}{4\pi k_1^2}\cdot\frac{E_0\,\delta(u-k_0)}{4\pi u^2} = \frac{E_0^2}{16\pi^2}\cdot\frac{1}{k_0^2}\cdot\frac{1}{k}\,\Theta(2k_0-k). \tag{D10}$$

With both momenta pinned at k_0, the geometric kernel evaluates to

$$\begin{aligned}\mathcal{K}(k,k_0,k_0) &= \frac{27}{6} + \frac{k^4}{12k_0^4} + \frac{k_0^2}{12k_0^2} + \frac{k_0^2}{12k_0^2} - \frac{k^2}{6k_0^2} - \frac{k^2}{6k_0^2}\\ &= \frac{27}{6} + \frac{p^4}{12} + \frac{1}{12} + \frac{1}{12} - \frac{p^2}{6} - \frac{p^2}{6}\\ &= \frac{27+1+1}{6} + \frac{p^4}{12} - \frac{p^2}{3} \;=\; \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12} \;\equiv\; \mathcal{K}_0(p).\end{aligned} \tag{D11}$$

Substituting (D10) and (D11) into (D9) gives

$$P(k) = \frac{E_0^2\,\mathcal{K}_0(p)}{(2\pi)^6\cdot 16\pi^2\,k\,k_0^2}\,\Theta(2-p). \tag{D12}$$

Inserting into (D6) and collecting the dimensional prefactor into an overall constant, we read off the dimensionless kernel

$$\boxed{\mathfrak{H}^{(\delta\tau,\delta k)}(p) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p), \qquad \mathcal{K}_0(p) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}.} \tag{D13}$$

There is no remaining integral, no frequency argument q, and no Mach number M: the kernel is a single algebraic expression in p. In p-shape it coincides with the Kraichnan monochromatic kernel of Sec. III evaluated at $q = 0$ — setting $q \to 0$ in e^{-q^2/M^2} trivially reproduces (D13) — because the impulsive source assigns equal weight to every on-shell frequency.

4. GW spectrum: peak, IR and UV slopes

Using $\Omega_{\rm GW}(p) \propto p^3\,\mathfrak{H}^{(\delta\tau,\delta k)}(p)$,

$$\Omega_{\rm GW}(p) \;\propto\; p^2\,\mathcal{K}_0(p)\,\Theta(2-p). \tag{D14}$$

a. Infrared slope. As $p \to 0$, $\mathcal{K}_0(p) = \frac{14}{3} + O(p^2)$ is constant, so

$$\Omega_{\rm GW}(p) \;\xrightarrow{p\to 0}\; \frac{14}{3}\,p^2. \tag{D15}$$

The infrared slope is p^2, not the causal p^3 of spatially extended sources. The extra power of $1/p$ in $\mathfrak{H}$ traces back to the angular residue in (D10): pinning both loop momenta to k_0 means the phase-space measure for small k grows as k rather than k^2, which demotes $k^3 \to k^2$.

b. Peak. Define $f(p) \equiv p^2\,\mathcal{K}_0(p) = \frac{14}{3}\,p^2 - \frac{p^4}{3} + \frac{p^6}{12}$. Its derivative is

$$f'(p) = p\left(\frac{28}{3} - \frac{4}{3}\,p^2 + \frac{p^4}{2}\right). \tag{D16}$$

Setting the bracket to zero requires $3p^4 - 8p^2 + 56 = 0$, a quadratic in p^2 with discriminant $\Delta = 64 - 4\cdot 3\cdot 56 = -608 < 0$: no real roots. Since the bracket is $56/3 > 0$ at $p = 0$ and $72 > 0$ at $p = 2$, and has no roots on $(0, 2)$, $f'(p) > 0$ throughout the support. The spectrum is strictly increasing and peaks at the triangle-inequality cutoff,

$$p_{\rm peak} = 2 \qquad \Longrightarrow \qquad k_{\rm GW}^{\rm peak} = 2k_0. \tag{D17}$$

Geometrically, the maximum GW wavenumber from two modes both at k_0 is $k_0 + k_0 = 2k_0$; the monotone kernel drives all power to this kinematic limit.

c. Ultraviolet. The Heaviside factor $\Theta(2 - p)$ enforces an exact hard cutoff: $\Omega_{\rm GW}(p) \equiv 0$ for $p > 2$. There is no UV power-law tail, in contrast to the smooth Gaussian (Kraichnan) or power-law (decay) falloffs of the stationary models.

5. Post-burst free propagation

For $\tau > \tau_s$ the source is off. The retarded Green's function gives

$$h_{ij}(\mathbf{k}, \tau) = \frac{16\pi G}{k} \int_{\tau_i}^{\tau} d\tau'\, \sin[k(\tau - \tau')]\, \Pi_{ij}^{\rm TT}(\mathbf{k}, \tau'). \tag{D18}$$

Substituting $\Pi_{ij}^{\rm TT}(\mathbf{k}, \tau') = A_{ij}^{\rm TT}(\mathbf{k})\, \delta(\tau' - \tau_s)$,

$$h_{ij}(\mathbf{k}, \tau) = \frac{16\pi G}{k}\, \sin[k(\tau - \tau_s)]\, A_{ij}^{\rm TT}(\mathbf{k}), \qquad \tau > \tau_s, \tag{D19}$$

with time derivative $\dot h_{ij}(\mathbf{k}, \tau) = 16\pi G \cos[k(\tau - \tau_s)]\, A_{ij}^{\rm TT}(\mathbf{k})$. The Isaacson energy density

$$\rho_{\rm GW} = \frac{\langle \dot h_{ij} \dot h_{ij}\rangle}{32\pi G} = \frac{(16\pi G)^2}{32\pi G} \cos^2[k(\tau - \tau_s)]\, P(k) \xrightarrow{\langle\cos^2\rangle = 1/2} \frac{(16\pi G)\, k^3}{2\pi}\, P(k)\, \frac{d\ln k}{1}, \tag{D20}$$

after integrating over k-shells, recovers the frozen spectrum (D14) without further time evolution. Each mode oscillates on-shell at $\omega = k$ with amplitude set at τ_s; the GW background decouples from the source permanently and carries a spectrum that never changes.

Appendix E: Impulsive source in real space

We keep every space integral in configuration space and transform in time alone. The GW spectrum comes out as a single frequency ω acting on a purely spatial stress correlator. With $a = 1$,

$$\Box\, h_{ij} = 16\pi G\, S_{ij}, \quad \Box = \partial_t^2 - \nabla^2; \qquad \dot h_{ij}(\mathbf{r}, t) = 4G \int d^3 r'\, \frac{\dot S_{ij}(\mathbf{r}', t - |\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|}. \tag{E1}$$

Following the standard turbulence–GW formalism [1, 2], introduce the unequal-time GW correlator at lag τ and its temporal transform,

$$L(\mathbf{r}, \tau) \equiv \frac{1}{32\pi G} \big\langle \dot h_{ij}(\mathbf{r}, t)\, \dot h_{ij}(\mathbf{r}, t + \tau) \big\rangle, \qquad \rho_{\rm GW}(\mathbf{r}) = L(\mathbf{r}, 0), \tag{E2}$$

$$I(\mathbf{r}, \omega) \equiv \frac{1}{2\pi} \int d\tau\, e^{i\omega\tau}\, L(\mathbf{r}, \tau), \qquad \rho_{\rm GW}(\mathbf{r}) = \int d\omega\, I(\mathbf{r}, \omega), \tag{E3}$$

so $I(\mathbf{r}, \omega)$ is the spectral energy density of the radiation and $d\rho_{\rm GW}/d\ln\omega = \omega\, I(\mathbf{r}, \omega)$ (at $\omega > 0$). Inserting the retarded field (E1) twice ($(4G)^2/32\pi G = G/2\pi$),

$$L(\mathbf{r}, \tau) = \frac{G}{2\pi} \int \frac{d^3 r_1\, d^3 r_2}{|\mathbf{r} - \mathbf{r}_1|\, |\mathbf{r} - \mathbf{r}_2|} \big\langle \dot S_{ij}(\mathbf{r}_1, t')\, \dot S_{ij}(\mathbf{r}_2, t'') \big\rangle, \quad t' = t - |\mathbf{r} - \mathbf{r}_1|, \quad t'' = t + \tau - |\mathbf{r} - \mathbf{r}_2|. \tag{E4}$$

For turbulence the stress obeys the identity [2]

$$\big\langle \dot S_{ij}(\mathbf{r}_1,t')\,\dot S_{ij}(\mathbf{r}_2,t'')\big\rangle = -\partial_\tau^2\,\big\langle S_{ij}(\mathbf{r}_1,t')\,S_{ij}(\mathbf{r}_2,t'')\big\rangle, \tag{E5}$$

so under the transform (E3) the two time-derivatives integrate by parts in τ to a factor ω^2,

$$I(\mathbf{r},\omega) = \frac{G\,\omega^2}{(2\pi)^2}\int \frac{d^3r_1\,d^3r_2}{|\mathbf{r}-\mathbf{r}_1|\,|\mathbf{r}-\mathbf{r}_2|}\int d\tau\, e^{i\omega\tau}\,\big\langle S_{ij}(\mathbf{r}_1,t')\,S_{ij}(\mathbf{r}_2,t'')\big\rangle. \tag{E6}$$

A single burst $S_{ij} = \mathcal{S}_{ij}(\mathbf{r})\,\delta(t-t_0)$ gives $\langle S_{ij}(\mathbf{r}_1,t')S_{ij}(\mathbf{r}_2,t'')\rangle = \Sigma(\mathbf{r}_1,\mathbf{r}_2)\,\delta(t'-t_0)\,\delta(t''-t_0)$ with $\Sigma(\mathbf{r}_1,\mathbf{r}_2) = \langle \mathcal{S}_{ij}(\mathbf{r}_1)\mathcal{S}_{ij}(\mathbf{r}_2)\rangle$; the δ's fix the retarded shells (so t_0 sets only the shell radius, not the spectral shape), the τ-integral leaves the phase $e^{-i\omega(|\mathbf{r}-\mathbf{r}_1|-|\mathbf{r}-\mathbf{r}_2|)}$, and with $\mathbf{r}=0$ ($|\mathbf{r}-\mathbf{r}_a| = r_a$), the isotropic correlator $\Sigma(r)$ ($r = |\mathbf{r}_1-\mathbf{r}_2|$), and the angular average $\langle e^{i\boldsymbol{\omega}\cdot\mathbf{r}}\rangle = \sin(\omega r)/(\omega r)$,

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega} = 8G\,\omega^2\int_0^\infty dr\,r\,\sin(\omega r)\,\Sigma(r).} \tag{E7}$$

The leading ω^2 is exactly the two time-derivatives of the quadrupole source, converted by the identity (E5).

1. The peak is set by the source, not by t_0

The firing time enters (E6) only through the δ's, which fix the radiating shell but leave no t_0 in the real spectrum (E7): *the spectrum is independent of* t_0. An instantaneous burst nonetheless radiates a broadband, structured spectrum, fixed entirely by the spatial stress correlator $\langle \mathcal{S}_{ij}(\mathbf{r}_1)\mathcal{S}_{ij}(\mathbf{r}_2)\rangle$. Let ℓ be its coherence length (the correlator decays for $|\mathbf{r}_1-\mathbf{r}_2| \gtrsim \ell$). The retardation phase $\cos[\omega(r_1-r_2)]$ stays coherent across that volume while $\omega\ell \lesssim 1$ and averages away above it, so the radial integral is white in the infrared and the causal rise $d\rho_{\rm GW}/d\ln\omega \propto \omega^3$ turns over at

$$\boxed{\omega_{\rm peak} \simeq \frac{1}{\ell}\quad (\text{independent of } t_0),\qquad f_{\rm peak} = \frac{\omega_{\rm peak}}{2\pi} \simeq \frac{1}{2\pi\ell}.} \tag{E8}$$

The peak GW frequency is the inverse coherence length of the burst, with no reference to when it fired.

2. Burst of finite duration Δt

Now let the stress switch on over a finite interval rather than an instant, $S_{ij}(\mathbf{r},t) = \mathcal{S}_{ij}(\mathbf{r})\,g(t-t_0)$, with g a temporal profile of width Δt (the instantaneous burst is $\Delta t \to 0$, $g \to \delta$). The source correlator in (E6) now carries $g(t'-t_0)\,g(t''-t_0)$ instead of the two δ's, so the τ-transform multiplies the spectrum by the single temporal filter $|\tilde g(\omega)|^2$, $\tilde g(\omega) = \int dt\, g(t)e^{i\omega t}$,

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega} = 8G\,\omega^2\,|\tilde g(\omega)|^2\int_0^\infty dr\,r\,\sin(\omega r)\,\Sigma(r).} \tag{E9}$$

The factor $|\tilde g(\omega)|^2$ is flat for $\omega\Delta t \lesssim 1$ and rolls off above — a low-pass at $\omega \sim 1/\Delta t$. The spectrum now carries two scales, the spatial coherence ($\omega \sim 1/\ell$) and the duration ($\omega \sim 1/\Delta t$), and peaks at the lower of the two,

$$\boxed{\omega_{\rm peak} \simeq \min\!\Big(\frac{1}{\ell},\,\frac{1}{\Delta t}\Big).} \tag{E10}$$

A slow burst ($\Delta t \gtrsim \ell$) is duration-limited, $\omega_{\rm peak} \simeq 1/\Delta t$; a fast burst ($\Delta t \lesssim \ell$) recovers the instantaneous, coherence-limited peak $\omega_{\rm peak} \simeq 1/\ell$ of (E8).

3. Single-scale limit: deriving the quadrupole factor of two

The peak tracks whatever scale the source correlations carry; the cleanest limit is a fluid coherent at a single length scale ℓ_0. Here the factor of two relating the GW scale to the fluid scale is not an assumption — it follows from the source being the *quadratic* Reynolds stress $\mathcal{S}_{ij} = w\, v_i v_j$, through Wick's theorem. For a Gaussian velocity field the connected stress two-point function is quadratic in the velocity correlator $R_{ab}(\mathbf{r}) = \langle v_a(\mathbf{r}_1) v_b(\mathbf{r}_2)\rangle$ (the disconnected pairing is the constant mean stress $\langle \mathcal{S}_{ij}\rangle^2$ and does not radiate):

$$\Sigma(\mathbf{r}) = \big\langle \mathcal{S}_{ij}(\mathbf{r}_1)\mathcal{S}_{ij}(\mathbf{r}_2)\big\rangle_{\rm c} = w^2\Big[R_{aa}(\mathbf{r})R_{bb}(\mathbf{r}) + \tfrac{1}{3}R_{ab}(\mathbf{r})R_{ab}(\mathbf{r})\Big] \;\propto\; R(\mathbf{r})^2, \tag{E11}$$

the TT-contracted form of Eq. (12); the key feature is that Σ is the *square* of the velocity correlator. A fluid coherent at the single scale ℓ_0 has a velocity correlator that oscillates on that scale: for the isotropic single-scale field the trace is $R_{aa}(r) \propto \sin(r/\ell_0)/(r/\ell_0)$.[72] Squaring *halves the scale*,

$$R(r)^2 \;\propto\; \frac{\sin^2(r/\ell_0)}{(r/\ell_0)^2} = \frac{1-\cos(2r/\ell_0)}{2\,(r/\ell_0)^2}, \tag{E12}$$

so the radiating part of $\Sigma(r)$ oscillates on the halved scale $\ell_0/2$ (the constant term is the non-radiating mean). The GW frequency is the inverse of the scale it samples, so the spectrum $d\rho_{\rm GW}/d\ln\omega \propto \omega^3 P(\omega)|\tilde{g}(\omega)|^2$ collapses to a single line at

$$\boxed{\omega_{\rm GW} = \frac{2}{\ell_0} \qquad \big(\ell_{\rm GW} = \tfrac{1}{2}\,\ell_0\big).} \tag{E13}$$

The factor of two is thus *derived*: a quadratic (quadrupole) source $v_i v_j$ radiates on half the scale — twice the frequency — of the field that drives it, Eq. (E12) being that halving written as the identity $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$. There is no infrared ω^3 rise and no ultraviolet falloff: the ω^3 phase space and the duration filter $|\tilde{g}(\omega)|^2$ fix only the line's height, so $\omega_{\rm GW}$ is independent of both the firing time t_0 and the duration Δt. This is the configuration-space image of the GW peak at half the source scale found for the monochromatic source in Eq. (D17): two velocity features of scale ℓ_0 overlap to produce stress structure on the scale $\ell_0/2$. (A strictly single-scale field in fact fills all scales $\geq \ell_0/2$ — the autoconvolution in Eq. (E11) — piling up at the $\ell_0/2$ edge as in Sec. D; the δ-line keeps only the halved scale.)

Appendix F: Numerical results: $f^{\rm GW}_{\rm peak}(\tau_*, M_0)$

We scan the peak GW frequency $f^{\rm GW}_{\rm peak} = {\rm argmax}_\omega[\omega\,\Omega_{\rm GW}(\omega)]/2\pi$ over the (τ_*, M_0) plane for a delta-shell decaying source evolved with the BK2016 self-similar closure and time-integrated over the emission window (`Notebooks/f_peak_vs_tau_mac`
Two temporal-correlation kernels are compared through their dimensionless transform $\mathcal{G}(q) = \widehat{R^2}(q)$: the random-phase BK2016 form (Model A) and the coherent eddy (Model B).

Across all four self-similar classes the ratio $f^A_{\rm peak}/f^B_{\rm peak}$ is *independent of* (τ_*, M_0): the self-similar closure makes both the eddy frequency ω_e and the sweeping rate η_0 scale as $(1+T/\tau_*)^{-1}$, so the two kernels inherit the same overall frequency scaling and differ only in their fixed dimensionless peak position. The model difference therefore collapses to a single per-class constant (Fig. 21): $f_A/f_B \simeq 0.37$, 0.36 for the two HD classes (Loitsiansky; the class realised in the simulations of Ref. [39]), 0.40 for non-helical MHD governed by the Hosking integral, and 0.78 for helical MHD—Model A always peaks lower. ~~0.83, 0.78 for the MHD (nonhelical, helical) classes~~ [recomputed after the Hosking swap: the old 0.83 belonged to the superseded $\beta = 1$, $q = 1/2$ class] The offset comes from the side-peak of the coherent kernel at the eddy frequency ($q \simeq \pm 2$, the $\cos\sigma$ oscillation of R_B), which wins the $\omega\,\Omega_{\rm GW}$ argmax for the steeper-decay classes but nearly coincides with the broad Model-A peak for helical MHD. Note that with the Hosking class in place the split is no longer HD-versus-MHD: at 0.40 the non-helical MHD class is indistinguishable from the two HD classes, and it is *helical* MHD alone that stands apart. Because the per-model maps differ only by this constant, we display the ratio rather than the eight individual heatmaps.

Appendix G: Differences among the models

With every kernel of Table I in hand we can now say precisely what distinguishes them, and—more usefully—which of the two axes of Sec. II D each difference belongs to. The summary is Table X; the reasoning follows. Throughout, $\Omega_{\rm GW}(p) \propto p^3 H[S,T](p,p)$ and all numbers are measured by `Notebooks/model_grid_survey.py` at $M = 1$, $R = 10^4$.

a. The infrared asymptote belongs to the spatial axis alone. Every kernel with a spatially extended source gives the causal k^3, and every kernel with a monochromatic source gives k^2, *irrespective of the temporal model*: we measure 3.00 for $H[S_{\rm K41}, T_{\rm sw}]$ and 2.97 for $H[S_{\rm K41}, T_{\rm dec}]$, against 2.00 and 1.97 for the two δk kernels and 2.00 for the impulsive one. This is not a coincidence of our kernels but the universal theorem of Sec. V C: the infrared is fixed by whether the source spectrum has finite width, and a δ-function source violates that hypothesis by exactly one power through the $1/p$ pole of $\mathcal{K}_0(p)/p$. The temporal factor cannot contribute, because as $p \to 0$ it tends to a constant—$\pi\,\mathcal{C}(0) = 6\pi$ for $T_{\rm dec}$ by Eq. (C9), unity for $T_{\rm sw}$—and a constant carries no slope. *Corollary:* the temporal axis cannot move the *asymptotic* infrared slope. For a stationary UETC this is immediate from the constant $p \to 0$ limit just quoted; for a source of finite duration it follows from Eq. (9), whose temporal factor is likewise flat below $1/\Delta t$; and in general it is the causality theorem of [27]. This is why no treatment in Table III reproduces the simulated k^1 of Sec. V. What the temporal history *does* change is the intermediate band: a finite duration Δt contributes $4\sin^2(k\Delta t/2)/(\Delta t\, k^2)$, flat below $1/\Delta t$ and falling as $2/(\Delta t\, k^2)$ above it, so two powers are lost in $1/\Delta t \ll k \ll k_0$ and none in the limit [9, 25, 26]. The simulated k^1 lies in that band, not in the limit.

b. The ultraviolet belongs to the temporal axis alone. This is not new with us. It is the central result of the aeroacoustic analysis of Refs. [68, 69], where the frequency spectrum of sound radiated by isotropic turbulence is shown to be fixed by the Eulerian space–time correlation rather than by the energy spectrum, and it is applied to the cosmological problem in Ref. [24]. Here the roles reverse, and the discriminant is sharp: it is whether the UETC has a *cusp* at zero lag. By Eq. (37) the temporal factor falls as $-2f'(0^+)/\omega^2$, so

- $T_{\rm sw}$ (Gaussian, $f'(0^+) = 0$): no cusp, and the cutoff is super-exponential. Fitting a power law to it is meaningless—the apparent "slope" merely records where the fit window was placed, which is why $H[S_{\delta r}, T_{\rm sw}]$ returns -46.6 and $H[S_{\rm K41}, T_{\rm sw}]$ returns -4.73 on the same window.
- $T_{\rm dec}$ (power law, $f'(0^+) = -4/3$): a genuine heavy tail $\propto \omega^{-2}$, with the coefficient $8/3$ of Eq. (C9). The measured GW slope is mildly M-dependent (-1.65 at $M = 0.05$ rising to -1.28 at the formal $M = 3$) because the tail sets in only above $q \sim M$.
- $T_{\rm imp}$, $T_{\rm burst}$: white and window-modulated respectively, the latter carrying the $\sin^2$ factor of Eq. (47).

This single dichotomy—cusp or no cusp—is what the caveat of Sec. IV B turns on, and it is worth emphasising that it is *not* a matter of degree: the two behaviours differ by fourteen orders of magnitude by $q = 4$.

c. The peak is where the two axes compete. The peak is the one property neither axis owns outright, which is why it has been the most contested quantity in the literature. With $T_{\rm sw}$ the spectrum is cut off at the sweeping frequency and the peak sits at $p \simeq 1.49M$ (measured 1.53 at $M = 1$), *below* the source scale for subsonic turbulence. With $T_{\rm dec}$ the heavy tail survives past that cutoff, and the peak is instead pinned by the spatial structure at $p \simeq 2.3$, essentially independent of M (measured 2.27, 2.27, 2.31, 2.31 at $M = 0.1$–0.8, drifting to 2.66 by the formal $M = 1.6$). ~~(measured 2.32 for $M \leq 1$, drifting to 2.79 by the formal $M = 3$)~~ [the quoted end-points are not produced by the cited script, whose Mach sweep stops at $M = 1.6$; replaced by the values it does produce] The coarse survey grid used for Table X resolves a peak only to about 15% and returns 2.70 at $M = 1$, consistent with the finer sweep and with the 2.4 of Eq. (36) at that resolution; the peak of this kernel should not be read to three digits from any of them. The two temporal models therefore disagree about the peak by a factor $\sim 1/M$, which for the subsonic simulations is an order of magnitude—this is exactly the Gogoberidze-versus-Roper Pol tension of Sec. IV. Note that the δk and δr kernels have no dynamical peak at all: the former is truncated at $p = 2$ by the triangle inequality, and the latter, having $E \propto k^2$, is ultraviolet-dominated and simply rises to the dissipation cutoff. Their tabulated "peaks" are cutoffs, not physics.

d. Positivity is a property of the temporal axis, and not automatic. The temporal factor must be non-negative at every frequency, since it is the power spectrum of a real correlation; a UETC that fails this yields negative gravitational-wave energy. This is not a hypothetical failure mode—it is the defect that led Caprini et al. to discard a Gaussian straining-time model in favour of a top-hat, and that motivates the positive-definite Gibbs kernel of [23]. Both temporal models used here are safe, and for $T_{\rm dec}$ provably so: $(1+s)^{-4/3}$ is completely monotone, hence by Bernstein's theorem a superposition of decaying exponentials, whose cosine transforms $2a/(a^2+\omega^2)$ are individually positive; products of completely monotone functions are completely monotone, so the two-scale factor Eq. (C16) inherits the property. We verify $\mathcal{C} > 0$ over $10^{-8} < q < 10^4$.

1. The Mach exponent of the peak as a regime diagnostic

The peak's dependence on the Mach number deserves separating out, because it is the one place where the two temporal models make a *qualitatively* different prediction rather than a quantitatively different one. Measuring $d\ln p_{\rm peak}/d\ln M$ at fixed k_0 over $M \in [0.1, 1.6]$ (the upper end formal, $M > 1$ being superluminal, but included to pin the exponent) gives

Temporal model	$d\ln p_{\rm peak}/d\ln M$
$T_{\rm sw}$ (Kraichnan sweeping)	+0.970
$T_{\rm dec}$ (BK2016 power law)	+0.048
$T_{\rm imp}$ (δ-in-time)	0.000

a clean split between unity and zero. The mechanism is transparent: $T_{\rm sw}$ multiplies the spectrum by e^{-q^2/M^2}, a cutoff at $q \sim M$, so the peak simply tracks that cutoff; $T_{\rm dec}$ and $T_{\rm imp}$ impose no such frequency scale, so the peak falls back on the spatial structure. Strikingly, the integrated energy is nearly *blind* to this distinction—we measure $E_{\rm tot} \propto M^{5.8}$ and $M^{5.6}$ for the two—so it is the peak, not the amplitude, that carries the information.

a. What is and is not new here. Both halves of this contrast are already in the literature, and neither is ours. Gogoberidze et al. [2] state that the peak frequency scales linearly with the characteristic velocity, i.e. $f_{\rm peak} \propto M k_0$—our stationary row, of which the coefficient 1.49 of Sec. IV is a quantitative refinement. Auclair et al. [23] find instead that the peak is pinned to the integral scale independently of the velocity—our decaying and impulsive rows. What appears not to have been said is that the two sit on a single axis and that the exponent itself is therefore a diagnostic: [23] in particular does not contrast the velocity dependence between decorrelating and coherent sources.

b. What the diagnostic actually diagnoses. It is tempting to read the table as evidence for the power-law UETC specifically, and that reading is wrong. The impulsive kernel contains no M anywhere, so its exponent is identically zero while having no heavy tail, no power law, and indeed no temporal structure at all; and [23] obtain a velocity-independent peak while using Kraichnan *sweeping*, because in a freely decaying flow the integral scale itself evolves. Velocity-independence of the peak is therefore a property of a whole class of sources, not a signature of Eq. (C7). The exponent answers "does the temporal factor impose a cutoff at $q \sim M$?"—not "which UETC is it?". Read that way it is a genuine diagnostic; read as support for the decaying model in particular it proves much less than it appears to, and the peak-based argument of Sec. IV B should be weighted accordingly.

c. The aeroacoustic precedent. That the Mach scaling is regime-dependent is classical, and the relevant distinction is precisely a temporal one. Lighthill's eighth-power law [42] ($P \propto U^8$, fixed velocity and geometry) and Proudman's [43] ($P \propto \rho\varepsilon M^5$, isotropic turbulence at fixed dissipation rate) differ in what is held fixed—the same ambiguity that flips our own peak scaling between $1.49M$ at fixed k_0 and $1.49/M^2$ at fixed ε. More pointedly, Proudman's M^5 rests on *neglecting retarded-time differences*, replacing the retarded covariance of the stress by the simultaneous one—that is, on treating the source as temporally coherent—and Lilley [73] later rederived the result retaining them. The coherent versus decorrelating axis that separates our two rows is thus the Proudman–Lilley question transposed into a cosmological setting, and our integrated $M^{5.6}$–$M^{5.8}$ sits near Proudman's M^5. The contribution here is not the existence of a regime dependence but its sharpening into a spectral observable: an exponent one reads off the peak rather than off the total power. (The two integrated exponents are approximate; the stationary kernel emits convergence warnings over this sweep.)

d. Analytic structure. Finally, the models differ in what must actually be computed, and the reduction is larger than the usual presentation suggests. The decaying kernels are conventionally written as frequency convolutions of the singular $\hat{g}$, which is both expensive and numerically treacherous; by Eqs. (C8) and (C16) the single-scale case is closed-form and the two-scale case is a single smooth Gauss–Jacobi quadrature over the closed form. The decaying kernels are therefore no harder than the Gaussian ones.

Kernel	IR slope	Peak p	UV	Positive	Literature counterpart
$H[S_{\rm K41}, T_{\rm sw}]$	3.00	$1.49M$	super-exp	yes	Gogoberidze et al. [2]
$H[S_{\rm K41}, T_{\rm dec}]$	2.97	$\simeq 2.4$	ω^{-2}	yes	Roper Pol et al. [28] (peak)
$H[S_{\delta k}, T_{\rm sw}]$	2.00	cutoff	none	yes	—
$H[S_{\delta k}, T_{\rm dec}]$	1.97	cutoff	none	yes	—
$H[S_{\delta r}, T_{\rm sw}]$	3.00	cutoff	super-exp	yes	—
$H[S_{\delta r}, T_{\rm dec}]$	2.97	cutoff	ω^{-2}	yes	—
$H[S_{\delta k}, T_{\rm imp}]$	2.00	cutoff	none	yes	—

TABLE X. Systematic comparison of the kernels of Table I, measured at $M = 1$, $R = 10^4$ on the diagonal $q = p$ by `Notebooks/model_grid_survey.py`. The infrared slope tracks the *spatial* axis only (3 for extended sources, 2 for monochromatic ones, per Sec. V C); the ultraviolet tracks the *temporal* axis only (a power-law tail iff the UETC has a cusp at zero lag, Eq. (37)); both statements are about the asymptotes, and every intermediate feature, the peak included, depends on both. ~~only the peak depends on both.~~ "cutoff" denotes models with no dynamical peak—the δk kernels are truncated at $p = 2$ by the triangle inequality and the δr kernels are ultraviolet-dominated. "super-exp" denotes a Gaussian cutoff, for which no power-law slope exists.

Appendix H: Spectrum survey for all derived models

In this section we plot, on a common convention, the dimensionless GW energy-density spectrum

$$\Omega_{\rm GW}(p) \;\propto\; p^3\, H(p,p), \qquad p = k_{\rm GW}/k_0, \tag{H1}$$

along the sound-cone diagonal $q = p$ for every model derived in this paper, in the uniform notation $H[S,T]$ of Sec. II D. The p^2 factor relative to the characteristic strain $h_c^2(p) = p\,H(p,p)$ is the phase-space measure d^3k; together with causality and a source of finite spectral width it predicts $\Omega_{\rm GW} \sim p^3$ in the IR, followed by a peak and a decay in the UV. A dotted grey line shows the p^3 reference.

These plots are the graphical form of Table X, and should be read with its two organising facts in mind: the infrared slope is fixed by the *spatial* model alone and the ultraviolet by the *temporal* model alone. In particular the monochromatic panels sit at p^2 rather than p^3 not because of anything in their temporal treatment but because a zero-width source carries a $1/p$ pole, $\mathcal{K}_0(p)/p$, which is exactly the δ-function exception to the universal infrared theorem reproduced in Sec. V C. The build log of `Notebooks/all_spectra.py` flags cases whose IR slope departs from 3 by more than ± 0.5; the monochromatic models are expected to be flagged, and are not anomalies.

1. All nine kernels side by side

Figure 25 places every entry of Table I on one figure, at matched parameters, with the abscissa in units of the initial-power peak k_0 throughout. The comparison is organised by a single control common to all nine panels, the source coherence time in units of the light-crossing time of the source scale,

$$\hat\tau \;\equiv\; \tau_c\, k_0, \qquad \omega\tau_c\big|_{\omega=k} = p\,\hat\tau, \tag{H2}$$

so that the dimensionless temporal argument of every kernel is the same function $p\hat\tau$ of the plotted abscissa. For the sweeping models $\tau_c = \tau_{k_0} = \sqrt{2\pi}/(Mk_0)$, i.e. $M = \sqrt{2\pi}/\hat\tau$; for the finite burst $\tau_c = \Delta t$, giving the low-pass $|\tilde g|^2 = e^{-p^2\hat\tau^2}$ of Eq. (E9). The stationary and time-independent kernels ($T_{\rm sw}$, $T_{\rm imp}$) carry a single line at $\hat\tau = 1$; the time-dependent ones ($T_{\rm dec}$, $T_{\rm burst}$) are shown at $\hat\tau = 0.01, 0.1, 1, 10, 100$.

Three things are visible at a glance, and each is the graphical form of a statement made analytically above.

The infrared is set by the spatial model alone. Every $S_{\rm K41}$, $S_{\rm full}$ and $S_{\delta r}$ panel sits on p^3 (2.91–3.00 measured) regardless of which temporal model multiplies it and regardless of $\hat\tau$, while every $S_{\delta k}$ panel sits on p^2 (1.91–2.00). The one power of difference is the $1/p$ pole of $\mathcal{K}_0(p)/p$ carried by a zero-width source, the δ-function exception to the universal infrared theorem of Sec. V C. Changing $\hat\tau$ by four decades moves the infrared amplitude but never its slope, which is Eq. (B26) restated: as $p \to 0$ the temporal factor tends to a constant [$\mathcal{C}(0) = 6$, Eq. (B86)] and cannot alter the infrared scaling.

The ultraviolet is set by the temporal model alone. Only the three $T_{\rm dec}$ panels have a power-law tail, inherited from the $\bar q^{-2}$ cusp tail of Eq. (C9); the sweeping and burst kernels cut off as Gaussians, for which a fitted log-log slope is a property of the fitting window and not of the model, and is therefore reported as "Gaussian" rather than as a number. The monochromatic panels terminate at $p = 2$ on the triangle inequality $\Theta(2 - p)$ instead.

The peak responds to both, and in opposite ways. At $\hat\tau = 1$ the peaks cluster at $p_{\rm pk} = 2.0$–3.2, i.e. at the source scale. Lengthening the coherence time drives them apart: the $T_{\rm burst}$ peak slides down as $p_{\rm pk} \simeq 1/\hat\tau$, the duration-limited branch of Eq. (E10), whereas the $T_{\rm dec}$ peaks stay pinned near the source scale because the heavy $\bar q^{-2}$ tail keeps supplying power at $\omega = k$ (Sec. IV B). $H[S_{\delta r}, T_{\rm dec}]$ has no peak at all in the plotted range: a white spatial source is ultraviolet dominated by construction ($E \propto k^2$) and its maximum is set by the dissipation cutoff $R^{3/4}$, not by any dynamical scale, which is why the panel is labelled UV-limited rather than assigned a peak.

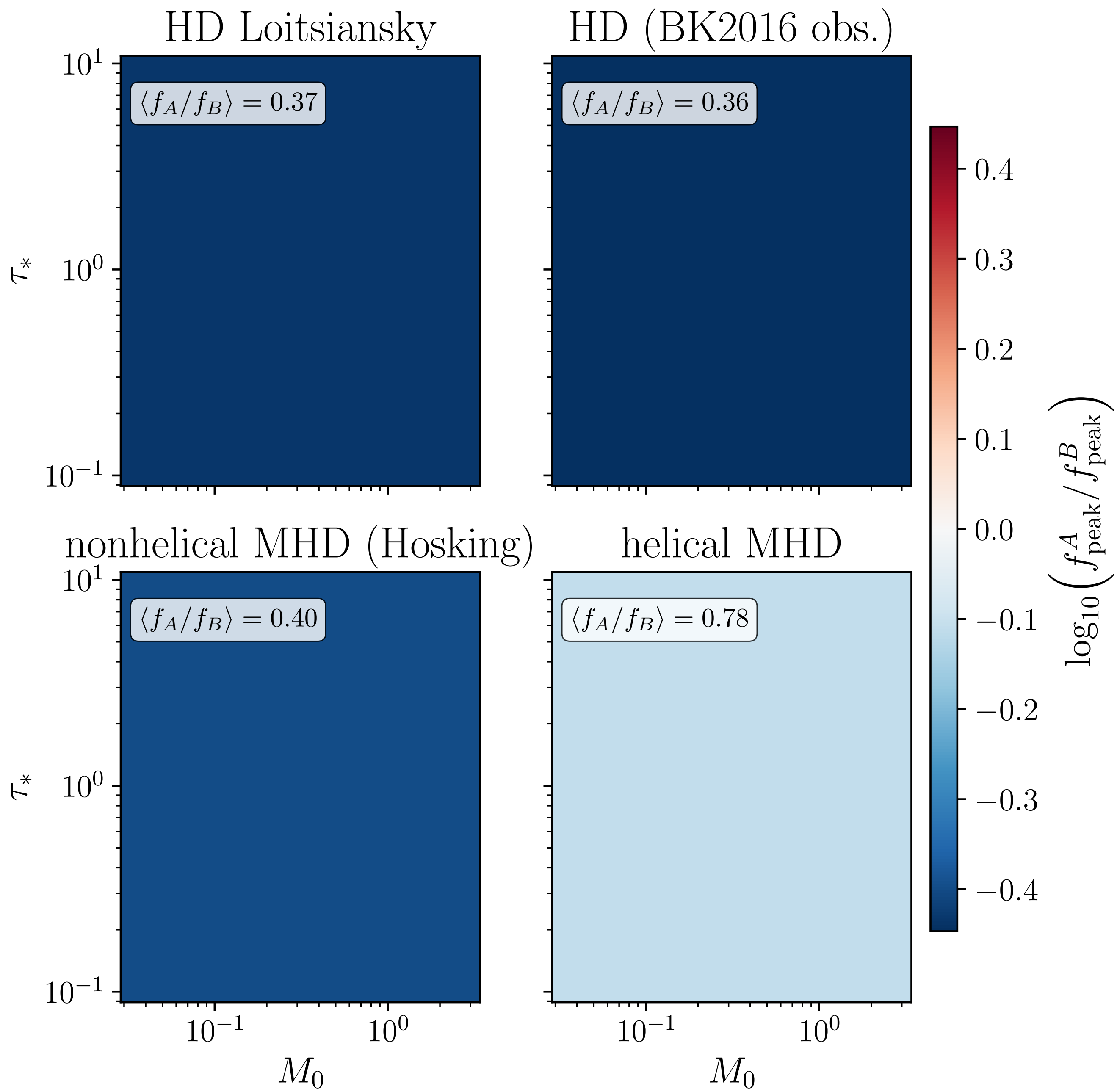
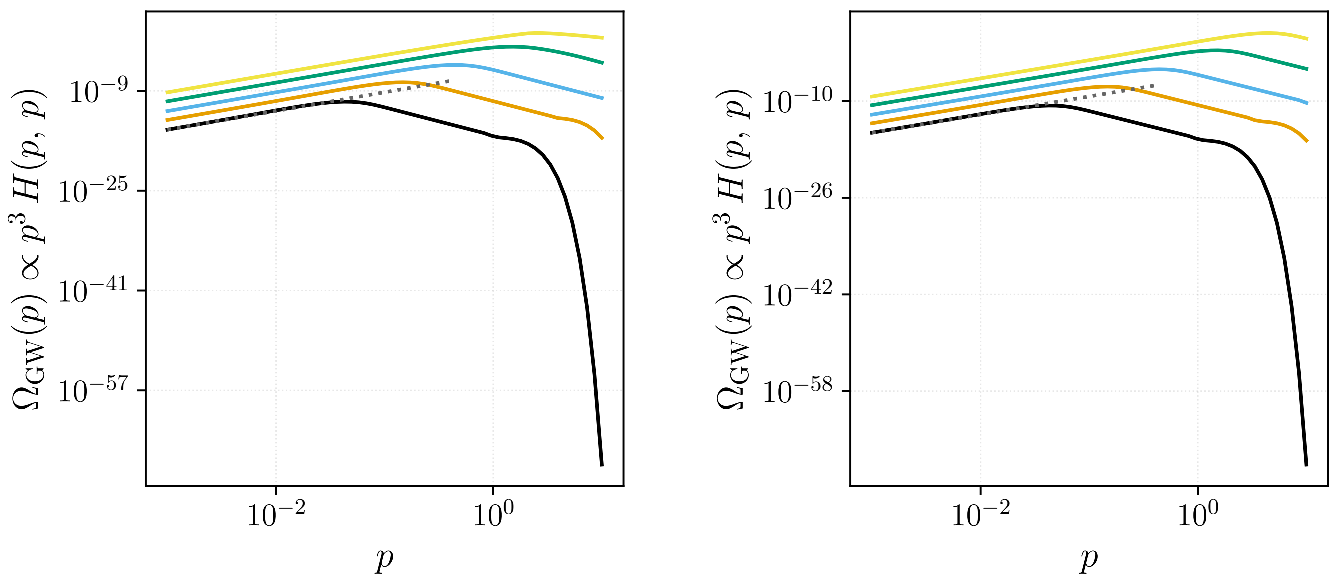
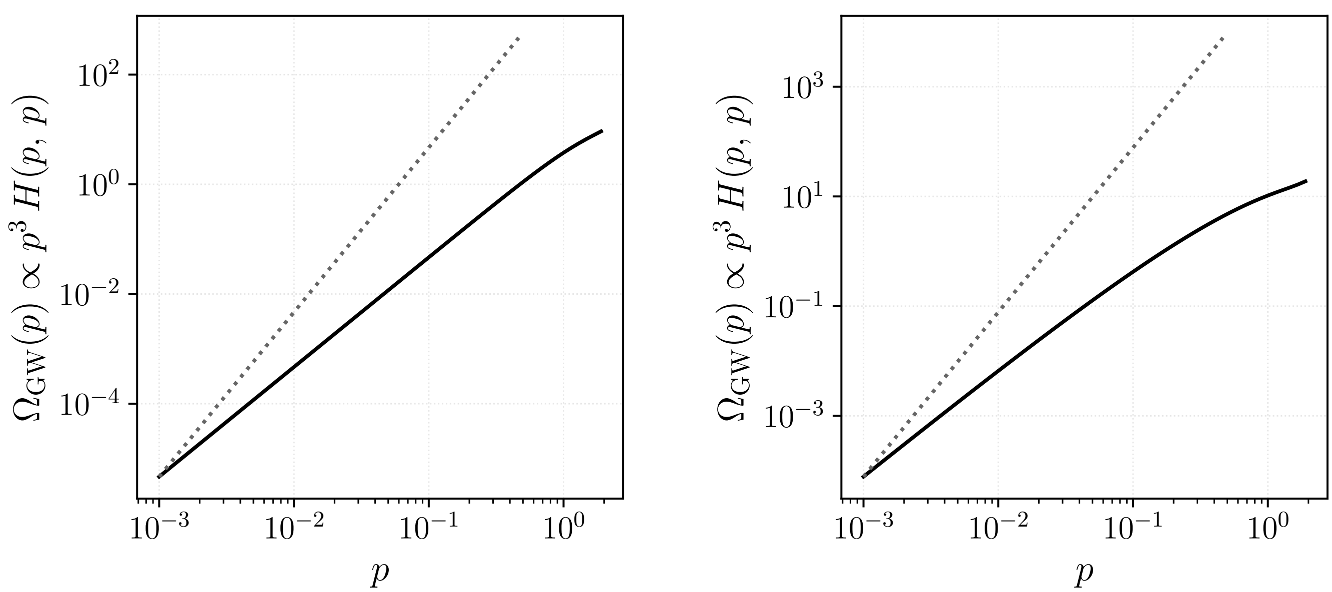


FIG. 21. Model A/B ratio of the peak GW frequency, $\log_{10}(f^A_{\rm peak}/f^B_{\rm peak})$, on the (τ_*, M_0) plane for the four BK2016 self-similar decay classes (HD Loitsiansky $q = 2/7$, $\beta = 4$; the HD class realised in the simulations of Ref. [39] $q = 1/3$, $\beta = 3$; nonhelical MHD $q = 4/9$, $\beta = 3/2$, the Hosking-integral class; helical MHD $q = 2/3$, $\beta = 0$). ~~HD Saffman $q = 1/3$, $\beta = 3$; nonhelical MHD $q = 1/2$, $\beta = 1$;~~ [the caption still documented the superseded $q = 1/2$, $\beta = 1$ magnetic class that the code no longer plots, and named the $\beta = 3$ panel "Saffman", which it is not ($\beta = 2$, $q = 2/5$)] Each panel is flat: the ratio is independent of (τ_*, M_0), fixed only by the kernel shapes, with the per-class value $\langle f_A/f_B\rangle$ annotated. Model A (random-phase) peaks below Model B (coherent eddy) everywhere. The offsets group as 0.37, 0.36, 0.40, 0.78: three classes together and helical MHD apart, not HD against MHD. ~~; the HD classes show the larger offset~~ [the old clause restated the HD-versus-MHD split that the text two paragraphs above now explicitly retracts]



(a) full H_{pq} *(b) $p \to 0$ limit*

FIG. 22. Kraichnan decorrelation + Kolmogorov spectrum. Five curves per panel correspond to $M = 0.03$, 0.1, 0.3, 1.0, 3.0 at $R = 10^4$. Panel (a) is the full numerical H_{pq} (Sec. III); panel (b) is the aeroacoustic $p \to 0$ limit (Kahniashvili et al. 2008). Both give the universal IR slope $\Omega_{\rm GW} \propto p^3$.



(a) Kraichnan *(b) Decay*

FIG. 23. Monochromatic spectrum $E(k) = E_0\delta(k - k_0)$ (no free parameters at the dimensionless level). Panel (a): Kraichnan decorrelation, closed form $\Omega_{\rm GW}(p) \propto p^3\, K_0(p)/p\, \Theta(2-p)\, e^{-p^2/(2\pi)}$ with IR slope **2** (flagged: $1/p$ pole from the monochromatic source removes one power of p from the universal scaling). Panel (b): decay temporal factor, Eq. (C8), IR slope **2** — the *same* slope as panel (a), as it must be, since the temporal factor tends to the constant $\pi\,\mathcal{C}(0)$ as $p \to 0$ and so cannot alter the infrared scaling.

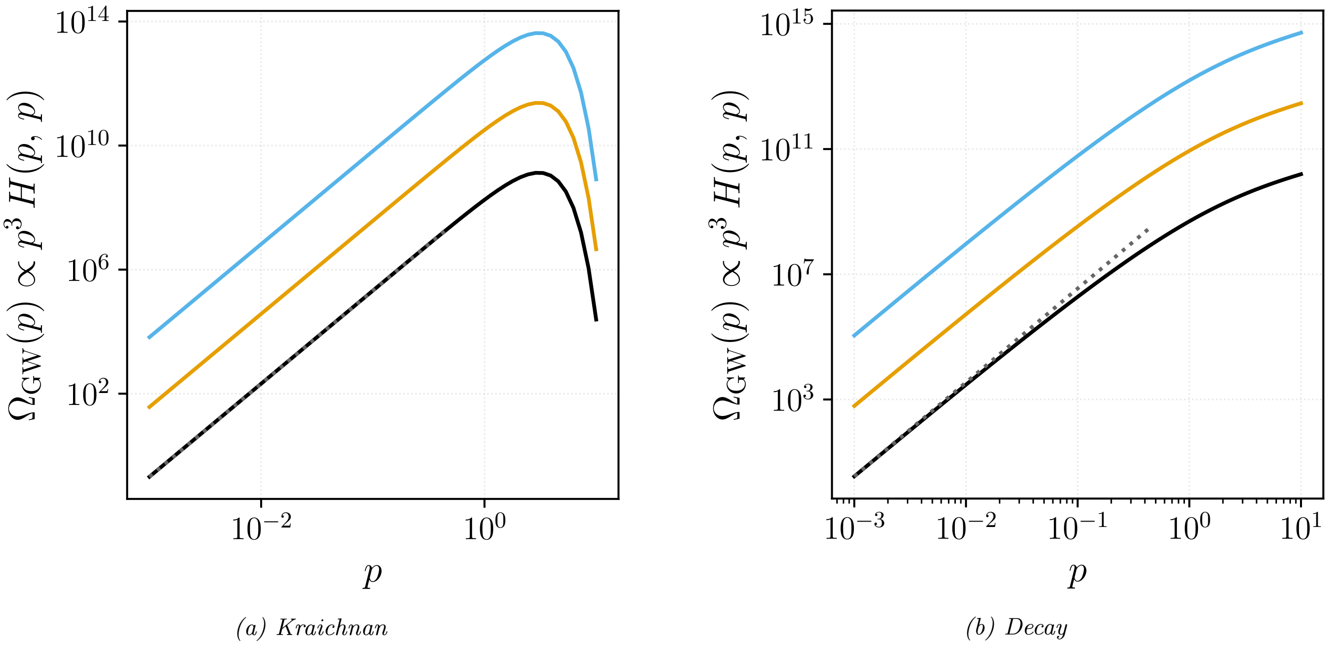


FIG. 24. White-noise real-space correlations $R_{ij} \propto \delta^{(3)}(\mathbf{r})$. Three curves per panel correspond to $R = 10^3$, 10^4, 10^5. Panel (a): Kraichnan decorrelation, matches the universal IR slope **3**. Panel (b): decay temporal model, IR slope **3** as well, for all three R — the decaying and Kraichnan models share the universal causal infrared, the temporal factor affecting only the amplitude and the ultraviolet tail.

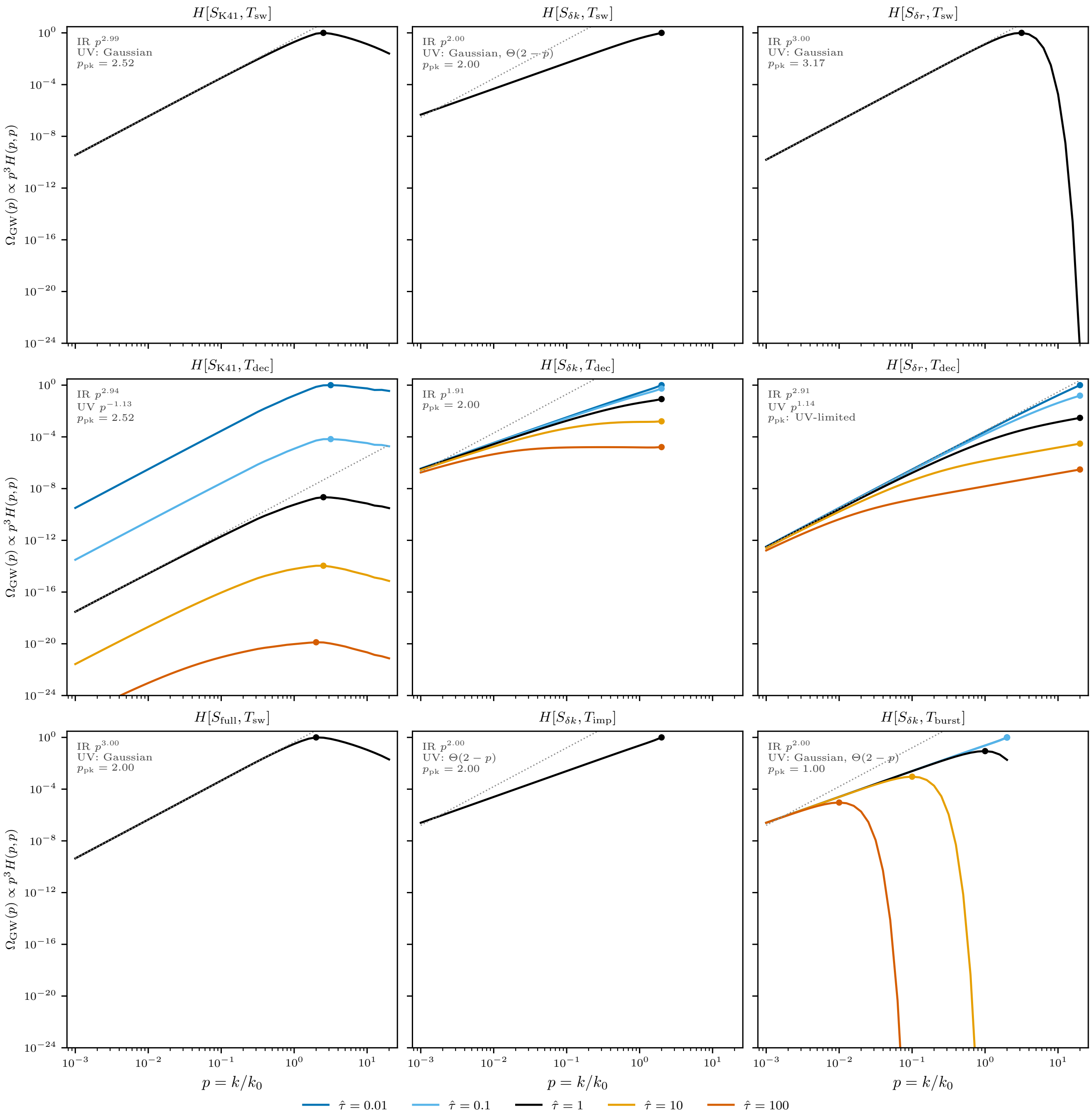


FIG. 25. The nine kernels of Table I at matched parameters, $\Omega_{\rm GW}(p) \propto p^3 H[S,T](p,p)$ on the sound-cone diagonal, $p = k/k_0$, at $R = 10^4$ (and $R_{\rm IR} = 10^2$ with a Batchelor infrared band for $S_{\rm full}$). Rows group the temporal model: sweeping (top), decaying (middle), and the full-spectrum, impulsive and burst cases (bottom). Curves are indexed by $\hat\tau = \tau_c k_0$ of Eq. (H2); stationary and time-independent kernels carry the single reference line $\hat\tau = 1$ (equivalently $M = \sqrt{2\pi} = 2.51$ for the sweeping panels), the time-dependent ones the five values $\hat\tau = 0.01, 0.1, 1, 10, 100$ (equivalently $M = 251, 25.1, 2.51, 0.251, 0.0251$; the small-$\hat\tau$ members are formal extrapolations past the luminal limit, retained as in Fig. 1 to expose the scaling). Each panel is normalised by its own maximum over all its curves, since the boxed kernels drop different dimensional prefactors and absolute amplitudes are not comparable between panels; within a panel the $\hat\tau$ dependence of the amplitude is preserved. All nine panels share identical axes, set from the data so that the peak of every curve is on scale: the floor lies four decades below the faintest peak (the $\hat\tau = 100$ member of $H[S_{\rm K41}, T_{\rm dec}]$, suppressed as M^4 by Eq. (38)), which costs 24 decades of ordinate. The deep dives of the Gaussian-cutoff kernels leave the frame by design and carry no information: the $\hat\tau = 100$ burst reaches 10^{-277} before the triangle inequality terminates it. Dots mark peaks, the dotted grey line is the causal p^3 reference, and the annotations give the measured infrared slope, the ultraviolet behaviour and the peak of the $\hat\tau = 1$ curve. Generated by `Notebooks/kernel_comparison_grid.py`.

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